Supplementary Information for

Accurate and Stable Computation of Gravitational Potential and Its Derivatives up to Third Order for Spherical Prisms

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The derivations support Section 2 of the main text; the numerical results supplement Section 3. The Fukushima QP checks are restricted to the quantities supported by the analytical comparisons. The independent mpmath reference evaluates all 20 gravitational quantities along the four profiles in Tests No. 11–14.

1 Detailed derivation of the radial-moment formulation

This section derives the radial moments and stable endpoint formulas used in Section 2.3 of the main text. Here Δ denotes differences of analytical primitives at the integration endpoints, not numerical differentiation of the potential. Throughout this section,

$$c = \cos\psi, \qquad s = \sin\psi, \qquad u = R - c, \qquad \ell^2 = u^2 + s^2, \tag{1}$$

and

$$u_1 = R_1 - c, \qquad u_2 = R_2 - c. \tag{2}$$

For convenience, define

$$\Delta J_n^{(k)}(u_1, u_2; s) = \int_{u_1}^{u_2} \frac{u^k}{(u^2 + s^2)^{n/2}}\, du, \tag{3}$$

and let $J_n^{(k)}$ denote any corresponding primitive. Additive constants in the primitives cancel from all finite differences.

1.1 From the radial functional to the two moment families

The normalized radial functional used in the main text is

$$\mathcal{J}_n[f] = \int_{R_1}^{R_2} \frac{R^2 f(R)}{\ell^n}\, dR. \tag{4}$$

The factor R^2 is inherited from the spherical volume element $r'^2 \cos\varphi'\, dr'\, d\varphi'\, d\lambda'$ after the normalization $R = r'/r$. For a monomial $f(R) = R^m$,

$$\mathcal{J}_n[R^m] = \int_{R_1}^{R_2} \frac{R^{m+2}}{\ell^n}\, dR \tag{5}$$

$$= M_n^{(m+2)}, \tag{6}$$

where

$$M_n^{(p)} = \int_{R_1}^{R_2} \frac{R^p}{\ell^n}\, dR. \tag{7}$$

Likewise, because $q_z = R\cos\psi - 1$,

$$\mathcal{J}_n[R^m q_z^\nu] = \int_{R_1}^{R_2} \frac{R^{m+2}(R\cos\psi - 1)^\nu}{\ell^n}\, dR \tag{8}$$

$$= Z_n^{(m+2,\nu)}, \tag{9}$$

with

$$Z_n^{(p,q)} = \int_{R_1}^{R_2} \frac{R^p(R\cos\psi - 1)^q}{\ell^n}\, dR. \tag{10}$$

Since T_x and T_y are independent of R, they can be taken outside the radial integral. Hence

$$\mathcal{J}_n[T_x^a T_y^b R^m q_z^\nu] = T_x^a T_y^b Z_n^{(m+2,\nu)}. \tag{11}$$

This explains both the shift $m \mapsto m+2$ and the use of the two moment families in the tensor assembly.

1.2 Derivation of the ordinary-moment expansion

With $R = u + c$ and $dR = du$, the ordinary moment becomes

$$M_n^{(p)} = \int_{u_1}^{u_2} \frac{(u+c)^p}{(u^2+s^2)^{n/2}}\, du. \tag{12}$$

The binomial theorem gives

$$(u+c)^p = \sum_{k=0}^{p} \binom{p}{k} u^k c^{p-k}. \tag{13}$$

Substitution into the integral and interchange of the finite sum and integration yield

$$M_n^{(p)} = \sum_{k=0}^{p} \binom{p}{k} c^{p-k} \int_{u_1}^{u_2} \frac{u^k}{(u^2+s^2)^{n/2}}\, du \tag{14}$$

$$= \sum_{k=0}^{p} \binom{p}{k} c^{p-k} \Delta J_n^{(k)}(u_1, u_2; s). \tag{15}$$

For example,

$$M_n^{(2)} = c^2 \Delta J_n^{(0)} + 2c\Delta J_n^{(1)} + \Delta J_n^{(2)}. \tag{16}$$

The combination coefficients therefore arise solely from the expansion of the shifted radial power $(u+c)^p$.

1.3 Derivation of the radial-displacement-weighted expansion

First,

$$R\cos\psi - 1 = Rc - 1 \tag{17}$$
$$= c(u+c) - 1 \tag{18}$$
$$= cu + c^2 - 1 \tag{19}$$
$$= cu - s^2, \tag{20}$$

where $c^2 + s^2 = 1$ has been used. Consequently,

$$Z_n^{(p,q)} = \int_{u_1}^{u_2} \frac{(u+c)^p (cu - s^2)^q}{(u^2+s^2)^{n/2}}\, du. \tag{21}$$

The two factors are expanded separately:

$$(u+c)^p = \sum_{k=0}^{p} \binom{p}{k} u^k c^{p-k}, \tag{22}$$

$$(cu - s^2)^q = \sum_{m=0}^{q} \binom{q}{m} (cu)^m (-s^2)^{q-m} = \sum_{m=0}^{q} \binom{q}{m} c^m u^m (-s^2)^{q-m}. \tag{23}$$

Their product is

$$(u+c)^p (cu - s^2)^q = \sum_{k=0}^{p} \sum_{m=0}^{q} \binom{p}{k} \binom{q}{m} c^{p-k+m} (-s^2)^{q-m} u^{k+m}. \tag{24}$$

Integrating term by term gives

$$Z_n^{(p,q)} = \sum_{k=0}^{p} \sum_{m=0}^{q} \binom{p}{k} \binom{q}{m} c^{p-k+m} (-s^2)^{q-m} \Delta J_n^{(k+m)}(u_1, u_2; s). \tag{25}$$

The order $k+m$ of the basic finite difference follows directly from $u^k u^m = u^{k+m}$. As a low-order check, for $p = q = 1$,

$$Z_n^{(1,1)} = c\Delta J_n^{(2)} + (c^2 - s^2)\Delta J_n^{(1)} - cs^2 \Delta J_n^{(0)}, \tag{26}$$

which agrees with the direct expansion $(u+c)(cu - s^2) = cu^2 + (c^2 - s^2)u - cs^2$.

1.4 Derivation of the three base primitives for $n = 1$

For $n = 1$ and $k = 0$,

$$J_1^{(0)} = \int \frac{du}{\sqrt{u^2 + s^2}}. \tag{27}$$

Using the hyperbolic substitution $u = s\sinh t$, one has $du = s\cosh t\, dt$ and

$$\sqrt{u^2 + s^2} = s\sqrt{1 + \sinh^2 t} = s\cosh t \tag{28}$$

for $s > 0$. Therefore

$$J_1^{(0)} = \int dt = t = \operatorname{asinh}\left(\frac{u}{s}\right). \tag{29}$$

For $k = 1$, let $w = u^2 + s^2$, so that $dw = 2u\,du$:

$$J_1^{(1)} = \int \frac{u}{\sqrt{u^2+s^2}}\,du \tag{30}$$

$$= \frac{1}{2}\int w^{-1/2}\,dw \tag{31}$$

$$= \sqrt{u^2+s^2}. \tag{32}$$

For $k = 2$, write $u^2 = (u^2 + s^2) - s^2$:

$$J_1^{(2)} = \int \frac{u^2}{\sqrt{u^2+s^2}}\,du \tag{33}$$

$$= \int \sqrt{u^2+s^2}\,du - s^2 J_1^{(0)}. \tag{34}$$

Using

$$\int \sqrt{u^2+s^2}\,du = \frac{1}{2}\left[u\sqrt{u^2+s^2} + s^2\,\mathrm{asinh}\left(\frac{u}{s}\right)\right], \tag{35}$$

we obtain

$$J_1^{(2)} = \frac{1}{2}\left[u\sqrt{u^2+s^2} - s^2\,\mathrm{asinh}\left(\frac{u}{s}\right)\right]. \tag{36}$$

Differentiation of these expressions recovers the corresponding integrands and provides a direct algebraic check.

1.5 Derivation of the recurrence for $J_n^{(0)}$

For odd $n \geq 3$, consider

$$F(u) = u(u^2+s^2)^{-(n-2)/2}. \tag{37}$$

Differentiation gives

$$F'(u) = (u^2+s^2)^{-(n-2)/2} - (n-2)u^2(u^2+s^2)^{-n/2} \tag{38}$$

$$= \left[(u^2+s^2) - (n-2)u^2\right](u^2+s^2)^{-n/2} \tag{39}$$

$$= \left[s^2 - (n-3)u^2\right](u^2+s^2)^{-n/2}. \tag{40}$$

Using $u^2 = (u^2 + s^2) - s^2$ in the last line yields

$$F'(u) = (n-2)s^2(u^2+s^2)^{-n/2} - (n-3)(u^2+s^2)^{-(n-2)/2}. \tag{41}$$

Solving for $(u^2+s^2)^{-n/2}$ and integrating gives

$$J_n^{(0)} = \frac{u}{(n-2)s^2(u^2+s^2)^{(n-2)/2}} + \frac{n-3}{(n-2)s^2}J_{n-2}^{(0)}. \tag{42}$$

This derivation explains the simultaneous appearance of the factors $n-2$, $n-3$, and s^{-2}.

1.6 Derivation of $J_n^{(1)}$

For odd $n \geq 3$,

$$J_n^{(1)} = \int u(u^2+s^2)^{-n/2}\,du. \tag{43}$$

With $w = u^2 + s^2$ and $dw = 2u\,du$,

$$J_n^{(1)} = \frac{1}{2}\int w^{-n/2}\,dw \tag{44}$$

$$= \frac{w^{1-n/2}}{2-n} \tag{45}$$

$$= -\frac{1}{(n-2)(u^2+s^2)^{(n-2)/2}}. \tag{46}$$

This formula supplies the first-order seed required by the recurrence for higher powers of u.

1.7 Derivation of the higher-order recurrence in k

For $k \geq 2$,

$$J_n^{(k)} = \int \frac{u^k}{(u^2+s^2)^{n/2}}\,du \tag{47}$$

$$= \int \frac{u^{k-2}u^2}{(u^2+s^2)^{n/2}}\,du. \tag{48}$$

Substituting $u^2 = (u^2+s^2) - s^2$ gives

$$J_n^{(k)} = \int \frac{u^{k-2}}{(u^2+s^2)^{(n-2)/2}}\,du - s^2\int \frac{u^{k-2}}{(u^2+s^2)^{n/2}}\,du \tag{49}$$

$$= J_{n-2}^{(k-2)} - s^2 J_n^{(k-2)}. \tag{50}$$

Thus both indices decrease by two in the first term, whereas only the power index decreases in the second term.

1.8 From primitive recurrences to finite-difference recurrences

For any primitive identity of the form

$$J_n^{(k)}(u) = A(u) + aJ_{n-2}^{(k-2)}(u) + bJ_n^{(k-2)}(u), \tag{51}$$

taking its value at u_2 minus its value at u_1 gives

$$\Delta J_n^{(k)} = \Delta A + a\Delta J_{n-2}^{(k-2)} + b\Delta J_n^{(k-2)}. \tag{52}$$

Applying this operation to the preceding primitive formulas yields

$$\Delta J_n^{(0)} = \frac{1}{(n-2)s^2}\Delta\left[\frac{u}{(u^2+s^2)^{(n-2)/2}}\right] + \frac{n-3}{(n-2)s^2}\Delta J_{n-2}^{(0)}, \tag{53}$$

$$\Delta J_n^{(1)} = -\frac{1}{n-2}\Delta\left[(u^2+s^2)^{-(n-2)/2}\right], \tag{54}$$

and

$$\Delta J_n^{(k)} = \Delta J_{n-2}^{(k-2)} - s^2\Delta J_n^{(k-2)}, \qquad k \geq 2. \tag{55}$$

Although these relations follow immediately from linearity, directly subtracting two separately rounded primitive values can lose many significant digits when $u_2 \approx u_1$. The remaining subsec-

tions derive equivalent endpoint-difference formulas that retain the small interval scale explicitly.

1.9 Cancellation-resistant evaluation of $\Delta J_1^{(0)}$

Because

$$\operatorname{asinh} x = \log\left(x + \sqrt{1+x^2}\right), \tag{56}$$

we may write, for $s > 0$,

$$J_1^{(0)}(u) = \log\left(\frac{u + \sqrt{u^2+s^2}}{s}\right). \tag{57}$$

Let $L_i = \sqrt{u_i^2 + s^2}$. If $0 \le u_1 < u_2$, the common factor s cancels and

$$\Delta J_1^{(0)} = \log\left(\frac{u_2 + L_2}{u_1 + L_1}\right). \tag{58}$$

For negative u, the sum $L + u$ can suffer severe cancellation because $L \approx |u|$. The identity

$$(L+u)(L-u) = L^2 - u^2 = s^2 \tag{59}$$

provides the stable replacement $L + u = s^2/(L-u)$. Thus, if $u_1 < u_2 \le 0$,

$$\Delta J_1^{(0)} = \log\left(\frac{u_2 + L_2}{u_1 + L_1}\right) \tag{60}$$

$$= \log\left(\frac{L_1 - u_1}{L_2 - u_2}\right). \tag{61}$$

For nearby endpoints, a ratio a_2/a_1 is evaluated as

$$\log\left(\frac{a_2}{a_1}\right) = \operatorname{log1p}\left(\frac{a_2 - a_1}{a_1}\right), \tag{62}$$

which avoids forming a ratio numerically indistinguishable from unity.

1.10 Rationalized square-root and product differences

The first-order base difference is

$$\Delta J_1^{(1)} = L_2 - L_1. \tag{63}$$

Multiplication by the conjugate gives

$$L_2 - L_1 = \frac{L_2^2 - L_1^2}{L_2 + L_1} \tag{64}$$

$$= \frac{u_2^2 - u_1^2}{L_2 + L_1} \tag{65}$$

$$= \frac{(u_2 - u_1)(u_2 + u_1)}{L_2 + L_1}. \tag{66}$$

The second-order base difference contains

$$\Delta(uL) = u_2 L_2 - u_1 L_1. \tag{67}$$

On a same-sign interval, set $A = u_2 L_2$ and $B = u_1 L_1$. Then

$$A - B = \frac{A^2 - B^2}{A + B}. \tag{68}$$

$$\begin{aligned} A^2 - B^2 &= u_2^2(u_2^2 + s^2) - u_1^2(u_1^2 + s^2) && (69)\\ &= (u_2^2 - u_1^2)(u_2^2 + u_1^2 + s^2) && (70)\\ &= (u_2 - u_1)(u_2 + u_1)(u_2^2 + u_1^2 + s^2). && (71) \end{aligned}$$

Hence

$$\Delta(uL) = \frac{(u_2 - u_1)(u_2 + u_1)(u_2^2 + u_1^2 + s^2)}{u_2 L_2 + u_1 L_1}, \tag{72}$$

and

$$\Delta J_1^{(2)} = \frac{1}{2}\left[\Delta(uL) - s^2 \Delta J_1^{(0)}\right]. \tag{73}$$

These rationalized forms expose the small factor $u_2 - u_1$ rather than obtaining it indirectly by subtracting two close square roots or products.

1.11 Why an interval crossing $u = 0$ is split

The same-sign formulas above are constructed so that the potentially cancelling quantities have a consistent sign and a well-scaled denominator. If $u_1 u_2 \le 0$, the interval contains the point at which

$$\ell^2 = u^2 + s^2 \tag{74}$$

attains its minimum. In a nearly collinear configuration, s can be very small, so the kernel may become sharply localized near $u = 0$. A same-sign rationalization used across this point may also produce a small or sign-changing denominator, for example $u_2 L_2 + u_1 L_1$ in the expression for $\Delta(uL)$. We therefore decompose

$$[u_1, u_2] = [u_1, 0] \cup [0, u_2] \tag{75}$$

whenever $u_1 < 0 < u_2$ and evaluate the two subintervals independently. At a zero endpoint, the corresponding same-sign expression remains valid for $s > 0$. In particular, $\Delta J_1^{(0)} = \operatorname{asinh}(u_2/s) + \operatorname{asinh}(-u_1/s)$ across zero. The resulting finite differences are then added. This split is algebraically exact and preserves the assumptions under which each stable same-sign expression was derived.

1.12 Derivation of the nearly collinear small-s expansion

Assume $u_1 u_2 > 0$ and define

$$\sigma = \operatorname{sign}(u_1) = \operatorname{sign}(u_2), \qquad q_1 = \min(|u_1|, |u_2|), \qquad q_2 = \max(|u_1|, |u_2|). \tag{76}$$

With $u = \sigma q$, the finite difference is

$$\Delta J_n^{(k)} = \sigma^k \int_{q_1}^{q_2} q^{k-n} \left(1 + \frac{s^2}{q^2}\right)^{-n/2} dq. \tag{77}$$

For $s/q_1 \ll 1$, apply the generalized binomial series

$$(1+x)^{\alpha} = \sum_{j=0}^{4} \binom{\alpha}{j} x^j + O(x^5) \tag{78}$$

with $\alpha = -n/2$ and $x = s^2/q^2$. This gives

$$\begin{aligned}\left(1+\frac{s^2}{q^2}\right)^{-n/2} &= 1 - \frac{n}{2}\frac{s^2}{q^2} + \frac{n(n+2)}{8}\frac{s^4}{q^4} \\ &\quad - \frac{n(n+2)(n+4)}{48}\frac{s^6}{q^6} + \frac{n(n+2)(n+4)(n+6)}{384}\frac{s^8}{q^8} + O\left(\frac{s^{10}}{q^{10}}\right).\end{aligned} \tag{79}$$

Termwise integration then yields

$$\begin{aligned}\Delta J_n^{(k)} = \sigma^k \Bigg[& P_{k-n} - \frac{n}{2}s^2 P_{k-n-2} + \frac{n(n+2)}{8}s^4 P_{k-n-4} \\ & - \frac{n(n+2)(n+4)}{48}s^6 P_{k-n-6} + \frac{n(n+2)(n+4)(n+6)}{384}s^8 P_{k-n-8} \\ & + O(s^{10} P_{k-n-10}) \Bigg].\end{aligned} \tag{80}$$

where

$$P_{\alpha}(q_1, q_2) = \int_{q_1}^{q_2} q^{\alpha}\, dq = \begin{cases} \log(q_2/q_1), & \alpha = -1, \\ \dfrac{q_2^{\alpha+1} - q_1^{\alpha+1}}{\alpha+1}, & \alpha \neq -1. \end{cases} \tag{81}$$

For nearby endpoints, define

$$\delta = \log\left(\frac{q_2}{q_1}\right) = \operatorname{log1p}\left(\frac{q_2 - q_1}{q_1}\right). \tag{82}$$

When $\alpha \neq -1$,

$$P_{\alpha} = q_1^{\alpha+1}\frac{\exp[(\alpha+1)\delta] - 1}{\alpha+1} \tag{83}$$

$$= q_1^{\alpha+1}\frac{\operatorname{expm1}[(\alpha+1)\delta]}{\alpha+1}. \tag{84}$$

The actual asymptotic control parameter is s/q_1, not s alone. Accordingly, the expansion is not applied when the interval approaches or crosses $u = 0$, because then q_1 is small and the series is not uniformly valid.

1.13 Summary of the stable evaluation sequence

The complete radial calculation can be organized as follows:

(1) Transform R to $u = R - c$ and compute u_1, u_2, c, and s.

(2) If $u_1 u_2 \leq 0$, split the interval at $u = 0$ and evaluate the two pieces separately.

(3) On a same-sign interval, use the small-s expansion through s^8 when $s/q_1 < 10^{-2}$; otherwise use the finite-difference recurrences.

(4) Evaluate the $n = 1$ base differences with logarithmic ratios, `log1p`, and rationalized endpoint differences.

(5) Build higher n and k values using the finite-difference recurrences rather than separately evaluated primitives.

(6) Assemble $M_n^{(p)}$ and $Z_n^{(p,q)}$ from the required $\Delta J_n^{(k)}$, and then construct the tensor kernels.

The rationalized identities and recurrences evaluate the defining radial integral while limiting endpoint cancellation. The small-s series provides an asymptotic approximation on same-sign intervals, controlled by s/q_1.

2 Morphology of the integral kernels

We examine the morphology of the integral kernels using a single spherical prism bounded radially by $r_1 = R_\mathrm{E} - 30$ km and $r_2 = R_\mathrm{E}$, latitudinally by $\varphi_1 = 30^\circ$ and $\varphi_2 = 50^\circ$, and longitudinally by $\lambda_1 = 110^\circ$ and $\lambda_2 = 130^\circ$. Here, R_E denotes the reference radius of the Earth.

Three source–observation configurations are considered. Test S1 places the observation point directly above the prism at $r = R_\mathrm{E} + 300$ km, $\varphi = 40^\circ$, and $\lambda = 120^\circ$. Test S2 moves the point to $r = R_\mathrm{E} + 50$ km at the same latitude and longitude. Test S3 retains this radius but places the point at $\varphi = 90^\circ$ and $\lambda = 0^\circ$, far from the prism in angular position. Figs. S1–S3 show the corresponding kernels.

The kernels become narrower and sharper as the observation point approaches the prism above its angular projection. They remain smooth for the angularly separated configuration. Resolving this localization requires angular refinement in addition to stable radial evaluation.

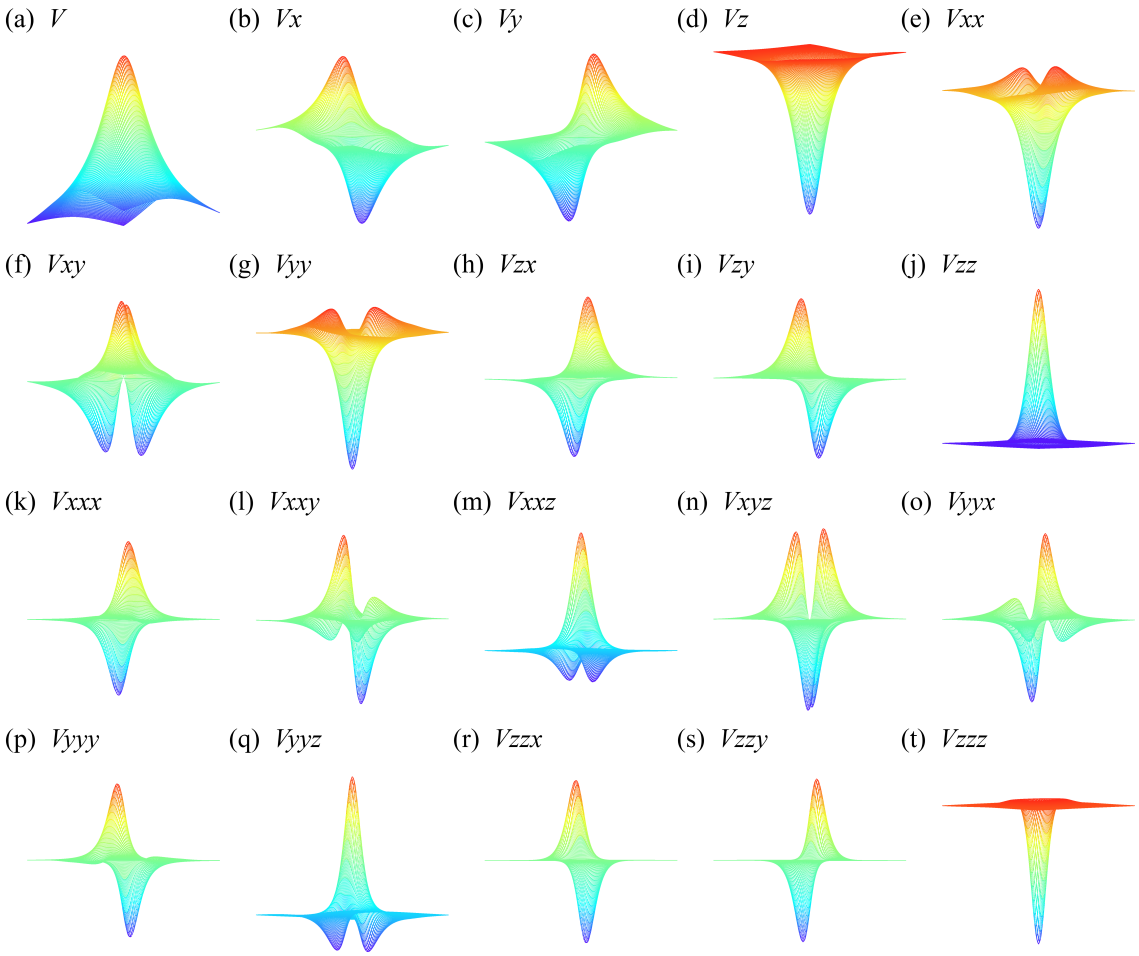


Figure S1: Test S1: integral-kernel morphology for an observation point 300 km directly above the spherical prism. The panels show the gravitational potential, acceleration components, gravity-gradient components, and third-order gravitational derivatives.

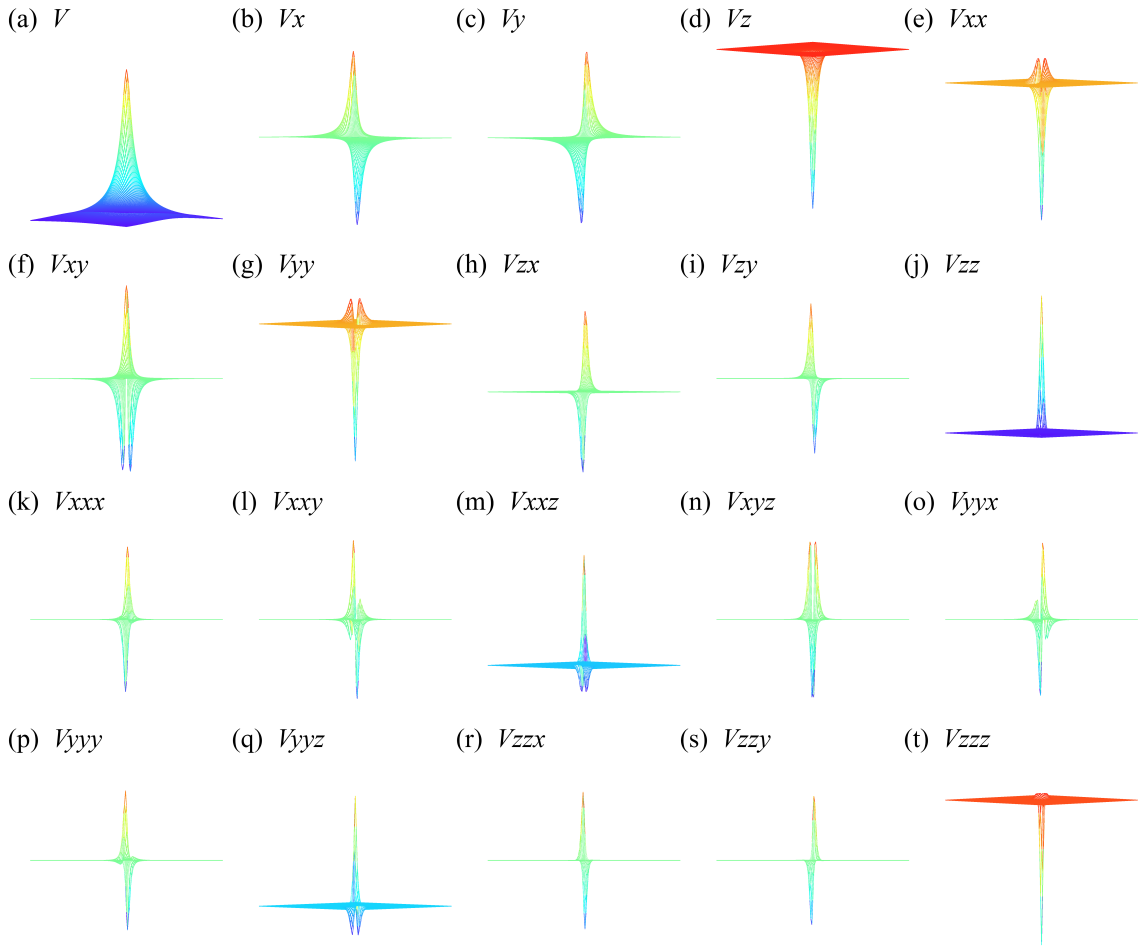


Figure S2: Test S2: integral-kernel morphology for an observation point 50 km directly above the spherical prism. Relative to Test S1, the near-zone features are substantially narrower and sharper.

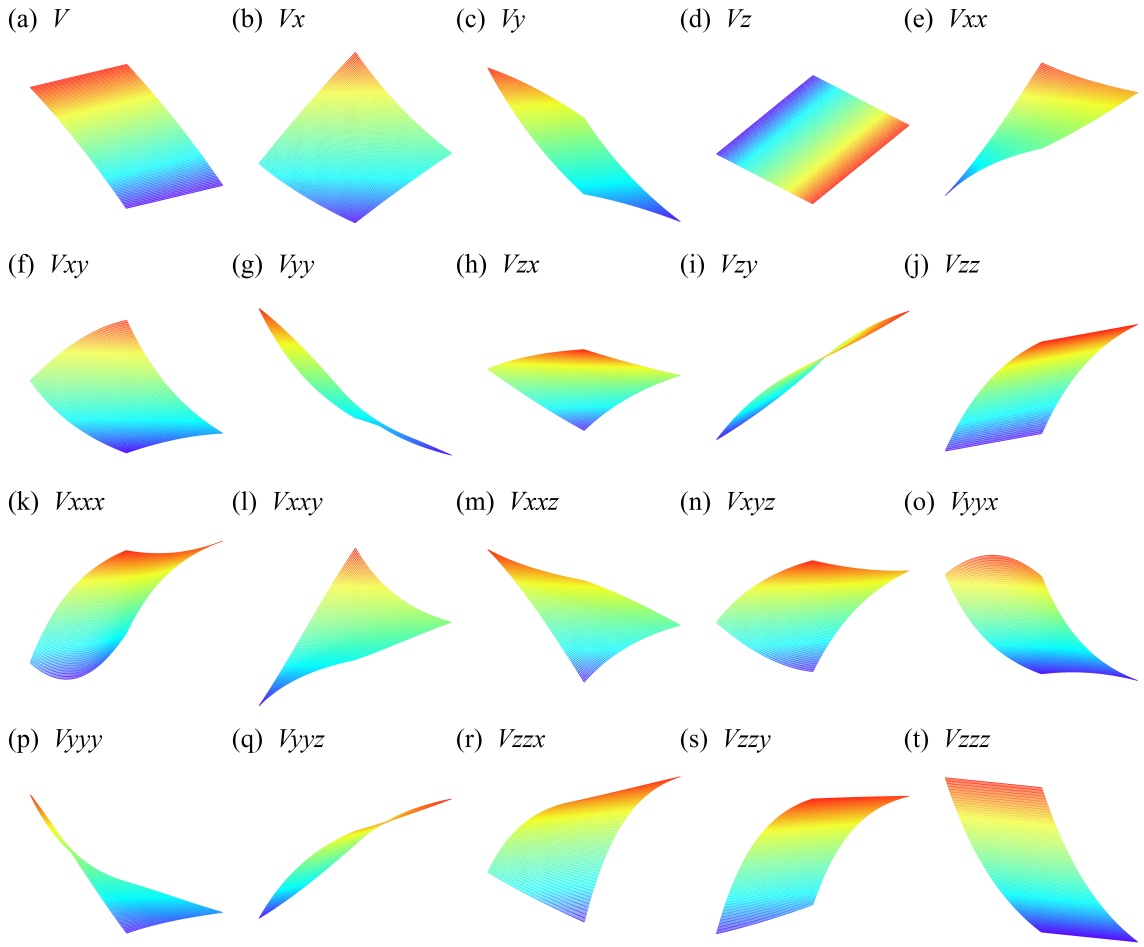


Figure S3: Test S3: integral-kernel morphology for an observation point 50 km above the reference sphere and angularly separated from the spherical prism. The kernels vary smoothly across the source domain.

3 Validation against analytical polar-axis solutions

The analytical references of Deng and Sneeuw (2023) are used in the four tests described in Section 3.1 of the main text. Four source geometries are considered: Tests No. 1 and 2 use the high-latitude interval $69.99° \leq \varphi' \leq 89.99°$ with longitude spans of 20° and 360°, respectively, whereas Tests No. 3 and 4 use the mid-latitude interval $35° \leq \varphi' \leq 55°$ with the same two longitude spans. In every test, $r_1 = R_E - 30$ km and $r_2 = R_E$, and the field is evaluated along the positive polar axis from $R_E - 1000$ km to $R_E + 1000$ km at a spacing of 1 km.

For each field component X, the absolute and relative errors are defined as

$$\epsilon_{\mathrm{abs}} = |X_{\mathrm{num}} - X_{\mathrm{ana}}|, \qquad \epsilon_{\mathrm{rel}} = \frac{|X_{\mathrm{num}} - X_{\mathrm{ana}}|}{|X_{\mathrm{ana}}|}. \tag{85}$$

Relative errors are omitted from the statistics where $X_{\mathrm{ana}} = 0$. The complete graphical comparisons and statistics below complement the two representative tests shown in the main text. As in Section 3.1 of the main text, all reported gravitational values and absolute errors use $G = 6.67430 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$ and $\rho = 1\ \mathrm{kg\,m^{-3}}$. The field-unit convention throughout the gravitational validation is

$$[V] = \mathrm{m^2\,s^{-2}}, \qquad [V_i] = \mathrm{mGal}, \qquad [V_{ij}] = \mathrm{E}, \qquad [V_{ijk}] = \mathrm{E\,km^{-1}}. \tag{86}$$

Absolute errors, standard deviations, and RMSE values carry the unit of the corresponding field component; relative errors are dimensionless. The second-order trace residual is in E and its differentiated residuals are in $\mathrm{E\,km^{-1}}$. The derivation and conversion coefficients are given in Section 8.5.

3.1 Relative and absolute errors

Figs. S4 and S5 show the relative errors for Tests No. 3 and 4, complementing Figs. 2 and 3 of the main text. Figs. S6–S9 show the absolute errors for all four tests. Together, these results distinguish genuine near-zone loss of accuracy from large relative errors caused by analytical values that approach zero.

3.2 Analytical and computed values

Figures S10–S13 compare the analytical field values with the three numerical results, showing their amplitudes and near-source extrema.

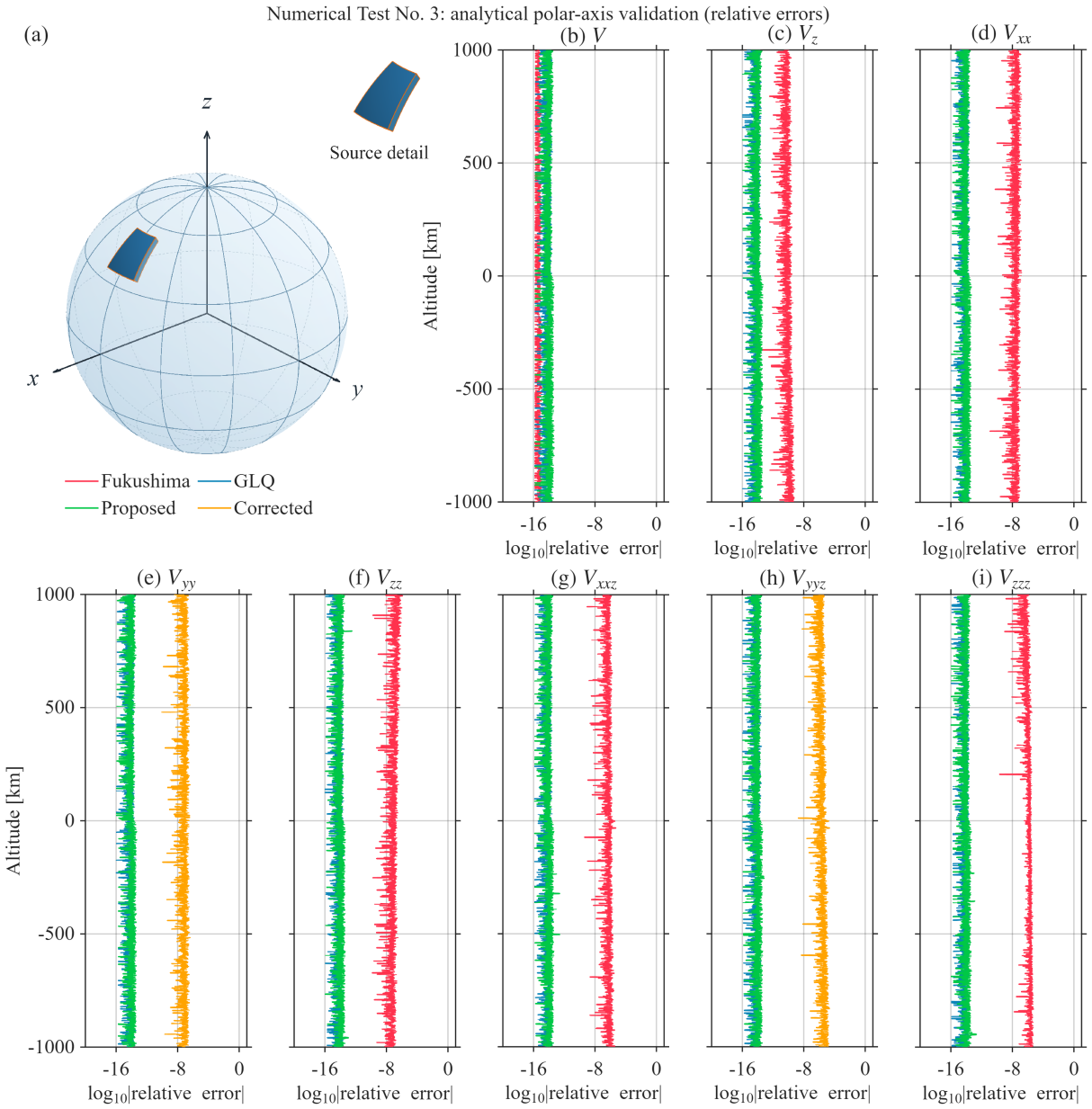


Figure S4: Numerical Test No. 3. Panel (a) shows the spherical-prism geometry. Panels (b)–(i) show the relative errors for V, V_z, V_{xx}, V_{yy}, V_{zz}, V_{xxz}, V_{yyz}, and V_{zzz}, respectively. Fukushima's method is shown in red, its Laplace-equation corrections in yellow, 30-node GLQ in blue, and the proposed method in green.

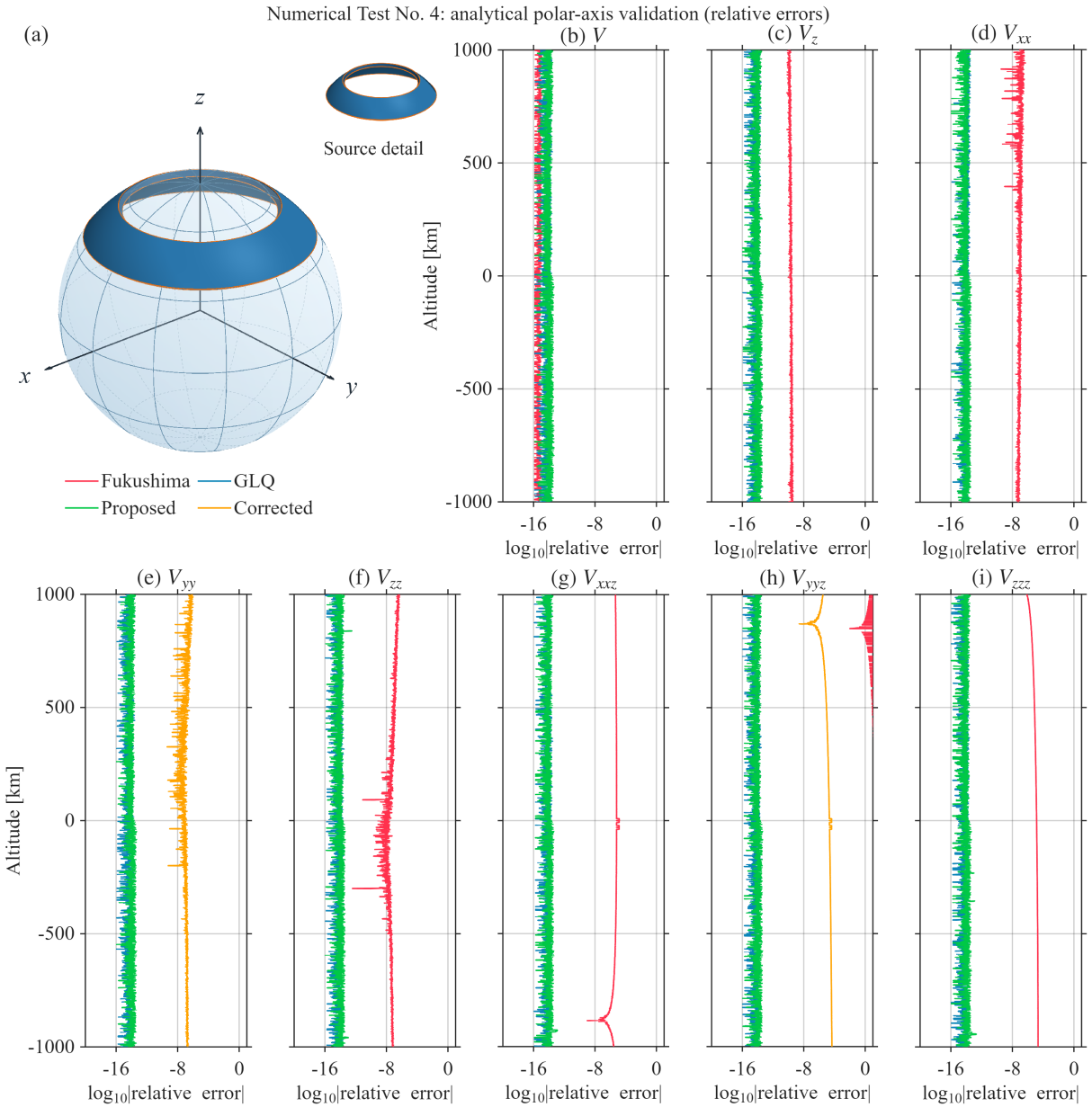


Figure S5: Numerical Test No. 4. Panel (a) shows the spherical-cap geometry. Panels (b)–(i) show the relative errors for V, V_z, V_{xx}, V_{yy}, V_{zz}, V_{xxz}, V_{yyz}, and V_{zzz}, respectively. Line colours and corrections are the same as in Fig. S4.

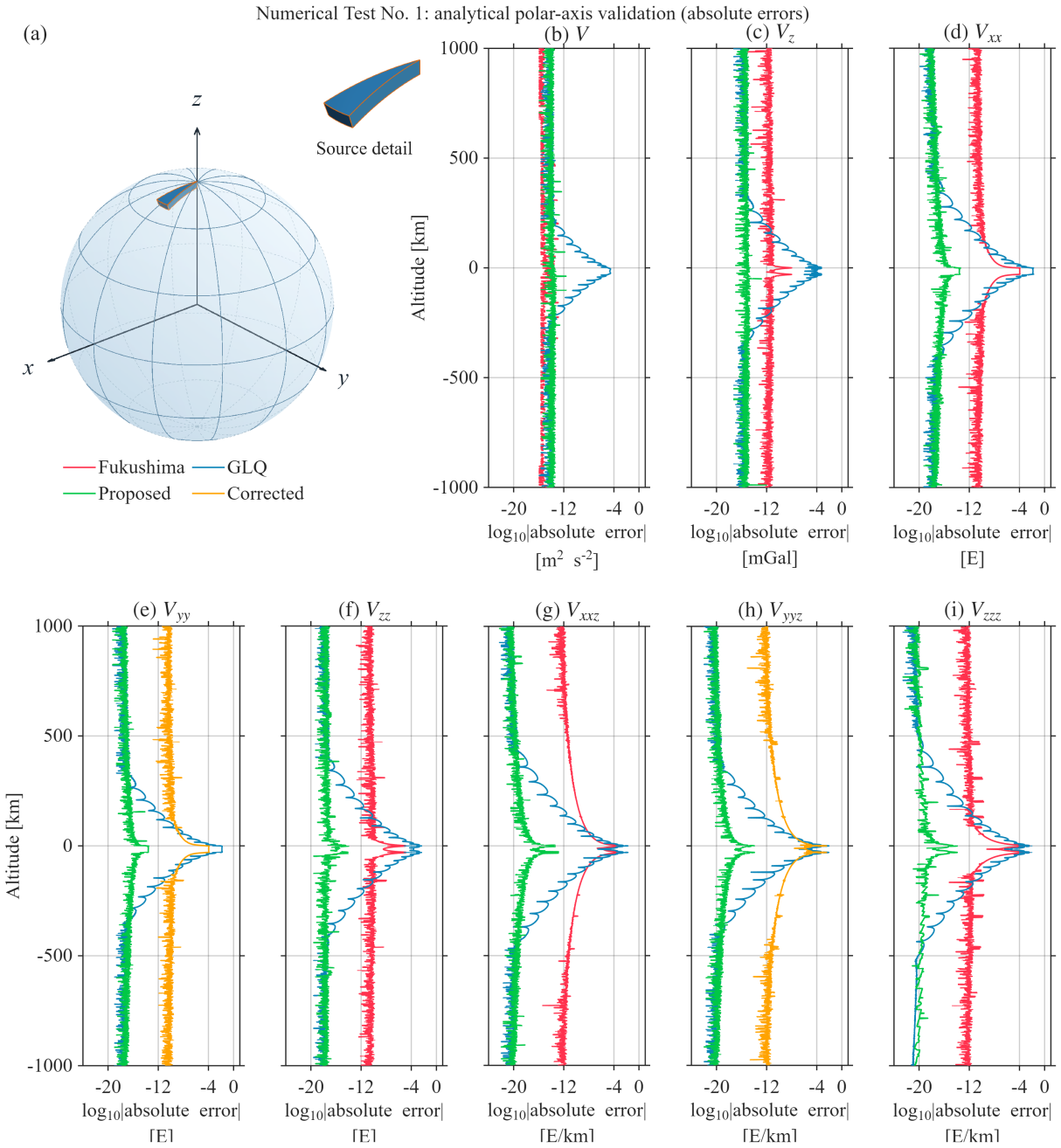


Figure S6: Numerical Test No. 1. Panel (a) shows the spherical-prism geometry. Panels (b)–(i) show the absolute errors for V, V_z, V_{xx}, V_{yy}, V_{zz}, V_{xxz}, V_{yyz}, and V_{zzz}, respectively. Fukushima's method is shown in red, its Laplace-equation corrections in yellow, 30-node GLQ in blue, and the proposed method in green. Physical units follow Eq. (86).

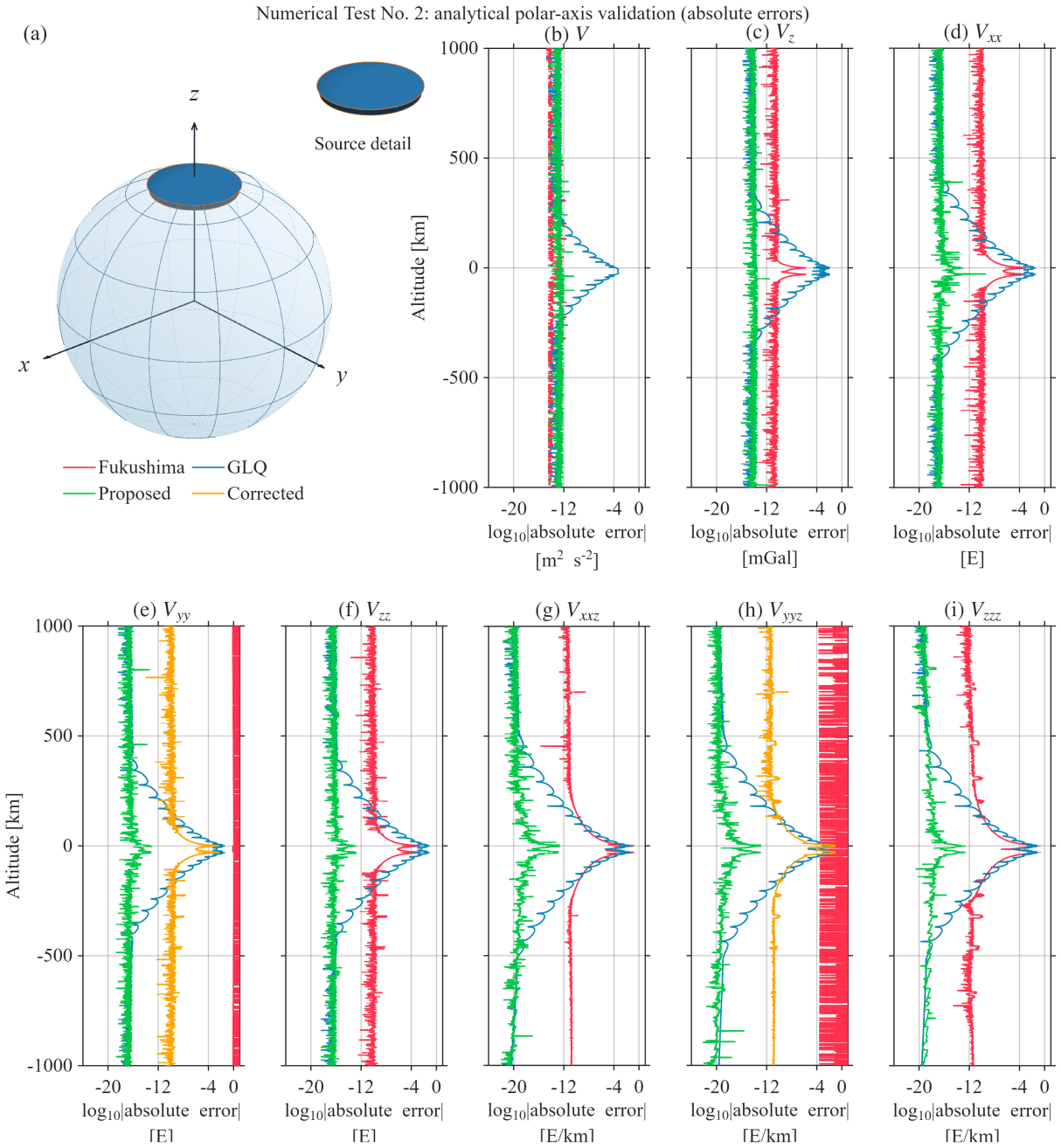


Figure S7: Numerical Test No. 2. Panel (a) shows the spherical-cap geometry; panels (b)–(i) show the absolute errors for the eight field components listed in Fig. S6. Line colours and corrections are the same as in Fig. S6. Physical units follow Eq. (86).

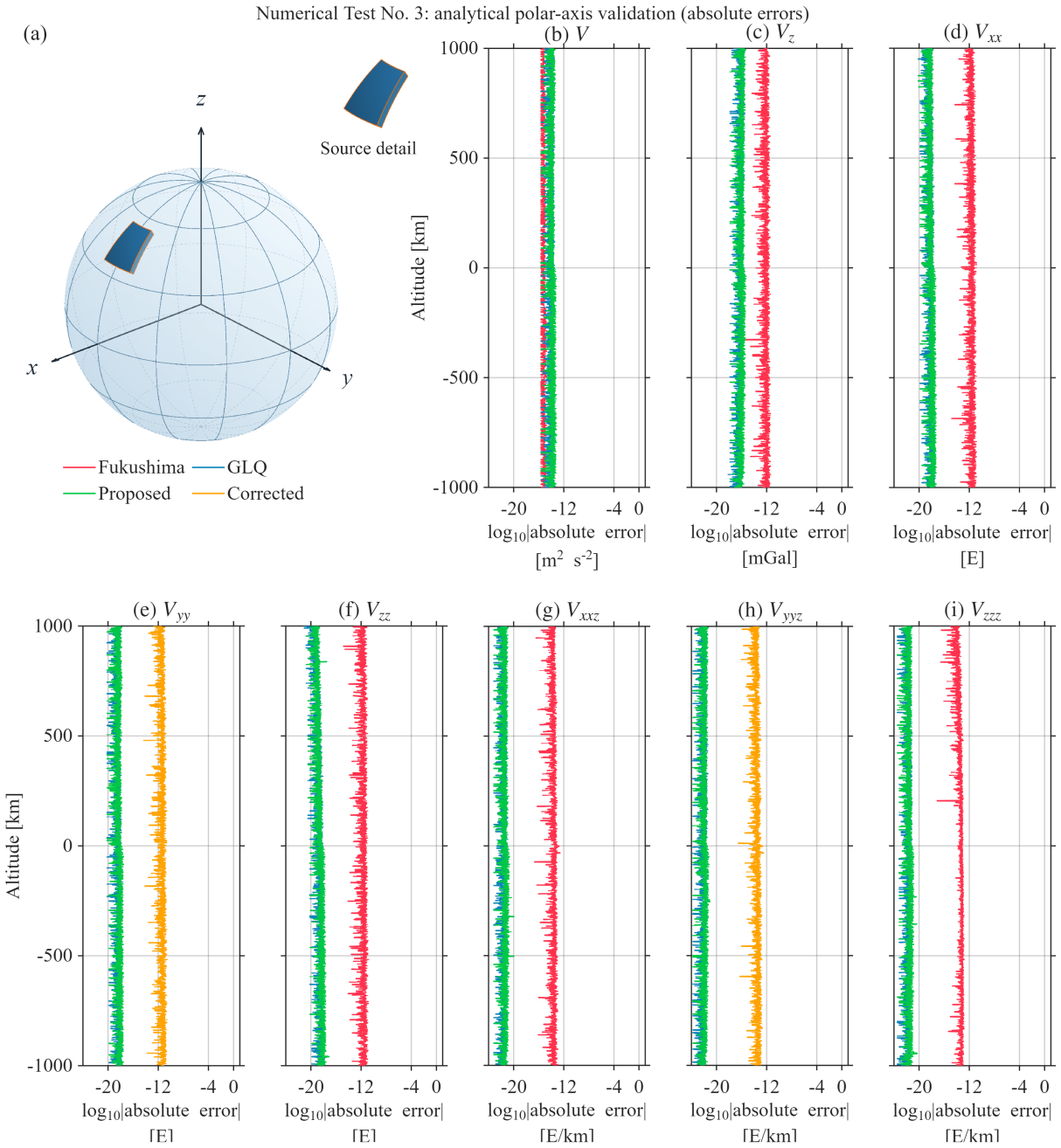


Figure S8: Numerical Test No. 3. Panel (a) shows the spherical-prism geometry; panels (b)–(i) show the absolute errors for the eight field components listed in Fig. S6. Line colours and corrections are the same as in Fig. S6. Physical units follow Eq. (86).

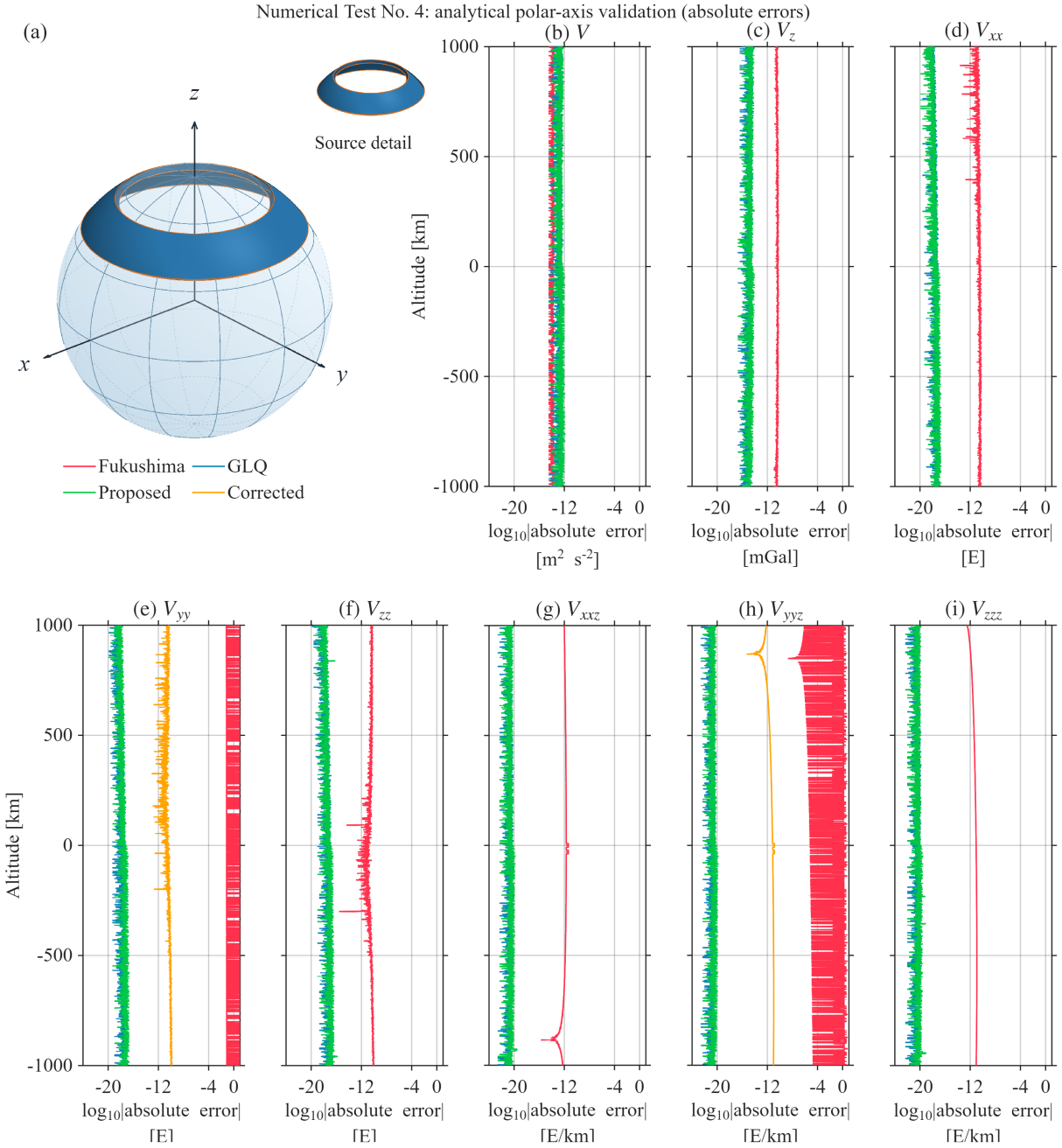


Figure S9: Numerical Test No. 4. Panel (a) shows the spherical-cap geometry; panels (b)–(i) show the absolute errors for the eight field components listed in Fig. S6. Line colours and corrections are the same as in Fig. S6. Physical units follow Eq. (86).

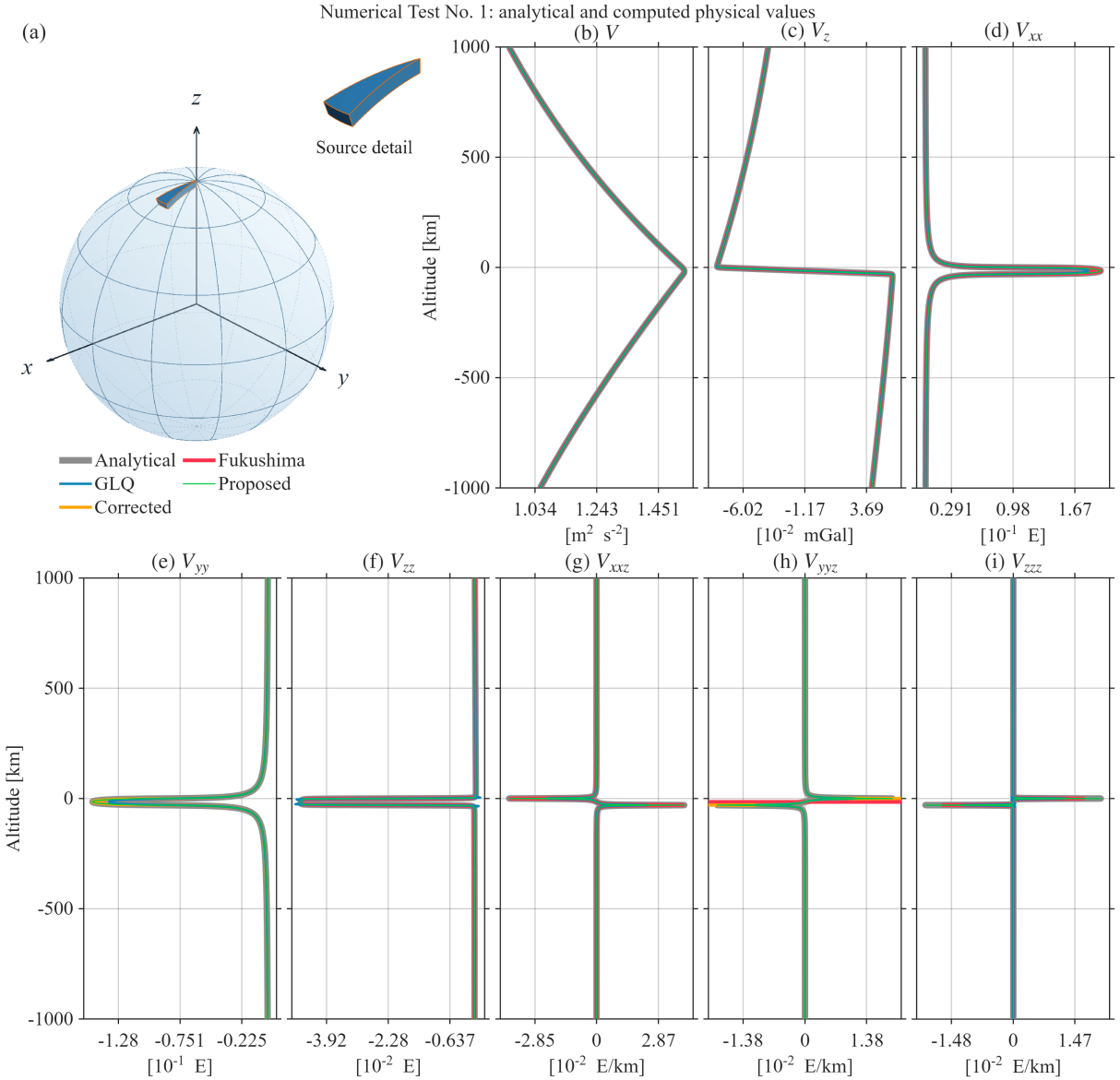


Figure S10: Numerical Test No. 1. Panel (a) shows the spherical-prism geometry. Panels (b)–(i) compare the analytical and computed values of V, V_z, V_{xx}, V_{yy}, V_{zz}, V_{xxz}, V_{yyz}, and V_{zzz}, respectively. Analytical values are grey, Fukushima's method red, its corrected components yellow, GLQ blue, and the proposed method green. Physical units follow Eq. (86).

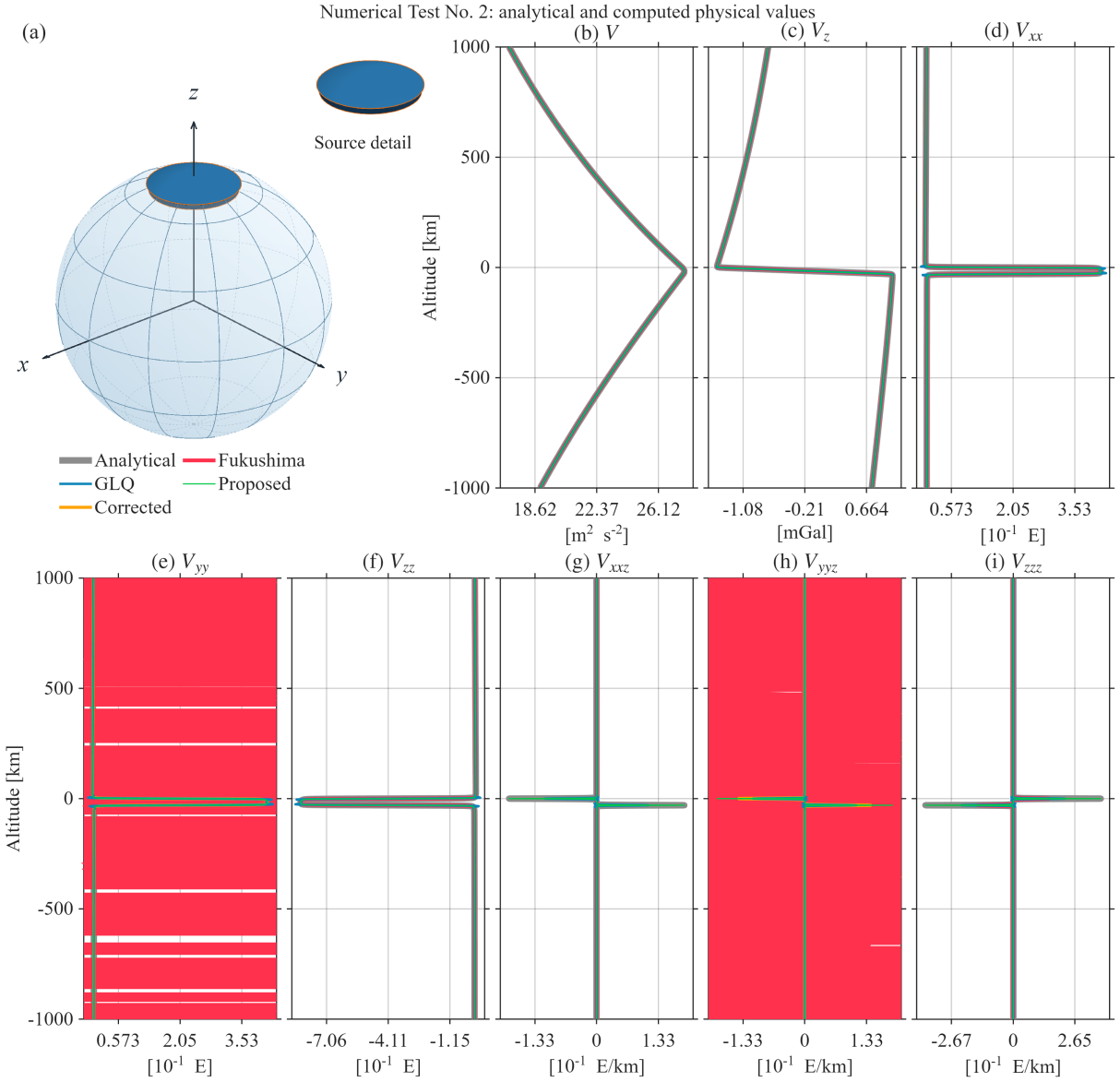


Figure S11: Numerical Test No. 2. Panel (a) shows the spherical-cap geometry; panels (b)–(i) compare the analytical and computed values of the eight field components listed in Fig. S10. Colours are the same as in Fig. S10. The red backgrounds in the V_{yy} and V_{yyz} panels are caused by large-amplitude oscillations in Fukushima's polar-axis results. Physical units follow Eq. (86).

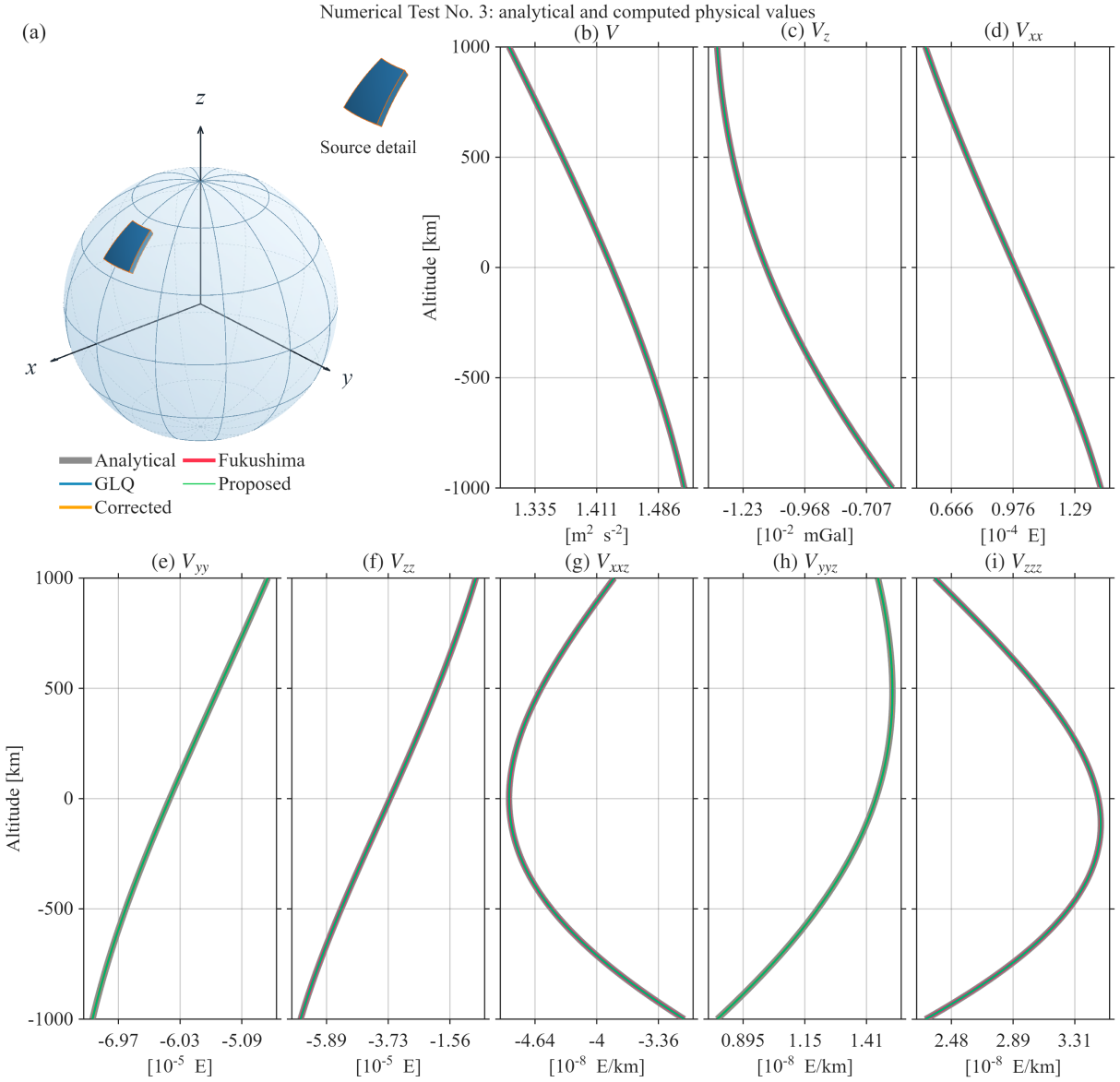


Figure S12: Numerical Test No. 3. Panel (a) shows the spherical-prism geometry; panels (b)–(i) compare the analytical and computed values of the eight field components listed in Fig. S10. Colours are the same as in Fig. S10. Physical units follow Eq. (86).

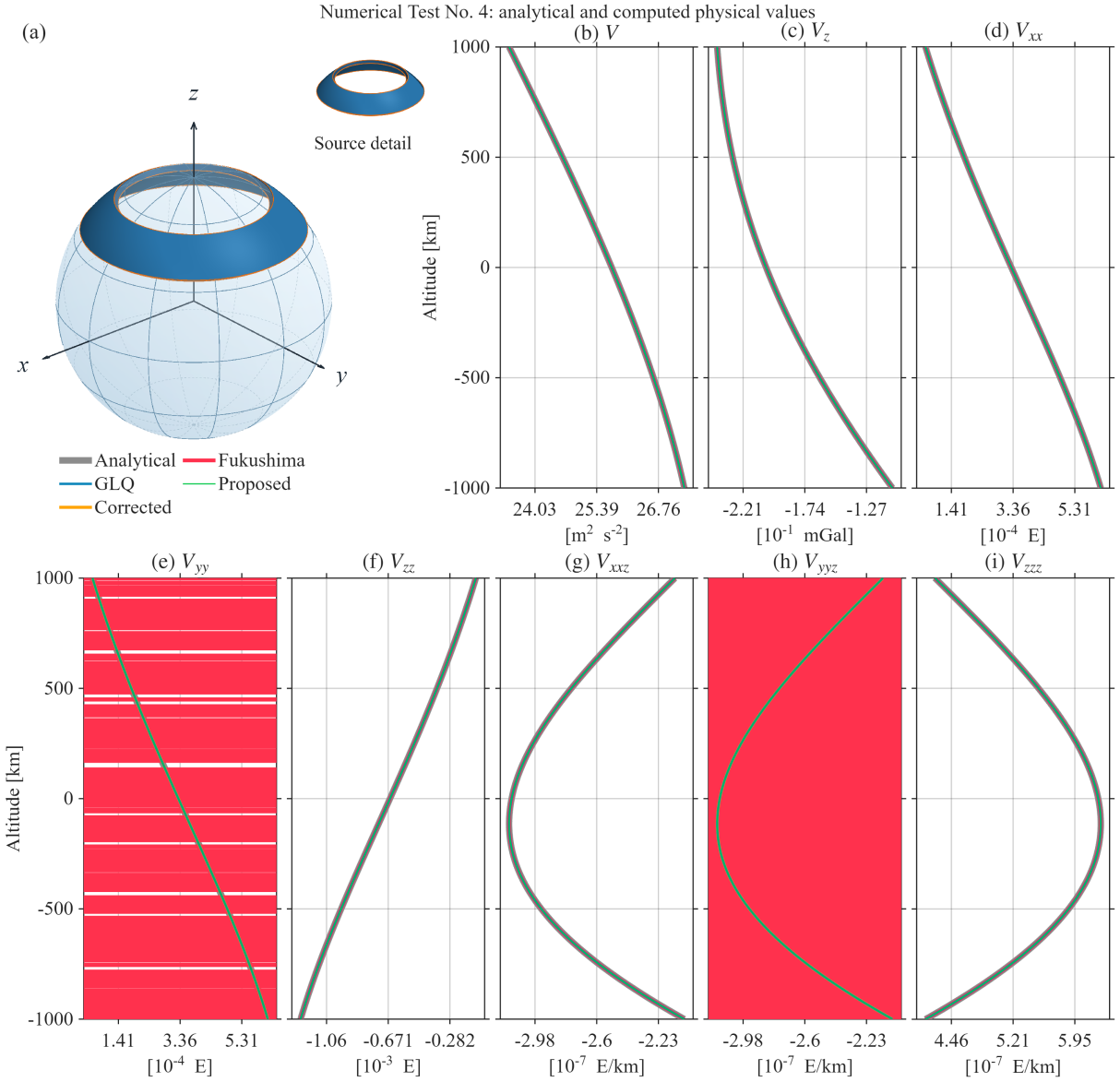


Figure S13: Numerical Test No. 4. Panel (a) shows the spherical-cap geometry; panels (b)–(i) compare the analytical and computed values of the eight field components listed in Fig. S10. Colours are the same as in Fig. S10. The red backgrounds in the V_{yy} and V_{yyz} panels are caused by large-amplitude oscillations in Fukushima's polar-axis results. Physical units follow Eq. (86).

3.3 Statistical summary

Tables S1–S4 summarize the analytical values by their minimum, maximum, arithmetic mean, and sample standard deviation. Tables S5–S8 and S9–S12 report the corresponding absolute- and relative-error statistics using the minimum, maximum, arithmetic mean, and root-mean-square error (RMSE). Each statistic is evaluated over 2,001 observation points. Rows labelled "Corrected" contain the Laplace-equation reconstructions of V_{yy} and V_{yyz} used for Fukushima's method.

Table S1: Analytical-value statistics for Test No. 1. Physical units follow Eq. (86).

Component	Minimum	Maximum	Mean	Std
V	$9.47 \cdot 10^{-1}$	$1.54 \cdot 10^{0}$	$1.25 \cdot 10^{0}$	$1.62 \cdot 10^{-1}$
V_z	$-8.07 \cdot 10^{-2}$	$5.73 \cdot 10^{-2}$	$-5.24 \cdot 10^{-3}$	$5.47 \cdot 10^{-2}$
V_{xx}	$8.15 \cdot 10^{-5}$	$1.96 \cdot 10^{-1}$	$5.24 \cdot 10^{-3}$	$2.28 \cdot 10^{-2}$
V_{yy}	$-1.50 \cdot 10^{-1}$	$-3.56 \cdot 10^{-4}$	$-4.83 \cdot 10^{-3}$	$1.75 \cdot 10^{-2}$
V_{zz}	$-4.62 \cdot 10^{-2}$	$5.40 \cdot 10^{-4}$	$-4.08 \cdot 10^{-4}$	$5.49 \cdot 10^{-3}$
V_{xxz}	$-4.05 \cdot 10^{-2}$	$4.07 \cdot 10^{-2}$	$-9.28 \cdot 10^{-8}$	$1.80 \cdot 10^{-3}$
V_{yyz}	$-1.96 \cdot 10^{-2}$	$1.96 \cdot 10^{-2}$	$5.39 \cdot 10^{-8}$	$1.09 \cdot 10^{-3}$
V_{zzz}	$-2.10 \cdot 10^{-2}$	$2.10 \cdot 10^{-2}$	$3.89 \cdot 10^{-8}$	$7.76 \cdot 10^{-4}$

Table S2: Analytical-value statistics for Test No. 2. Physical units follow Eq. (86).

Component	Minimum	Maximum	Mean	Std
V	$1.70 \cdot 10^{1}$	$2.77 \cdot 10^{1}$	$2.24 \cdot 10^{1}$	$2.92 \cdot 10^{0}$
V_z	$-1.45 \cdot 10^{0}$	$1.03 \cdot 10^{0}$	$-9.44 \cdot 10^{-2}$	$9.84 \cdot 10^{-1}$
V_{xx}	$-4.86 \cdot 10^{-3}$	$4.15 \cdot 10^{-1}$	$3.67 \cdot 10^{-3}$	$4.94 \cdot 10^{-2}$
V_{yy}	$-4.86 \cdot 10^{-3}$	$4.15 \cdot 10^{-1}$	$3.67 \cdot 10^{-3}$	$4.94 \cdot 10^{-2}$
V_{zz}	$-8.31 \cdot 10^{-1}$	$9.73 \cdot 10^{-3}$	$-7.34 \cdot 10^{-3}$	$9.89 \cdot 10^{-2}$
V_{xxz}	$-1.89 \cdot 10^{-1}$	$1.89 \cdot 10^{-1}$	$-3.50 \cdot 10^{-7}$	$6.99 \cdot 10^{-3}$
V_{yyz}	$-1.89 \cdot 10^{-1}$	$1.89 \cdot 10^{-1}$	$-3.50 \cdot 10^{-7}$	$6.99 \cdot 10^{-3}$
V_{zzz}	$-3.79 \cdot 10^{-1}$	$3.77 \cdot 10^{-1}$	$7.00 \cdot 10^{-7}$	$1.40 \cdot 10^{-2}$

Table S3: Analytical-value statistics for Test No. 3. Physical units follow Eq. (86).

Component	Minimum	Maximum	Mean	Std
V	$1.30 \cdot 10^{0}$	$1.52 \cdot 10^{0}$	$1.42 \cdot 10^{0}$	$6.36 \cdot 10^{-2}$
V_z	$-1.34 \cdot 10^{-2}$	$-5.98 \cdot 10^{-3}$	$-1.08 \cdot 10^{-2}$	$2.19 \cdot 10^{-3}$
V_{xx}	$5.35 \cdot 10^{-5}$	$1.42 \cdot 10^{-4}$	$9.84 \cdot 10^{-5}$	$2.66 \cdot 10^{-5}$
V_{yy}	$-7.36 \cdot 10^{-5}$	$-4.70 \cdot 10^{-5}$	$-6.14 \cdot 10^{-5}$	$7.97 \cdot 10^{-6}$
V_{zz}	$-6.80 \cdot 10^{-5}$	$-6.54 \cdot 10^{-6}$	$-3.70 \cdot 10^{-5}$	$1.87 \cdot 10^{-5}$
V_{xxz}	$-4.91 \cdot 10^{-8}$	$-3.09 \cdot 10^{-8}$	$-4.41 \cdot 10^{-8}$	$4.63 \cdot 10^{-9}$
V_{yyz}	$7.87 \cdot 10^{-9}$	$1.51 \cdot 10^{-8}$	$1.33 \cdot 10^{-8}$	$2.10 \cdot 10^{-9}$
V_{zzz}	$2.30 \cdot 10^{-8}$	$3.48 \cdot 10^{-8}$	$3.07 \cdot 10^{-8}$	$3.49 \cdot 10^{-9}$

Table S4: Analytical-value statistics for Test No. 4. Physical units follow Eq. (86).

Component	Minimum	Maximum	Mean	Std
V	$2.35 \cdot 10^{1}$	$2.73 \cdot 10^{1}$	$2.56 \cdot 10^{1}$	$1.15 \cdot 10^{0}$
V_z	$-2.41 \cdot 10^{-1}$	$-1.08 \cdot 10^{-1}$	$-1.94 \cdot 10^{-1}$	$3.94 \cdot 10^{-2}$
V_{xx}	$5.89 \cdot 10^{-5}$	$6.12 \cdot 10^{-4}$	$3.33 \cdot 10^{-4}$	$1.68 \cdot 10^{-4}$
V_{yy}	$5.89 \cdot 10^{-5}$	$6.12 \cdot 10^{-4}$	$3.33 \cdot 10^{-4}$	$1.68 \cdot 10^{-4}$
V_{zz}	$-1.22 \cdot 10^{-3}$	$-1.18 \cdot 10^{-4}$	$-6.66 \cdot 10^{-4}$	$3.36 \cdot 10^{-4}$
V_{xxz}	$-3.13 \cdot 10^{-7}$	$-2.07 \cdot 10^{-7}$	$-2.77 \cdot 10^{-7}$	$3.15 \cdot 10^{-8}$
V_{yyz}	$-3.13 \cdot 10^{-7}$	$-2.07 \cdot 10^{-7}$	$-2.77 \cdot 10^{-7}$	$3.15 \cdot 10^{-8}$
V_{zzz}	$4.14 \cdot 10^{-7}$	$6.27 \cdot 10^{-7}$	$5.53 \cdot 10^{-7}$	$6.29 \cdot 10^{-8}$

Table S5: Absolute-error statistics for Test No. 1. Physical units follow Eq. (86).

Component	Method	Minimum	Maximum	Mean	RMSE
V	Fukushima	$0.00 \cdot 10^{0}$	$1.35 \cdot 10^{-13}$	$2.79 \cdot 10^{-15}$	$1.28 \cdot 10^{-14}$
V	GLQ	$0.00 \cdot 10^{0}$	$2.96 \cdot 10^{-5}$	$4.19 \cdot 10^{-7}$	$3.03 \cdot 10^{-6}$
V	Proposed method	$0.00 \cdot 10^{0}$	$2.36 \cdot 10^{-12}$	$1.84 \cdot 10^{-14}$	$8.29 \cdot 10^{-14}$
V_z	Fukushima	$1.14 \cdot 10^{-15}$	$9.58 \cdot 10^{-9}$	$2.54 \cdot 10^{-11}$	$3.54 \cdot 10^{-10}$
V_z	GLQ	$0.00 \cdot 10^{0}$	$5.47 \cdot 10^{-4}$	$5.02 \cdot 10^{-6}$	$4.00 \cdot 10^{-5}$
V_z	Proposed method	$0.00 \cdot 10^{0}$	$8.76 \cdot 10^{-13}$	$1.17 \cdot 10^{-15}$	$1.99 \cdot 10^{-14}$
V_{xx}	Fukushima	$2.19 \cdot 10^{-14}$	$1.19 \cdot 10^{-4}$	$1.70 \cdot 10^{-6}$	$1.36 \cdot 10^{-5}$
V_{xx}	GLQ	$0.00 \cdot 10^{0}$	$1.76 \cdot 10^{-2}$	$2.25 \cdot 10^{-4}$	$1.74 \cdot 10^{-3}$
V_{xx}	Proposed method	$0.00 \cdot 10^{0}$	$4.03 \cdot 10^{-14}$	$4.50 \cdot 10^{-16}$	$3.27 \cdot 10^{-15}$
V_{yy}	Fukushima	$8.19 \cdot 10^{11}$	$1.00 \cdot 10^{13}$	$2.76 \cdot 10^{12}$	$3.25 \cdot 10^{12}$
V_{yy}	GLQ	$0.00 \cdot 10^{0}$	$1.55 \cdot 10^{-2}$	$2.19 \cdot 10^{-4}$	$1.74 \cdot 10^{-3}$
V_{yy}	Proposed method	$0.00 \cdot 10^{0}$	$3.06 \cdot 10^{-14}$	$4.38 \cdot 10^{-16}$	$3.13 \cdot 10^{-15}$
V_{yy}	Corrected	$1.50 \cdot 10^{-13}$	$1.31 \cdot 10^{-4}$	$1.73 \cdot 10^{-6}$	$1.38 \cdot 10^{-5}$
V_{zz}	Fukushima	$1.82 \cdot 10^{-13}$	$1.23 \cdot 10^{-5}$	$3.32 \cdot 10^{-8}$	$5.64 \cdot 10^{-7}$
V_{zz}	GLQ	$0.00 \cdot 10^{0}$	$3.99 \cdot 10^{-3}$	$2.60 \cdot 10^{-5}$	$2.32 \cdot 10^{-4}$
V_{zz}	Proposed method	$0.00 \cdot 10^{0}$	$1.01 \cdot 10^{-14}$	$3.60 \cdot 10^{-17}$	$4.10 \cdot 10^{-16}$
V_{xxz}	Fukushima	$4.47 \cdot 10^{-16}$	$1.78 \cdot 10^{-3}$	$5.40 \cdot 10^{-6}$	$8.34 \cdot 10^{-5}$
V_{xxz}	GLQ	$0.00 \cdot 10^{0}$	$1.70 \cdot 10^{-2}$	$3.50 \cdot 10^{-5}$	$5.68 \cdot 10^{-4}$
V_{xxz}	Proposed method	$0.00 \cdot 10^{0}$	$1.43 \cdot 10^{-13}$	$2.21 \cdot 10^{-16}$	$4.68 \cdot 10^{-15}$
V_{yyz}	Fukushima	$1.15 \cdot 10^{9}$	$2.17 \cdot 10^{11}$	$8.83 \cdot 10^{9}$	$2.13 \cdot 10^{10}$
V_{yyz}	GLQ	$0.00 \cdot 10^{0}$	$8.38 \cdot 10^{-3}$	$2.03 \cdot 10^{-5}$	$3.20 \cdot 10^{-4}$
V_{yyz}	Proposed method	$0.00 \cdot 10^{0}$	$1.15 \cdot 10^{-14}$	$2.72 \cdot 10^{-17}$	$4.16 \cdot 10^{-16}$
V_{yyz}	Corrected	$8.58 \cdot 10^{-16}$	$3.66 \cdot 10^{-3}$	$5.18 \cdot 10^{-6}$	$1.18 \cdot 10^{-4}$
V_{zzz}	Fukushima	$6.45 \cdot 10^{-16}$	$4.21 \cdot 10^{-3}$	$8.05 \cdot 10^{-6}$	$1.45 \cdot 10^{-4}$
V_{zzz}	GLQ	$0.00 \cdot 10^{0}$	$8.63 \cdot 10^{-3}$	$2.55 \cdot 10^{-5}$	$3.26 \cdot 10^{-4}$
V_{zzz}	Proposed method	$2.83 \cdot 10^{-23}$	$1.22 \cdot 10^{-14}$	$2.97 \cdot 10^{-17}$	$4.65 \cdot 10^{-16}$

Table S6: Absolute-error statistics for Test No. 2. Physical units follow Eq. (86).

Component	Method	Minimum	Maximum	Mean	RMSE
V	Fukushima	$0.00 \cdot 10^{0}$	$2.42 \cdot 10^{-12}$	$5.13 \cdot 10^{-14}$	$2.30 \cdot 10^{-13}$
V	GLQ	$0.00 \cdot 10^{0}$	$5.32 \cdot 10^{-4}$	$7.55 \cdot 10^{-6}$	$5.46 \cdot 10^{-5}$
V	Proposed method	$0.00 \cdot 10^{0}$	$4.24 \cdot 10^{-11}$	$3.31 \cdot 10^{-13}$	$1.49 \cdot 10^{-12}$
V_z	Fukushima	$1.06 \cdot 10^{-14}$	$1.81 \cdot 10^{-6}$	$4.04 \cdot 10^{-9}$	$6.66 \cdot 10^{-8}$
V_z	GLQ	$0.00 \cdot 10^{0}$	$9.85 \cdot 10^{-3}$	$9.03 \cdot 10^{-5}$	$7.20 \cdot 10^{-4}$
V_z	Proposed method	$0.00 \cdot 10^{0}$	$1.58 \cdot 10^{-11}$	$2.11 \cdot 10^{-14}$	$3.58 \cdot 10^{-13}$
V_{xx}	Fukushima	$5.23 \cdot 10^{-14}$	$8.65 \cdot 10^{-4}$	$2.33 \cdot 10^{-6}$	$3.98 \cdot 10^{-5}$
V_{xx}	GLQ	$0.00 \cdot 10^{0}$	$3.59 \cdot 10^{-2}$	$2.34 \cdot 10^{-4}$	$2.09 \cdot 10^{-3}$
V_{xx}	Proposed method	$0.00 \cdot 10^{0}$	$4.19 \cdot 10^{-10}$	$2.10 \cdot 10^{-13}$	$9.36 \cdot 10^{-12}$
V_{yy}	Fukushima	$7.30 \cdot 10^{-1}$	$8.09 \cdot 10^{2}$	$1.11 \cdot 10^{2}$	$2.01 \cdot 10^{2}$
V_{yy}	GLQ	$0.00 \cdot 10^{0}$	$3.59 \cdot 10^{-2}$	$2.34 \cdot 10^{-4}$	$2.09 \cdot 10^{-3}$
V_{yy}	Proposed method	$0.00 \cdot 10^{0}$	$9.02 \cdot 10^{-14}$	$4.25 \cdot 10^{-16}$	$4.24 \cdot 10^{-15}$
V_{yy}	Corrected	$1.18 \cdot 10^{-14}$	$3.18 \cdot 10^{-3}$	$8.58 \cdot 10^{-6}$	$1.46 \cdot 10^{-4}$
V_{zz}	Fukushima	$2.39 \cdot 10^{-14}$	$2.31 \cdot 10^{-3}$	$6.25 \cdot 10^{-6}$	$1.06 \cdot 10^{-4}$
V_{zz}	GLQ	$0.00 \cdot 10^{0}$	$7.19 \cdot 10^{-2}$	$4.68 \cdot 10^{-4}$	$4.18 \cdot 10^{-3}$
V_{zz}	Proposed method	$0.00 \cdot 10^{0}$	$1.84 \cdot 10^{-13}$	$6.51 \cdot 10^{-16}$	$7.45 \cdot 10^{-15}$
V_{xxz}	Fukushima	$2.24 \cdot 10^{-16}$	$1.45 \cdot 10^{-1}$	$3.59 \cdot 10^{-4}$	$5.82 \cdot 10^{-3}$
V_{xxz}	GLQ	$0.00 \cdot 10^{0}$	$7.77 \cdot 10^{-2}$	$2.30 \cdot 10^{-4}$	$2.93 \cdot 10^{-3}$
V_{xxz}	Proposed method	$0.00 \cdot 10^{0}$	$2.13 \cdot 10^{-13}$	$5.98 \cdot 10^{-16}$	$8.66 \cdot 10^{-15}$
V_{yyz}	Fukushima	$2.07 \cdot 10^{-4}$	$1.02 \cdot 10^{2}$	$1.62 \cdot 10^{1}$	$2.24 \cdot 10^{1}$
V_{yyz}	GLQ	$3.17 \cdot 10^{-21}$	$7.77 \cdot 10^{-2}$	$2.30 \cdot 10^{-4}$	$2.93 \cdot 10^{-3}$
V_{yyz}	Proposed method	$0.00 \cdot 10^{0}$	$1.15 \cdot 10^{-13}$	$3.41 \cdot 10^{-16}$	$5.03 \cdot 10^{-15}$
V_{yyz}	Corrected	$2.05 \cdot 10^{-14}$	$7.89 \cdot 10^{-2}$	$5.07 \cdot 10^{-4}$	$5.64 \cdot 10^{-3}$
V_{zzz}	Fukushima	$3.32 \cdot 10^{-15}$	$2.24 \cdot 10^{-1}$	$4.94 \cdot 10^{-4}$	$7.79 \cdot 10^{-3}$
V_{zzz}	GLQ	$0.00 \cdot 10^{0}$	$1.55 \cdot 10^{-1}$	$4.60 \cdot 10^{-4}$	$5.86 \cdot 10^{-3}$
V_{zzz}	Proposed method	$9.05 \cdot 10^{-22}$	$2.20 \cdot 10^{-13}$	$5.35 \cdot 10^{-16}$	$8.37 \cdot 10^{-15}$

Table S7: Absolute-error statistics for Test No. 3. Physical units follow Eq. (86).

Component	Method	Minimum	Maximum	Mean	RMSE
V	Fukushima	$0.00 \cdot 10^{0}$	$1.94 \cdot 10^{-15}$	$5.91 \cdot 10^{-16}$	$7.04 \cdot 10^{-16}$
V	GLQ	$0.00 \cdot 10^{0}$	$3.98 \cdot 10^{-14}$	$1.03 \cdot 10^{-14}$	$1.27 \cdot 10^{-14}$
V	Proposed method	$0.00 \cdot 10^{0}$	$6.77 \cdot 10^{-14}$	$1.73 \cdot 10^{-14}$	$2.19 \cdot 10^{-14}$
V_z	Fukushima	$1.87 \cdot 10^{-16}$	$4.19 \cdot 10^{-12}$	$9.73 \cdot 10^{-13}$	$1.23 \cdot 10^{-12}$
V_z	GLQ	$0.00 \cdot 10^{0}$	$3.57 \cdot 10^{-16}$	$7.71 \cdot 10^{-17}$	$9.67 \cdot 10^{-17}$
V_z	Proposed method	$0.00 \cdot 10^{0}$	$5.07 \cdot 10^{-16}$	$1.24 \cdot 10^{-16}$	$1.53 \cdot 10^{-16}$
V_{xx}	Fukushima	$1.67 \cdot 10^{-15}$	$1.43 \cdot 10^{-11}$	$3.27 \cdot 10^{-12}$	$4.14 \cdot 10^{-12}$
V_{xx}	GLQ	$0.00 \cdot 10^{0}$	$3.70 \cdot 10^{-18}$	$7.39 \cdot 10^{-19}$	$9.31 \cdot 10^{-19}$
V_{xx}	Proposed method	$0.00 \cdot 10^{0}$	$5.44 \cdot 10^{-18}$	$1.21 \cdot 10^{-18}$	$1.62 \cdot 10^{-18}$
V_{yy}	Fukushima	$4.61 \cdot 10^{11}$	$1.00 \cdot 10^{12}$	$7.14 \cdot 10^{11}$	$7.31 \cdot 10^{11}$
V_{yy}	GLQ	$0.00 \cdot 10^{0}$	$2.58 \cdot 10^{-18}$	$4.61 \cdot 10^{-19}$	$5.79 \cdot 10^{-19}$
V_{yy}	Proposed method	$0.00 \cdot 10^{0}$	$2.91 \cdot 10^{-18}$	$7.45 \cdot 10^{-19}$	$9.58 \cdot 10^{-19}$
V_{yy}	Corrected	$4.93 \cdot 10^{-15}$	$2.76 \cdot 10^{-11}$	$5.72 \cdot 10^{-12}$	$7.21 \cdot 10^{-12}$
V_{zz}	Fukushima	$1.63 \cdot 10^{-15}$	$1.65 \cdot 10^{-11}$	$3.14 \cdot 10^{-12}$	$3.95 \cdot 10^{-12}$
V_{zz}	GLQ	$0.00 \cdot 10^{0}$	$1.53 \cdot 10^{-18}$	$2.72 \cdot 10^{-19}$	$3.72 \cdot 10^{-19}$
V_{zz}	Proposed method	$0.00 \cdot 10^{0}$	$7.55 \cdot 10^{-18}$	$4.91 \cdot 10^{-19}$	$7.37 \cdot 10^{-19}$
V_{xxz}	Fukushima	$2.26 \cdot 10^{-17}$	$2.43 \cdot 10^{-13}$	$2.89 \cdot 10^{-14}$	$3.58 \cdot 10^{-14}$
V_{xxz}	GLQ	$0.00 \cdot 10^{0}$	$1.36 \cdot 10^{-21}$	$3.22 \cdot 10^{-22}$	$3.97 \cdot 10^{-22}$
V_{xxz}	Proposed method	$0.00 \cdot 10^{0}$	$1.09 \cdot 10^{-20}$	$5.29 \cdot 10^{-22}$	$7.41 \cdot 10^{-22}$
V_{yyz}	Fukushima	$2.06 \cdot 10^{8}$	$3.01 \cdot 10^{8}$	$2.69 \cdot 10^{8}$	$2.71 \cdot 10^{8}$
V_{yyz}	GLQ	$0.00 \cdot 10^{0}$	$4.33 \cdot 10^{-22}$	$9.79 \cdot 10^{-23}$	$1.21 \cdot 10^{-22}$
V_{yyz}	Proposed method	$0.00 \cdot 10^{0}$	$8.89 \cdot 10^{-22}$	$1.54 \cdot 10^{-22}$	$1.93 \cdot 10^{-22}$
V_{yyz}	Corrected	$2.95 \cdot 10^{-17}$	$3.10 \cdot 10^{-13}$	$4.11 \cdot 10^{-14}$	$5.33 \cdot 10^{-14}$
V_{zzz}	Fukushima	$7.80 \cdot 10^{-18}$	$1.47 \cdot 10^{-13}$	$4.88 \cdot 10^{-14}$	$5.70 \cdot 10^{-14}$
V_{zzz}	GLQ	$0.00 \cdot 10^{0}$	$1.02 \cdot 10^{-21}$	$2.28 \cdot 10^{-22}$	$2.85 \cdot 10^{-22}$
V_{zzz}	Proposed method	$0.00 \cdot 10^{0}$	$4.91 \cdot 10^{-21}$	$3.78 \cdot 10^{-22}$	$5.20 \cdot 10^{-22}$

Table S8: Absolute-error statistics for Test No. 4. Physical units follow Eq. (86).

Component	Method	Minimum	Maximum	Mean	RMSE
V	Fukushima	$0.00 \cdot 10^{0}$	$3.88 \cdot 10^{-14}$	$1.30 \cdot 10^{-14}$	$1.48 \cdot 10^{-14}$
V	GLQ	$0.00 \cdot 10^{0}$	$7.19 \cdot 10^{-13}$	$1.84 \cdot 10^{-13}$	$2.28 \cdot 10^{-13}$
V	Proposed method	$0.00 \cdot 10^{0}$	$1.22 \cdot 10^{-12}$	$3.11 \cdot 10^{-13}$	$3.94 \cdot 10^{-13}$
V_z	Fukushima	$1.09 \cdot 10^{-11}$	$7.02 \cdot 10^{-11}$	$3.98 \cdot 10^{-11}$	$4.07 \cdot 10^{-11}$
V_z	GLQ	$0.00 \cdot 10^{0}$	$6.40 \cdot 10^{-15}$	$1.38 \cdot 10^{-15}$	$1.73 \cdot 10^{-15}$
V_z	Proposed method	$0.00 \cdot 10^{0}$	$9.06 \cdot 10^{-15}$	$2.23 \cdot 10^{-15}$	$2.76 \cdot 10^{-15}$
V_{xx}	Fukushima	$2.68 \cdot 10^{-14}$	$6.20 \cdot 10^{-11}$	$2.83 \cdot 10^{-11}$	$3.11 \cdot 10^{-11}$
V_{xx}	GLQ	$0.00 \cdot 10^{0}$	$1.64 \cdot 10^{-17}$	$3.37 \cdot 10^{-18}$	$4.28 \cdot 10^{-18}$
V_{xx}	Proposed method	$0.00 \cdot 10^{0}$	$2.35 \cdot 10^{-17}$	$4.31 \cdot 10^{-18}$	$6.25 \cdot 10^{-18}$
V_{yy}	Fukushima	$6.59 \cdot 10^{-2}$	$2.17 \cdot 10^{2}$	$2.40 \cdot 10^{1}$	$5.08 \cdot 10^{1}$
V_{yy}	GLQ	$0.00 \cdot 10^{0}$	$1.27 \cdot 10^{-17}$	$1.94 \cdot 10^{-18}$	$2.75 \cdot 10^{-18}$
V_{yy}	Proposed method	$0.00 \cdot 10^{0}$	$2.33 \cdot 10^{-17}$	$4.30 \cdot 10^{-18}$	$6.23 \cdot 10^{-18}$
V_{yy}	Corrected	$1.00 \cdot 10^{-13}$	$1.63 \cdot 10^{-10}$	$4.67 \cdot 10^{-11}$	$5.81 \cdot 10^{-11}$
V_{zz}	Fukushima	$3.33 \cdot 10^{-16}$	$1.06 \cdot 10^{-10}$	$3.49 \cdot 10^{-11}$	$4.04 \cdot 10^{-11}$
V_{zz}	GLQ	$0.00 \cdot 10^{0}$	$2.83 \cdot 10^{-17}$	$4.86 \cdot 10^{-18}$	$6.64 \cdot 10^{-18}$
V_{zz}	Proposed method	$0.00 \cdot 10^{0}$	$1.36 \cdot 10^{-16}$	$8.81 \cdot 10^{-18}$	$1.32 \cdot 10^{-17}$
V_{xxz}	Fukushima	$2.30 \cdot 10^{-16}$	$4.64 \cdot 10^{-12}$	$1.47 \cdot 10^{-12}$	$1.61 \cdot 10^{-12}$
V_{xxz}	GLQ	$0.00 \cdot 10^{0}$	$9.61 \cdot 10^{-21}$	$2.35 \cdot 10^{-21}$	$2.85 \cdot 10^{-21}$
V_{xxz}	Proposed method	$0.00 \cdot 10^{0}$	$2.82 \cdot 10^{-20}$	$3.34 \cdot 10^{-21}$	$4.37 \cdot 10^{-21}$
V_{yyz}	Fukushima	$2.37 \cdot 10^{-9}$	$6.00 \cdot 10^{0}$	$1.16 \cdot 10^{0}$	$1.59 \cdot 10^{0}$
V_{yyz}	GLQ	$0.00 \cdot 10^{0}$	$1.03 \cdot 10^{-20}$	$1.82 \cdot 10^{-21}$	$2.31 \cdot 10^{-21}$
V_{yyz}	Proposed method	$0.00 \cdot 10^{0}$	$1.37 \cdot 10^{-20}$	$3.26 \cdot 10^{-21}$	$4.14 \cdot 10^{-21}$
V_{yyz}	Corrected	$5.80 \cdot 10^{-16}$	$1.35 \cdot 10^{-11}$	$6.20 \cdot 10^{-12}$	$7.27 \cdot 10^{-12}$
V_{zzz}	Fukushima	$3.30 \cdot 10^{-13}$	$1.15 \cdot 10^{-11}$	$7.44 \cdot 10^{-12}$	$8.29 \cdot 10^{-12}$
V_{zzz}	GLQ	$0.00 \cdot 10^{0}$	$1.82 \cdot 10^{-20}$	$4.07 \cdot 10^{-21}$	$5.08 \cdot 10^{-21}$
V_{zzz}	Proposed method	$0.00 \cdot 10^{0}$	$8.84 \cdot 10^{-20}$	$6.81 \cdot 10^{-21}$	$9.38 \cdot 10^{-21}$

Table S9: Relative-error statistics for Test No. 1.

Component	Method	Minimum	Maximum	Mean	RMSE
V	Fukushima	$0.00 \cdot 10^{0}$	$9.60 \cdot 10^{-14}$	$2.06 \cdot 10^{-15}$	$9.24 \cdot 10^{-15}$
V	GLQ	$0.00 \cdot 10^{0}$	$1.93 \cdot 10^{-5}$	$2.73 \cdot 10^{-7}$	$1.97 \cdot 10^{-6}$
V	Proposed method	$0.00 \cdot 10^{0}$	$1.58 \cdot 10^{-12}$	$1.43 \cdot 10^{-14}$	$5.59 \cdot 10^{-14}$
V_z	Fukushima	$2.78 \cdot 10^{-14}$	$1.74 \cdot 10^{-7}$	$4.17 \cdot 10^{-10}$	$5.59 \cdot 10^{-9}$
V_z	GLQ	$0.00 \cdot 10^{0}$	$3.83 \cdot 10^{-2}$	$1.33 \cdot 10^{-4}$	$1.26 \cdot 10^{-3}$
V_z	Proposed method	$0.00 \cdot 10^{0}$	$2.12 \cdot 10^{-11}$	$2.49 \cdot 10^{-14}$	$4.78 \cdot 10^{-13}$
V_{xx}	Fukushima	$2.44 \cdot 10^{-11}$	$8.10 \cdot 10^{-4}$	$9.75 \cdot 10^{-6}$	$7.59 \cdot 10^{-5}$
V_{xx}	GLQ	$0.00 \cdot 10^{0}$	$1.20 \cdot 10^{-1}$	$1.42 \cdot 10^{-3}$	$1.00 \cdot 10^{-2}$
V_{xx}	Proposed method	$0.00 \cdot 10^{0}$	$2.75 \cdot 10^{-13}$	$1.64 \cdot 10^{-14}$	$2.87 \cdot 10^{-14}$
V_{yy}	Fukushima	$6.68 \cdot 10^{13}$	$3.32 \cdot 10^{15}$	$1.86 \cdot 10^{15}$	$2.02 \cdot 10^{15}$
V_{yy}	GLQ	$0.00 \cdot 10^{0}$	$1.30 \cdot 10^{-1}$	$1.68 \cdot 10^{-3}$	$1.29 \cdot 10^{-2}$
V_{yy}	Proposed method	$0.00 \cdot 10^{0}$	$1.99 \cdot 10^{-12}$	$1.65 \cdot 10^{-14}$	$5.50 \cdot 10^{-14}$
V_{yy}	Corrected	$1.55 \cdot 10^{-10}$	$1.21 \cdot 10^{-3}$	$1.32 \cdot 10^{-5}$	$1.03 \cdot 10^{-4}$
V_{zz}	Fukushima	$4.88 \cdot 10^{-10}$	$8.61 \cdot 10^{-3}$	$1.17 \cdot 10^{-5}$	$2.12 \cdot 10^{-4}$
V_{zz}	GLQ	$0.00 \cdot 10^{0}$	$1.07 \cdot 10^{2}$	$1.09 \cdot 10^{-1}$	$2.60 \cdot 10^{0}$
V_{zz}	Proposed method	$0.00 \cdot 10^{0}$	$4.59 \cdot 10^{-11}$	$7.11 \cdot 10^{-14}$	$1.19 \cdot 10^{-12}$
V_{xxz}	Fukushima	$3.21 \cdot 10^{-10}$	$7.75 \cdot 10^{-2}$	$3.99 \cdot 10^{-4}$	$4.13 \cdot 10^{-3}$
V_{xxz}	GLQ	$0.00 \cdot 10^{0}$	$4.18 \cdot 10^{-1}$	$3.39 \cdot 10^{-3}$	$2.46 \cdot 10^{-2}$
V_{xxz}	Proposed method	$0.00 \cdot 10^{0}$	$2.01 \cdot 10^{-11}$	$4.36 \cdot 10^{-14}$	$6.43 \cdot 10^{-13}$
V_{yyz}	Fukushima	$1.08 \cdot 10^{13}$	$2.08 \cdot 10^{15}$	$1.13 \cdot 10^{15}$	$1.28 \cdot 10^{15}$
V_{yyz}	GLQ	$0.00 \cdot 10^{0}$	$4.26 \cdot 10^{-1}$	$2.01 \cdot 10^{-3}$	$1.99 \cdot 10^{-2}$
V_{yyz}	Proposed method	$0.00 \cdot 10^{0}$	$5.08 \cdot 10^{-12}$	$1.80 \cdot 10^{-14}$	$1.18 \cdot 10^{-13}$
V_{yyz}	Corrected	$9.39 \cdot 10^{-10}$	$1.86 \cdot 10^{-1}$	$4.22 \cdot 10^{-4}$	$6.65 \cdot 10^{-3}$
V_{zzz}	Fukushima	$2.17 \cdot 10^{-9}$	$2.16 \cdot 10^{0}$	$3.17 \cdot 10^{-3}$	$5.10 \cdot 10^{-2}$
V_{zzz}	GLQ	$0.00 \cdot 10^{0}$	$6.61 \cdot 10^{2}$	$4.76 \cdot 10^{-1}$	$1.48 \cdot 10^{1}$
V_{zzz}	Proposed method	$1.13 \cdot 10^{-16}$	$1.96 \cdot 10^{-9}$	$1.77 \cdot 10^{-12}$	$4.40 \cdot 10^{-11}$

Table S10: Relative-error statistics for Test No. 2.

Component	Method	Minimum	Maximum	Mean	RMSE
V	Fukushima	$0.00 \cdot 10^{0}$	$9.50 \cdot 10^{-14}$	$2.11 \cdot 10^{-15}$	$9.21 \cdot 10^{-15}$
V	GLQ	$0.00 \cdot 10^{0}$	$1.93 \cdot 10^{-5}$	$2.73 \cdot 10^{-7}$	$1.97 \cdot 10^{-6}$
V	Proposed method	$0.00 \cdot 10^{0}$	$1.58 \cdot 10^{-12}$	$1.43 \cdot 10^{-14}$	$5.59 \cdot 10^{-14}$
V_z	Fukushima	$1.37 \cdot 10^{-14}$	$1.82 \cdot 10^{-6}$	$3.58 \cdot 10^{-9}$	$5.84 \cdot 10^{-8}$
V_z	GLQ	$0.00 \cdot 10^{0}$	$3.83 \cdot 10^{-2}$	$1.33 \cdot 10^{-4}$	$1.26 \cdot 10^{-3}$
V_z	Proposed method	$0.00 \cdot 10^{0}$	$2.12 \cdot 10^{-11}$	$2.49 \cdot 10^{-14}$	$4.78 \cdot 10^{-13}$
V_{xx}	Fukushima	$3.10 \cdot 10^{-11}$	$6.73 \cdot 10^{-2}$	$8.93 \cdot 10^{-5}$	$1.66 \cdot 10^{-3}$
V_{xx}	GLQ	$0.00 \cdot 10^{0}$	$1.07 \cdot 10^{2}$	$1.09 \cdot 10^{-1}$	$2.60 \cdot 10^{0}$
V_{xx}	Proposed method	$0.00 \cdot 10^{0}$	$1.07 \cdot 10^{-9}$	$6.53 \cdot 10^{-13}$	$2.39 \cdot 10^{-11}$
V_{yy}	Fukushima	$1.85 \cdot 10^{0}$	$9.35 \cdot 10^{6}$	$5.75 \cdot 10^{4}$	$2.62 \cdot 10^{5}$
V_{yy}	GLQ	$0.00 \cdot 10^{0}$	$1.07 \cdot 10^{2}$	$1.09 \cdot 10^{-1}$	$2.60 \cdot 10^{0}$
V_{yy}	Proposed method	$0.00 \cdot 10^{0}$	$4.66 \cdot 10^{-11}$	$8.36 \cdot 10^{-14}$	$1.21 \cdot 10^{-12}$
V_{yy}	Corrected	$3.98 \cdot 10^{-12}$	$2.48 \cdot 10^{-1}$	$3.29 \cdot 10^{-4}$	$6.11 \cdot 10^{-3}$
V_{zz}	Fukushima	$4.32 \cdot 10^{-12}$	$9.04 \cdot 10^{-2}$	$1.20 \cdot 10^{-4}$	$2.23 \cdot 10^{-3}$
V_{zz}	GLQ	$0.00 \cdot 10^{0}$	$1.07 \cdot 10^{2}$	$1.09 \cdot 10^{-1}$	$2.60 \cdot 10^{0}$
V_{zz}	Proposed method	$0.00 \cdot 10^{0}$	$4.59 \cdot 10^{-11}$	$7.12 \cdot 10^{-14}$	$1.19 \cdot 10^{-12}$
V_{xxz}	Fukushima	$8.41 \cdot 10^{-11}$	$7.35 \cdot 10^{1}$	$6.12 \cdot 10^{-2}$	$1.65 \cdot 10^{0}$
V_{xxz}	GLQ	$0.00 \cdot 10^{0}$	$6.61 \cdot 10^{2}$	$4.76 \cdot 10^{-1}$	$1.48 \cdot 10^{1}$
V_{xxz}	Proposed method	$0.00 \cdot 10^{0}$	$1.57 \cdot 10^{-9}$	$1.86 \cdot 10^{-12}$	$4.38 \cdot 10^{-11}$
V_{yyz}	Fukushima	$3.91 \cdot 10^{-1}$	$6.89 \cdot 10^{10}$	$1.03 \cdot 10^{8}$	$1.67 \cdot 10^{9}$
V_{yyz}	GLQ	$1.62 \cdot 10^{-15}$	$6.61 \cdot 10^{2}$	$4.76 \cdot 10^{-1}$	$1.48 \cdot 10^{1}$
V_{yyz}	Proposed method	$0.00 \cdot 10^{0}$	$3.67 \cdot 10^{-9}$	$4.55 \cdot 10^{-12}$	$1.15 \cdot 10^{-10}$
V_{yyz}	Corrected	$7.15 \cdot 10^{-9}$	$2.78 \cdot 10^{1}$	$6.80 \cdot 10^{-2}$	$7.74 \cdot 10^{-1}$
V_{zzz}	Fukushima	$8.34 \cdot 10^{-10}$	$2.29 \cdot 10^{1}$	$3.35 \cdot 10^{-2}$	$5.42 \cdot 10^{-1}$
V_{zzz}	GLQ	$0.00 \cdot 10^{0}$	$6.61 \cdot 10^{2}$	$4.76 \cdot 10^{-1}$	$1.48 \cdot 10^{1}$
V_{zzz}	Proposed method	$1.86 \cdot 10^{-16}$	$1.99 \cdot 10^{-9}$	$1.79 \cdot 10^{-12}$	$4.48 \cdot 10^{-11}$

Table S11: Relative-error statistics for Test No. 3.

Component	Method	Minimum	Maximum	Mean	RMSE
V	Fukushima	$0.00 \cdot 10^{0}$	$1.32 \cdot 10^{-15}$	$4.14 \cdot 10^{-16}$	$4.91 \cdot 10^{-16}$
V	GLQ	$0.00 \cdot 10^{0}$	$2.73 \cdot 10^{-14}$	$7.25 \cdot 10^{-15}$	$8.94 \cdot 10^{-15}$
V	Proposed method	$0.00 \cdot 10^{0}$	$4.50 \cdot 10^{-14}$	$1.21 \cdot 10^{-14}$	$1.51 \cdot 10^{-14}$
V_z	Fukushima	$1.88 \cdot 10^{-14}$	$6.57 \cdot 10^{-10}$	$9.78 \cdot 10^{-11}$	$1.32 \cdot 10^{-10}$
V_z	GLQ	$0.00 \cdot 10^{0}$	$2.90 \cdot 10^{-14}$	$7.16 \cdot 10^{-15}$	$8.80 \cdot 10^{-15}$
V_z	Proposed method	$0.00 \cdot 10^{0}$	$4.54 \cdot 10^{-14}$	$1.19 \cdot 10^{-14}$	$1.50 \cdot 10^{-14}$
V_{xx}	Fukushima	$1.28 \cdot 10^{-11}$	$1.51 \cdot 10^{-7}$	$3.44 \cdot 10^{-8}$	$4.34 \cdot 10^{-8}$
V_{xx}	GLQ	$0.00 \cdot 10^{0}$	$3.05 \cdot 10^{-14}$	$7.51 \cdot 10^{-15}$	$9.15 \cdot 10^{-15}$
V_{xx}	Proposed method	$0.00 \cdot 10^{0}$	$4.38 \cdot 10^{-14}$	$1.17 \cdot 10^{-14}$	$1.47 \cdot 10^{-14}$
V_{yy}	Fukushima	$9.82 \cdot 10^{15}$	$1.36 \cdot 10^{16}$	$1.15 \cdot 10^{16}$	$1.15 \cdot 10^{16}$
V_{yy}	GLQ	$0.00 \cdot 10^{0}$	$3.53 \cdot 10^{-14}$	$7.45 \cdot 10^{-15}$	$9.18 \cdot 10^{-15}$
V_{yy}	Proposed method	$0.00 \cdot 10^{0}$	$4.44 \cdot 10^{-14}$	$1.19 \cdot 10^{-14}$	$1.49 \cdot 10^{-14}$
V_{yy}	Corrected	$9.00 \cdot 10^{-11}$	$4.01 \cdot 10^{-7}$	$9.30 \cdot 10^{-8}$	$1.16 \cdot 10^{-7}$
V_{zz}	Fukushima	$1.54 \cdot 10^{-10}$	$1.15 \cdot 10^{-6}$	$1.13 \cdot 10^{-7}$	$1.63 \cdot 10^{-7}$
V_{zz}	GLQ	$0.00 \cdot 10^{0}$	$3.06 \cdot 10^{-14}$	$7.43 \cdot 10^{-15}$	$9.18 \cdot 10^{-15}$
V_{zz}	Proposed method	$0.00 \cdot 10^{0}$	$3.10 \cdot 10^{-13}$	$1.22 \cdot 10^{-14}$	$1.68 \cdot 10^{-14}$
V_{xxz}	Fukushima	$4.61 \cdot 10^{-10}$	$4.95 \cdot 10^{-6}$	$6.59 \cdot 10^{-7}$	$8.19 \cdot 10^{-7}$
V_{xxz}	GLQ	$0.00 \cdot 10^{0}$	$2.98 \cdot 10^{-14}$	$7.34 \cdot 10^{-15}$	$9.01 \cdot 10^{-15}$
V_{xxz}	Proposed method	$0.00 \cdot 10^{0}$	$2.33 \cdot 10^{-13}$	$1.21 \cdot 10^{-14}$	$1.67 \cdot 10^{-14}$
V_{yyz}	Fukushima	$1.42 \cdot 10^{16}$	$3.80 \cdot 10^{16}$	$2.12 \cdot 10^{16}$	$2.20 \cdot 10^{16}$
V_{yyz}	GLQ	$0.00 \cdot 10^{0}$	$2.87 \cdot 10^{-14}$	$7.33 \cdot 10^{-15}$	$8.94 \cdot 10^{-15}$
V_{yyz}	Proposed method	$0.00 \cdot 10^{0}$	$6.61 \cdot 10^{-14}$	$1.19 \cdot 10^{-14}$	$1.50 \cdot 10^{-14}$
V_{yyz}	Corrected	$2.04 \cdot 10^{-9}$	$2.16 \cdot 10^{-5}$	$3.31 \cdot 10^{-6}$	$4.50 \cdot 10^{-6}$
V_{zzz}	Fukushima	$2.32 \cdot 10^{-10}$	$5.42 \cdot 10^{-6}$	$1.56 \cdot 10^{-6}$	$1.83 \cdot 10^{-6}$
V_{zzz}	GLQ	$0.00 \cdot 10^{0}$	$3.06 \cdot 10^{-14}$	$7.39 \cdot 10^{-15}$	$9.11 \cdot 10^{-15}$
V_{zzz}	Proposed method	$0.00 \cdot 10^{0}$	$2.01 \cdot 10^{-13}$	$1.23 \cdot 10^{-14}$	$1.73 \cdot 10^{-14}$

Table S12: Relative-error statistics for Test No. 4.

Component	Method	Minimum	Maximum	Mean	RMSE
V	Fukushima	$0.00 \cdot 10^{0}$	$1.60 \cdot 10^{-15}$	$5.09 \cdot 10^{-16}$	$5.81 \cdot 10^{-16}$
V	GLQ	$0.00 \cdot 10^{0}$	$2.74 \cdot 10^{-14}$	$7.20 \cdot 10^{-15}$	$8.88 \cdot 10^{-15}$
V	Proposed method	$0.00 \cdot 10^{0}$	$4.52 \cdot 10^{-14}$	$1.21 \cdot 10^{-14}$	$1.51 \cdot 10^{-14}$
V_z	Fukushima	$6.06 \cdot 10^{-11}$	$4.47 \cdot 10^{-10}$	$2.15 \cdot 10^{-10}$	$2.24 \cdot 10^{-10}$
V_z	GLQ	$0.00 \cdot 10^{0}$	$2.90 \cdot 10^{-14}$	$7.10 \cdot 10^{-15}$	$8.73 \cdot 10^{-15}$
V_z	Proposed method	$0.00 \cdot 10^{0}$	$4.51 \cdot 10^{-14}$	$1.20 \cdot 10^{-14}$	$1.50 \cdot 10^{-14}$
V_{xx}	Fukushima	$3.46 \cdot 10^{-10}$	$4.94 \cdot 10^{-7}$	$9.72 \cdot 10^{-8}$	$1.10 \cdot 10^{-7}$
V_{xx}	GLQ	$0.00 \cdot 10^{0}$	$3.77 \cdot 10^{-14}$	$1.11 \cdot 10^{-14}$	$1.29 \cdot 10^{-14}$
V_{xx}	Proposed method	$0.00 \cdot 10^{0}$	$4.53 \cdot 10^{-14}$	$1.18 \cdot 10^{-14}$	$1.49 \cdot 10^{-14}$
V_{yy}	Fukushima	$1.25 \cdot 10^{2}$	$2.24 \cdot 10^{6}$	$9.79 \cdot 10^{4}$	$2.43 \cdot 10^{5}$
V_{yy}	GLQ	$0.00 \cdot 10^{0}$	$2.51 \cdot 10^{-14}$	$5.77 \cdot 10^{-15}$	$7.23 \cdot 10^{-15}$
V_{yy}	Proposed method	$0.00 \cdot 10^{0}$	$4.58 \cdot 10^{-14}$	$1.18 \cdot 10^{-14}$	$1.48 \cdot 10^{-14}$
V_{yy}	Corrected	$4.33 \cdot 10^{-10}$	$1.06 \cdot 10^{-6}$	$1.57 \cdot 10^{-7}$	$2.05 \cdot 10^{-7}$
V_{zz}	Fukushima	$3.93 \cdot 10^{-13}$	$5.00 \cdot 10^{-7}$	$8.06 \cdot 10^{-8}$	$1.17 \cdot 10^{-7}$
V_{zz}	GLQ	$0.00 \cdot 10^{0}$	$3.07 \cdot 10^{-14}$	$7.38 \cdot 10^{-15}$	$9.12 \cdot 10^{-15}$
V_{zz}	Proposed method	$0.00 \cdot 10^{0}$	$3.10 \cdot 10^{-13}$	$1.22 \cdot 10^{-14}$	$1.67 \cdot 10^{-14}$
V_{xxz}	Fukushima	$9.85 \cdot 10^{-10}$	$1.49 \cdot 10^{-5}$	$5.16 \cdot 10^{-6}$	$5.55 \cdot 10^{-6}$
V_{xxz}	GLQ	$0.00 \cdot 10^{0}$	$3.08 \cdot 10^{-14}$	$8.48 \cdot 10^{-15}$	$1.02 \cdot 10^{-14}$
V_{xxz}	Proposed method	$0.00 \cdot 10^{0}$	$1.26 \cdot 10^{-13}$	$1.21 \cdot 10^{-14}$	$1.60 \cdot 10^{-14}$
V_{yyz}	Fukushima	$1.02 \cdot 10^{-2}$	$2.35 \cdot 10^{7}$	$4.26 \cdot 10^{6}$	$5.85 \cdot 10^{6}$
V_{yyz}	GLQ	$0.00 \cdot 10^{0}$	$3.29 \cdot 10^{-14}$	$6.51 \cdot 10^{-15}$	$8.15 \cdot 10^{-15}$
V_{yyz}	Proposed method	$0.00 \cdot 10^{0}$	$4.37 \cdot 10^{-14}$	$1.17 \cdot 10^{-14}$	$1.47 \cdot 10^{-14}$
V_{yyz}	Corrected	$2.52 \cdot 10^{-9}$	$4.64 \cdot 10^{-5}$	$2.21 \cdot 10^{-5}$	$2.63 \cdot 10^{-5}$
V_{zzz}	Fukushima	$7.75 \cdot 10^{-7}$	$2.18 \cdot 10^{-5}$	$1.32 \cdot 10^{-5}$	$1.48 \cdot 10^{-5}$
V_{zzz}	GLQ	$0.00 \cdot 10^{0}$	$3.02 \cdot 10^{-14}$	$7.31 \cdot 10^{-15}$	$9.02 \cdot 10^{-15}$
V_{zzz}	Proposed method	$0.00 \cdot 10^{0}$	$2.01 \cdot 10^{-13}$	$1.23 \cdot 10^{-14}$	$1.73 \cdot 10^{-14}$

4 Validation against a spherical-shell solution

This section supplements Section 3.2 of the main text. A 30-km-thick spherical shell with outer radius R_E is assembled from $10^\circ \times 10^\circ$ spherical prisms. The field is evaluated at 64,800 points on a $1^\circ \times 1^\circ$ longitude–latitude grid. Tests No. 5, 6, and 7 place the grid at $R_E + 1$ km, $R_E - 31$ km, and $R_E - 15$ km, respectively. For the proposed method, the target-shell and dummy-shell terms use independent peak caches and orthogonal initial bisections of otherwise identical outer subregions, while retaining the same quadrature rule and tolerances. This design prevents the test from reducing to subtraction along two identical numerical paths. Figs. 4 and 5 of the main text show Test No. 5; Figs. S14–S17 show Tests No. 6 and 7. Every error panel shows $|X_{\text{num}} - X_{\text{ana}}|$ in the unit of its component, following Eq. (86), on a logarithmic colour scale.

4.1 Statistical summary and interpretation

Tables S13–S18 report the minimum, maximum, mean, and RMSE of the absolute errors over all 64,800 points, with component units given by Eq. (86). For the potential, Fukushima's RMSE is 1.14×10^{-13}, 1.30×10^{-13}, and 1.30×10^{-13} $\mathrm{m^2\,s^{-2}}$ in Tests No. 5–7, compared with 5.45×10^{-11}, 1.19×10^{-10}, and 2.95×10^{-10} $\mathrm{m^2\,s^{-2}}$ for the proposed method. The ranking reverses for the derivatives. The largest acceleration RMSE of the proposed method is 1.23×10^{-12}, 3.33×10^{-12}, and 7.45×10^{-13} mGal, approximately 58, 21, and 88 times smaller than the corresponding Fukushima values. For the gravity-gradient tensor, the worst proposed-method RMSE is 4.21×10^{-14}, 5.98×10^{-15}, and 7.60×10^{-14} E, giving improvements of approximately 7.0×10^4, 5.0×10^5, and 4.0×10^4.

For the third-order tensor, the worst componentwise RMSE values are 8.45×10^{-17}, 6.78×10^{-16}, and 6.89×10^{-17} $\mathrm{E\,km^{-1}}$ for the proposed method, but 3.09×10^{-9}, 2.90×10^{-9}, and 1.66×10^{-7} $\mathrm{E\,km^{-1}}$ for Fukushima's method. These differences correspond to factors of approximately 3.7×10^7, 4.3×10^6, and 2.4×10^9. The largest Fukushima error occurs for V_{yyx} in Test No. 7, with an RMSE of 1.66×10^{-7} $\mathrm{E\,km^{-1}}$ and a maximum absolute error of 1.86×10^{-6} $\mathrm{E\,km^{-1}}$; the worst proposed-method component in all three tests is V_{xxx}.

Errors among prism contributions can cancel during symmetric shell assembly (Deng and Sneeuw, 2023). These statistics assess the assembled shell response; individual-prism accuracy is assessed by the analytical and independent profile comparisons. Direct integration avoids finite-difference truncation, while quadrature and shell-subtraction errors remain.

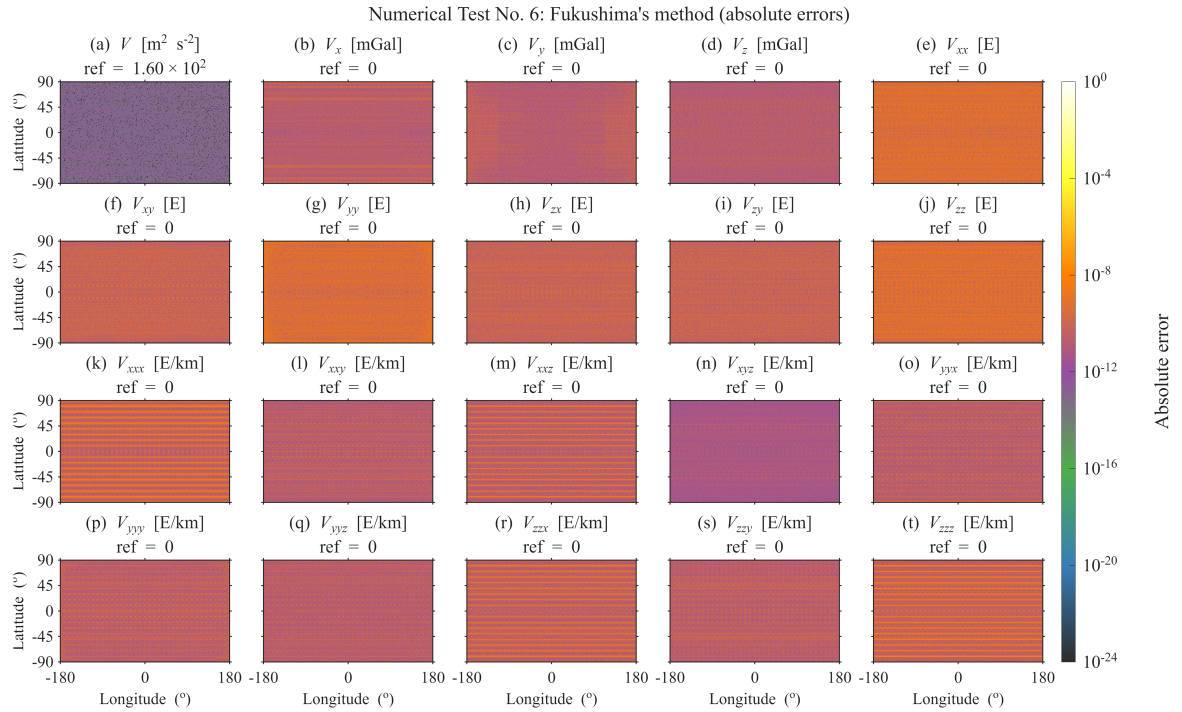


Figure S14: Numerical Test No. 6: absolute errors of Fukushima's method on the observation grid 31 km below the outer shell surface. Panels (a)–(t) correspond to V, V_x, V_y, V_z, V_{xx}, V_{xy}, V_{yy}, V_{zx}, V_{zy}, V_{zz}, V_{xxx}, V_{xxy}, V_{xxz}, V_{xyz}, V_{yyx}, V_{yyy}, V_{yyz}, V_{zzx}, V_{zzy}, and V_{zzz}, respectively. Physical units follow Eq. (86).

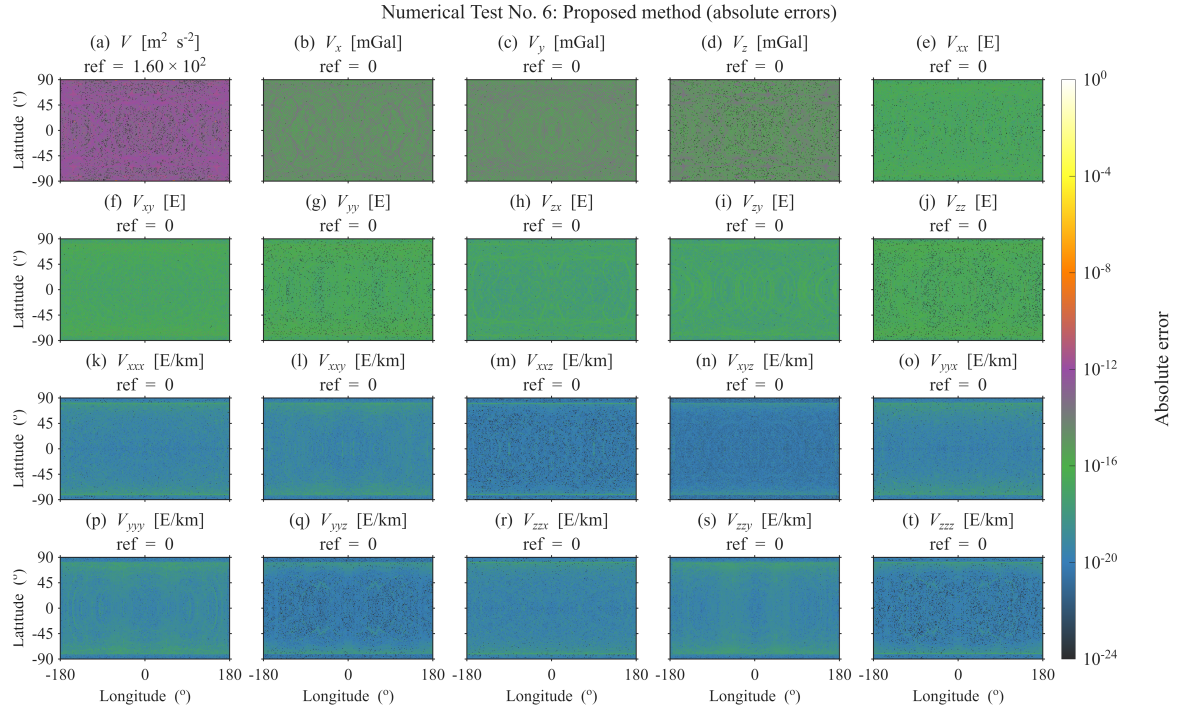


Figure S15: Numerical Test No. 6: absolute errors of the proposed method on the observation grid 31 km below the outer shell surface. Panel order is the same as in Fig. S14. Physical units follow Eq. (86).

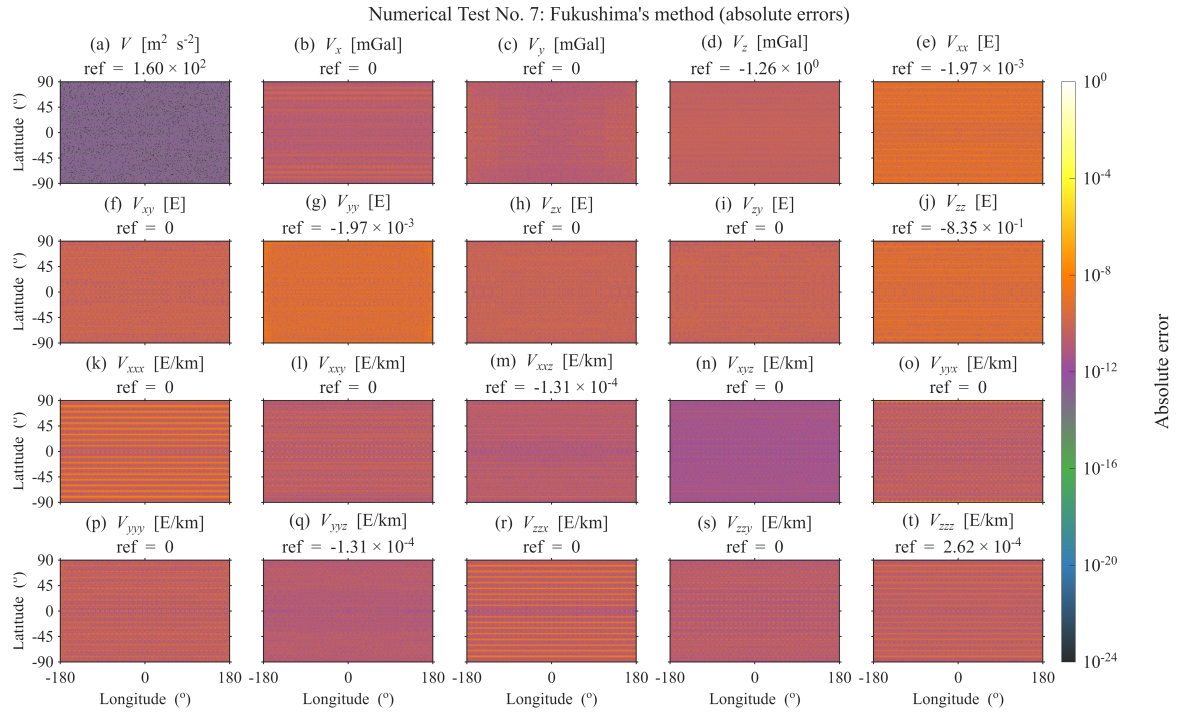


Figure S16: Numerical Test No. 7: absolute errors of Fukushima's method on the observation grid within the spherical shell at $R_E - 15$ km. Panel order is the same as in Fig. S14. Physical units follow Eq. (86).

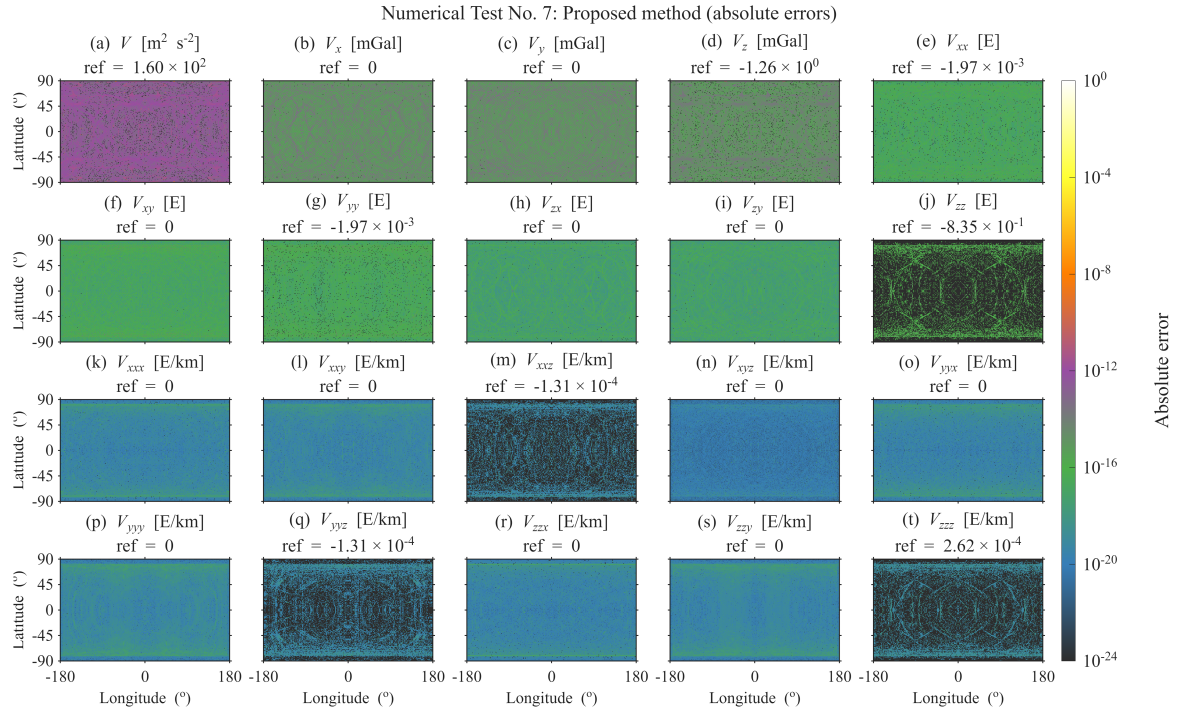


Figure S17: Numerical Test No. 7: absolute errors of the proposed method on the observation grid within the spherical shell at $R_E - 15$ km. Panel order is the same as in Fig. S14. Physical units follow Eq. (86).

Table S13: Absolute-error statistics for Fukushima's method in Test No. 5. Physical units follow Eq. (86).

Component	Minimum	Maximum	Mean	RMSE
V	$0.00 \cdot 10^{0}$	$4.97 \cdot 10^{-13}$	$9.04 \cdot 10^{-14}$	$1.14 \cdot 10^{-13}$
V_x	$3.58 \cdot 10^{-16}$	$3.66 \cdot 10^{-10}$	$4.86 \cdot 10^{-11}$	$7.12 \cdot 10^{-11}$
V_y	$4.57 \cdot 10^{-16}$	$7.08 \cdot 10^{-10}$	$4.26 \cdot 10^{-11}$	$6.35 \cdot 10^{-11}$
V_z	$0.00 \cdot 10^{0}$	$1.52 \cdot 10^{-10}$	$3.11 \cdot 10^{-11}$	$3.89 \cdot 10^{-11}$
V_{xx}	$7.66 \cdot 10^{-16}$	$1.00 \cdot 10^{-8}$	$6.94 \cdot 10^{-10}$	$9.93 \cdot 10^{-10}$
V_{xy}	$3.58 \cdot 10^{-14}$	$1.58 \cdot 10^{-9}$	$1.33 \cdot 10^{-10}$	$2.04 \cdot 10^{-10}$
V_{yy}	$1.04 \cdot 10^{-15}$	$2.87 \cdot 10^{-8}$	$9.79 \cdot 10^{-10}$	$2.94 \cdot 10^{-9}$
V_{zx}	$1.93 \cdot 10^{-16}$	$2.62 \cdot 10^{-9}$	$1.48 \cdot 10^{-10}$	$2.49 \cdot 10^{-10}$
V_{zy}	$3.89 \cdot 10^{-16}$	$2.76 \cdot 10^{-9}$	$1.45 \cdot 10^{-10}$	$2.50 \cdot 10^{-10}$
V_{zz}	$4.23 \cdot 10^{-14}$	$1.22 \cdot 10^{-8}$	$9.92 \cdot 10^{-10}$	$1.44 \cdot 10^{-9}$
V_{xxx}	$1.90 \cdot 10^{-15}$	$7.15 \cdot 10^{-9}$	$8.33 \cdot 10^{-10}$	$1.84 \cdot 10^{-9}$
V_{xxy}	$7.09 \cdot 10^{-16}$	$8.83 \cdot 10^{-10}$	$6.25 \cdot 10^{-11}$	$1.35 \cdot 10^{-10}$
V_{xxz}	$1.97 \cdot 10^{-15}$	$4.73 \cdot 10^{-9}$	$4.41 \cdot 10^{-10}$	$1.02 \cdot 10^{-9}$
V_{xyz}	$9.34 \cdot 10^{-16}$	$8.26 \cdot 10^{-9}$	$2.99 \cdot 10^{-11}$	$3.90 \cdot 10^{-10}$
V_{yyx}	$1.78 \cdot 10^{-16}$	$1.50 \cdot 10^{-8}$	$2.36 \cdot 10^{-10}$	$1.57 \cdot 10^{-9}$
V_{yyy}	$6.58 \cdot 10^{-16}$	$1.35 \cdot 10^{-9}$	$8.24 \cdot 10^{-11}$	$1.76 \cdot 10^{-10}$
V_{yyz}	$3.30 \cdot 10^{-15}$	$9.70 \cdot 10^{-10}$	$6.26 \cdot 10^{-11}$	$1.28 \cdot 10^{-10}$
V_{zzx}	$4.52 \cdot 10^{-16}$	$2.83 \cdot 10^{-9}$	$3.09 \cdot 10^{-10}$	$6.27 \cdot 10^{-10}$
V_{zzy}	$9.61 \cdot 10^{-17}$	$6.50 \cdot 10^{-10}$	$6.05 \cdot 10^{-11}$	$9.57 \cdot 10^{-11}$
V_{zzz}	$9.52 \cdot 10^{-15}$	$1.37 \cdot 10^{-8}$	$1.32 \cdot 10^{-9}$	$3.09 \cdot 10^{-9}$

Table S14: Absolute-error statistics for the proposed method in Test No. 5. Physical units follow Eq. (86).

Component	Minimum	Maximum	Mean	RMSE
V	$0.00 \cdot 10^{0}$	$3.03 \cdot 10^{-9}$	$2.00 \cdot 10^{-12}$	$5.45 \cdot 10^{-11}$
V_x	$0.00 \cdot 10^{0}$	$1.37 \cdot 10^{-11}$	$1.54 \cdot 10^{-14}$	$1.30 \cdot 10^{-13}$
V_y	$0.00 \cdot 10^{0}$	$1.77 \cdot 10^{-11}$	$1.82 \cdot 10^{-14}$	$2.43 \cdot 10^{-13}$
V_z	$0.00 \cdot 10^{0}$	$1.02 \cdot 10^{-10}$	$2.39 \cdot 10^{-14}$	$1.23 \cdot 10^{-12}$
V_{xx}	$0.00 \cdot 10^{0}$	$3.27 \cdot 10^{-13}$	$2.31 \cdot 10^{-16}$	$4.42 \cdot 10^{-15}$
V_{xy}	$0.00 \cdot 10^{0}$	$1.66 \cdot 10^{-13}$	$1.95 \cdot 10^{-16}$	$2.58 \cdot 10^{-15}$
V_{yy}	$0.00 \cdot 10^{0}$	$3.40 \cdot 10^{-13}$	$1.81 \cdot 10^{-16}$	$2.93 \cdot 10^{-15}$
V_{zx}	$0.00 \cdot 10^{0}$	$7.03 \cdot 10^{-14}$	$4.92 \cdot 10^{-17}$	$6.93 \cdot 10^{-16}$
V_{zy}	$0.00 \cdot 10^{0}$	$8.37 \cdot 10^{-13}$	$1.09 \cdot 10^{-16}$	$6.61 \cdot 10^{-15}$
V_{zz}	$0.00 \cdot 10^{0}$	$5.31 \cdot 10^{-12}$	$5.22 \cdot 10^{-16}$	$4.21 \cdot 10^{-14}$
V_{xxx}	$0.00 \cdot 10^{0}$	$7.57 \cdot 10^{-15}$	$1.58 \cdot 10^{-18}$	$8.45 \cdot 10^{-17}$
V_{xxy}	$0.00 \cdot 10^{0}$	$5.57 \cdot 10^{-15}$	$1.50 \cdot 10^{-18}$	$5.26 \cdot 10^{-17}$
V_{xxz}	$2.26 \cdot 10^{-22}$	$1.25 \cdot 10^{-16}$	$2.39 \cdot 10^{-19}$	$1.95 \cdot 10^{-18}$
V_{xyz}	$0.00 \cdot 10^{0}$	$1.27 \cdot 10^{-16}$	$1.23 \cdot 10^{-19}$	$1.54 \cdot 10^{-18}$
V_{yyx}	$0.00 \cdot 10^{0}$	$2.08 \cdot 10^{-15}$	$1.05 \cdot 10^{-18}$	$2.52 \cdot 10^{-17}$
V_{yyy}	$0.00 \cdot 10^{0}$	$8.83 \cdot 10^{-16}$	$1.09 \cdot 10^{-18}$	$1.43 \cdot 10^{-17}$
V_{yyz}	$2.26 \cdot 10^{-22}$	$4.31 \cdot 10^{-16}$	$2.00 \cdot 10^{-19}$	$3.66 \cdot 10^{-18}$
V_{zzx}	$0.00 \cdot 10^{0}$	$3.11 \cdot 10^{-16}$	$3.92 \cdot 10^{-19}$	$4.27 \cdot 10^{-18}$
V_{zzy}	$0.00 \cdot 10^{0}$	$1.29 \cdot 10^{-15}$	$6.12 \cdot 10^{-19}$	$1.24 \cdot 10^{-17}$
V_{zzz}	$4.52 \cdot 10^{-22}$	$7.85 \cdot 10^{-16}$	$4.39 \cdot 10^{-19}$	$8.22 \cdot 10^{-18}$

Table S15: Absolute-error statistics for Fukushima's method in Test No. 6. Physical units follow Eq. (86).

Component	Minimum	Maximum	Mean	RMSE
V	$0.00 \cdot 10^{0}$	$5.59 \cdot 10^{-13}$	$1.05 \cdot 10^{-13}$	$1.30 \cdot 10^{-13}$
V_x	$2.46 \cdot 10^{-15}$	$3.71 \cdot 10^{-10}$	$4.61 \cdot 10^{-11}$	$6.94 \cdot 10^{-11}$
V_y	$1.47 \cdot 10^{-15}$	$6.40 \cdot 10^{-10}$	$4.03 \cdot 10^{-11}$	$5.92 \cdot 10^{-11}$
V_z	$8.96 \cdot 10^{-17}$	$1.89 \cdot 10^{-10}$	$2.86 \cdot 10^{-11}$	$3.65 \cdot 10^{-11}$
V_{xx}	$2.66 \cdot 10^{-14}$	$9.37 \cdot 10^{-9}$	$7.51 \cdot 10^{-10}$	$1.07 \cdot 10^{-9}$
V_{xy}	$4.77 \cdot 10^{-15}$	$2.39 \cdot 10^{-9}$	$1.45 \cdot 10^{-10}$	$2.39 \cdot 10^{-10}$
V_{yy}	$5.74 \cdot 10^{-14}$	$2.78 \cdot 10^{-8}$	$9.94 \cdot 10^{-10}$	$2.98 \cdot 10^{-9}$
V_{zx}	$1.43 \cdot 10^{-15}$	$2.36 \cdot 10^{-9}$	$1.45 \cdot 10^{-10}$	$2.53 \cdot 10^{-10}$
V_{zy}	$3.99 \cdot 10^{-17}$	$2.22 \cdot 10^{-9}$	$1.40 \cdot 10^{-10}$	$2.33 \cdot 10^{-10}$
V_{zz}	$1.32 \cdot 10^{-13}$	$8.91 \cdot 10^{-9}$	$8.66 \cdot 10^{-10}$	$1.35 \cdot 10^{-9}$
V_{xxx}	$4.89 \cdot 10^{-15}$	$7.02 \cdot 10^{-9}$	$8.40 \cdot 10^{-10}$	$1.85 \cdot 10^{-9}$
V_{xxy}	$8.16 \cdot 10^{-16}$	$8.72 \cdot 10^{-10}$	$6.16 \cdot 10^{-11}$	$1.37 \cdot 10^{-10}$
V_{xxz}	$8.17 \cdot 10^{-16}$	$4.44 \cdot 10^{-9}$	$4.15 \cdot 10^{-10}$	$9.72 \cdot 10^{-10}$
V_{xyz}	$5.77 \cdot 10^{-16}$	$3.29 \cdot 10^{-10}$	$1.28 \cdot 10^{-11}$	$2.69 \cdot 10^{-11}$
V_{yyx}	$1.67 \cdot 10^{-15}$	$1.54 \cdot 10^{-8}$	$2.43 \cdot 10^{-10}$	$1.62 \cdot 10^{-9}$
V_{yyy}	$1.69 \cdot 10^{-16}$	$1.46 \cdot 10^{-9}$	$8.71 \cdot 10^{-11}$	$1.91 \cdot 10^{-10}$
V_{yyz}	$2.93 \cdot 10^{-15}$	$8.16 \cdot 10^{-10}$	$5.41 \cdot 10^{-11}$	$1.03 \cdot 10^{-10}$
V_{zzx}	$2.55 \cdot 10^{-16}$	$2.67 \cdot 10^{-9}$	$3.00 \cdot 10^{-10}$	$6.07 \cdot 10^{-10}$
V_{zzy}	$1.66 \cdot 10^{-16}$	$6.49 \cdot 10^{-10}$	$6.22 \cdot 10^{-11}$	$1.06 \cdot 10^{-10}$
V_{zzz}	$2.54 \cdot 10^{-14}$	$1.29 \cdot 10^{-8}$	$1.21 \cdot 10^{-9}$	$2.90 \cdot 10^{-9}$

Table S16: Absolute-error statistics for the proposed method in Test No. 6. Physical units follow Eq. (86).

Component	Minimum	Maximum	Mean	RMSE
V	$0.00 \cdot 10^{0}$	$1.35 \cdot 10^{-8}$	$2.84 \cdot 10^{-12}$	$1.19 \cdot 10^{-10}$
V_x	$0.00 \cdot 10^{0}$	$4.23 \cdot 10^{-10}$	$4.13 \cdot 10^{-14}$	$3.33 \cdot 10^{-12}$
V_y	$0.00 \cdot 10^{0}$	$1.25 \cdot 10^{-11}$	$1.74 \cdot 10^{-14}$	$2.08 \cdot 10^{-13}$
V_z	$0.00 \cdot 10^{0}$	$4.24 \cdot 10^{-11}$	$1.21 \cdot 10^{-14}$	$4.21 \cdot 10^{-13}$
V_{xx}	$0.00 \cdot 10^{0}$	$1.22 \cdot 10^{-13}$	$1.65 \cdot 10^{-16}$	$1.82 \cdot 10^{-15}$
V_{xy}	$0.00 \cdot 10^{0}$	$5.12 \cdot 10^{-13}$	$2.49 \cdot 10^{-16}$	$5.30 \cdot 10^{-15}$
V_{yy}	$0.00 \cdot 10^{0}$	$7.30 \cdot 10^{-13}$	$2.23 \cdot 10^{-16}$	$5.98 \cdot 10^{-15}$
V_{zx}	$0.00 \cdot 10^{0}$	$3.75 \cdot 10^{-14}$	$4.36 \cdot 10^{-17}$	$4.82 \cdot 10^{-16}$
V_{zy}	$0.00 \cdot 10^{0}$	$8.35 \cdot 10^{-14}$	$4.31 \cdot 10^{-17}$	$7.53 \cdot 10^{-16}$
V_{zz}	$0.00 \cdot 10^{0}$	$6.99 \cdot 10^{-14}$	$1.56 \cdot 10^{-16}$	$1.32 \cdot 10^{-15}$
V_{xxx}	$0.00 \cdot 10^{0}$	$8.63 \cdot 10^{-14}$	$6.02 \cdot 10^{-18}$	$6.78 \cdot 10^{-16}$
V_{xxy}	$0.00 \cdot 10^{0}$	$1.14 \cdot 10^{-15}$	$8.79 \cdot 10^{-19}$	$1.68 \cdot 10^{-17}$
V_{xxz}	$0.00 \cdot 10^{0}$	$8.11 \cdot 10^{-17}$	$2.05 \cdot 10^{-19}$	$1.68 \cdot 10^{-18}$
V_{xyz}	$0.00 \cdot 10^{0}$	$2.96 \cdot 10^{-15}$	$4.58 \cdot 10^{-19}$	$3.30 \cdot 10^{-17}$
V_{yyx}	$0.00 \cdot 10^{0}$	$1.62 \cdot 10^{-15}$	$6.87 \cdot 10^{-19}$	$1.39 \cdot 10^{-17}$
V_{yyy}	$0.00 \cdot 10^{0}$	$1.02 \cdot 10^{-15}$	$1.06 \cdot 10^{-18}$	$1.52 \cdot 10^{-17}$
V_{yyz}	$0.00 \cdot 10^{0}$	$2.35 \cdot 10^{-16}$	$1.60 \cdot 10^{-19}$	$2.54 \cdot 10^{-18}$
V_{zzx}	$0.00 \cdot 10^{0}$	$5.87 \cdot 10^{-16}$	$4.18 \cdot 10^{-19}$	$5.91 \cdot 10^{-18}$
V_{zzy}	$0.00 \cdot 10^{0}$	$1.34 \cdot 10^{-15}$	$5.66 \cdot 10^{-19}$	$1.12 \cdot 10^{-17}$
V_{zzz}	$0.00 \cdot 10^{0}$	$2.36 \cdot 10^{-16}$	$3.09 \cdot 10^{-19}$	$2.92 \cdot 10^{-18}$

Table S17: Absolute-error statistics for Fukushima's method in Test No. 7. Physical units follow Eq. (86).

Component	Minimum	Maximum	Mean	RMSE
V	$0.00 \cdot 10^{0}$	$4.97 \cdot 10^{-13}$	$1.06 \cdot 10^{-13}$	$1.30 \cdot 10^{-13}$
V_x	$9.18 \cdot 10^{-16}$	$3.57 \cdot 10^{-10}$	$4.38 \cdot 10^{-11}$	$6.55 \cdot 10^{-11}$
V_y	$5.91 \cdot 10^{-15}$	$7.45 \cdot 10^{-10}$	$4.11 \cdot 10^{-11}$	$6.08 \cdot 10^{-11}$
V_z	$9.48 \cdot 10^{-16}$	$1.87 \cdot 10^{-10}$	$5.80 \cdot 10^{-11}$	$6.52 \cdot 10^{-11}$
V_{xx}	$1.42 \cdot 10^{-14}$	$1.51 \cdot 10^{-8}$	$9.08 \cdot 10^{-10}$	$1.43 \cdot 10^{-9}$
V_{xy}	$7.12 \cdot 10^{-15}$	$3.22 \cdot 10^{-8}$	$1.79 \cdot 10^{-10}$	$7.82 \cdot 10^{-10}$
V_{yy}	$2.46 \cdot 10^{-14}$	$2.73 \cdot 10^{-8}$	$1.09 \cdot 10^{-9}$	$3.06 \cdot 10^{-9}$
V_{zx}	$5.84 \cdot 10^{-15}$	$6.20 \cdot 10^{-10}$	$1.08 \cdot 10^{-10}$	$1.38 \cdot 10^{-10}$
V_{zy}	$6.08 \cdot 10^{-16}$	$5.61 \cdot 10^{-10}$	$1.11 \cdot 10^{-10}$	$1.40 \cdot 10^{-10}$
V_{zz}	$1.36 \cdot 10^{-14}$	$4.16 \cdot 10^{-9}$	$6.85 \cdot 10^{-10}$	$9.46 \cdot 10^{-10}$
V_{xxx}	$5.89 \cdot 10^{-15}$	$1.53 \cdot 10^{-8}$	$1.63 \cdot 10^{-9}$	$3.80 \cdot 10^{-9}$
V_{xxy}	$1.28 \cdot 10^{-16}$	$8.03 \cdot 10^{-9}$	$1.38 \cdot 10^{-10}$	$6.48 \cdot 10^{-10}$
V_{xxz}	$1.68 \cdot 10^{-15}$	$1.97 \cdot 10^{-10}$	$3.31 \cdot 10^{-11}$	$4.61 \cdot 10^{-11}$
V_{xyz}	$1.53 \cdot 10^{-16}$	$3.15 \cdot 10^{-10}$	$7.26 \cdot 10^{-12}$	$2.22 \cdot 10^{-11}$
V_{yyx}	$1.06 \cdot 10^{-16}$	$1.86 \cdot 10^{-6}$	$1.75 \cdot 10^{-8}$	$1.66 \cdot 10^{-7}$
V_{yyy}	$9.81 \cdot 10^{-17}$	$6.55 \cdot 10^{-7}$	$1.59 \cdot 10^{-10}$	$5.16 \cdot 10^{-9}$
V_{yyz}	$1.18 \cdot 10^{-16}$	$2.82 \cdot 10^{-10}$	$2.68 \cdot 10^{-11}$	$3.86 \cdot 10^{-11}$
V_{zzx}	$2.89 \cdot 10^{-15}$	$5.45 \cdot 10^{-9}$	$5.51 \cdot 10^{-10}$	$1.27 \cdot 10^{-9}$
V_{zzy}	$1.38 \cdot 10^{-16}$	$5.23 \cdot 10^{-10}$	$4.80 \cdot 10^{-11}$	$9.02 \cdot 10^{-11}$
V_{zzz}	$7.10 \cdot 10^{-15}$	$7.46 \cdot 10^{-10}$	$9.74 \cdot 10^{-11}$	$1.70 \cdot 10^{-10}$

Table S18: Absolute-error statistics for the proposed method in Test No. 7. Physical units follow Eq. (86).

Component	Minimum	Maximum	Mean	RMSE
V	$0.00 \cdot 10^{0}$	$2.93 \cdot 10^{-8}$	$5.63 \cdot 10^{-12}$	$2.95 \cdot 10^{-10}$
V_x	$0.00 \cdot 10^{0}$	$9.33 \cdot 10^{-11}$	$2.11 \cdot 10^{-14}$	$7.45 \cdot 10^{-13}$
V_y	$0.00 \cdot 10^{0}$	$3.83 \cdot 10^{-11}$	$2.15 \cdot 10^{-14}$	$4.61 \cdot 10^{-13}$
V_z	$0.00 \cdot 10^{0}$	$3.03 \cdot 10^{-11}$	$2.07 \cdot 10^{-14}$	$5.38 \cdot 10^{-13}$
V_{xx}	$0.00 \cdot 10^{0}$	$9.78 \cdot 10^{-14}$	$1.47 \cdot 10^{-16}$	$1.17 \cdot 10^{-15}$
V_{xy}	$0.00 \cdot 10^{0}$	$3.55 \cdot 10^{-13}$	$1.77 \cdot 10^{-16}$	$3.80 \cdot 10^{-15}$
V_{yy}	$0.00 \cdot 10^{0}$	$4.99 \cdot 10^{-13}$	$2.13 \cdot 10^{-16}$	$4.53 \cdot 10^{-15}$
V_{zx}	$0.00 \cdot 10^{0}$	$2.70 \cdot 10^{-13}$	$7.75 \cdot 10^{-17}$	$2.83 \cdot 10^{-15}$
V_{zy}	$0.00 \cdot 10^{0}$	$9.14 \cdot 10^{-12}$	$9.05 \cdot 10^{-16}$	$7.60 \cdot 10^{-14}$
V_{zz}	$0.00 \cdot 10^{0}$	$9.37 \cdot 10^{-13}$	$2.28 \cdot 10^{-16}$	$8.96 \cdot 10^{-15}$
V_{xxx}	$0.00 \cdot 10^{0}$	$6.29 \cdot 10^{-15}$	$1.81 \cdot 10^{-18}$	$6.89 \cdot 10^{-17}$
V_{xxy}	$0.00 \cdot 10^{0}$	$2.28 \cdot 10^{-15}$	$1.11 \cdot 10^{-18}$	$3.29 \cdot 10^{-17}$
V_{xxz}	$0.00 \cdot 10^{0}$	$3.29 \cdot 10^{-17}$	$7.48 \cdot 10^{-20}$	$5.61 \cdot 10^{-19}$
V_{xyz}	$0.00 \cdot 10^{0}$	$3.43 \cdot 10^{-17}$	$6.72 \cdot 10^{-20}$	$6.63 \cdot 10^{-19}$
V_{yyx}	$0.00 \cdot 10^{0}$	$6.81 \cdot 10^{-15}$	$1.25 \cdot 10^{-18}$	$5.54 \cdot 10^{-17}$
V_{yyy}	$0.00 \cdot 10^{0}$	$1.02 \cdot 10^{-15}$	$1.26 \cdot 10^{-18}$	$2.02 \cdot 10^{-17}$
V_{yyz}	$0.00 \cdot 10^{0}$	$1.93 \cdot 10^{-16}$	$1.09 \cdot 10^{-19}$	$1.87 \cdot 10^{-18}$
V_{zzx}	$0.00 \cdot 10^{0}$	$6.07 \cdot 10^{-16}$	$5.19 \cdot 10^{-19}$	$7.75 \cdot 10^{-18}$
V_{zzy}	$0.00 \cdot 10^{0}$	$8.87 \cdot 10^{-16}$	$6.52 \cdot 10^{-19}$	$1.06 \cdot 10^{-17}$
V_{zzz}	$0.00 \cdot 10^{0}$	$5.17 \cdot 10^{-16}$	$1.53 \cdot 10^{-19}$	$4.56 \cdot 10^{-18}$

5 Validation using the Laplace and Poisson equations

This section supplements Section 3.3 of the main text. A spherical prism bounded by $R_E - 30$ km $\leq r' \leq R_E$, $30° \leq \varphi' \leq 50°$, and $110° \leq \lambda' \leq 130°$ is evaluated on a 401×401 grid spanning 100.05°–140.05° in longitude and 20.05°–60.05° in latitude. Tests No. 8, 9, and 10 place this grid at $R_E + 1$ km, $R_E - 31$ km, and $R_E - 15$ km, respectively.

The four tested identities are the second-order trace $V_{xx} + V_{yy} + V_{zz}$ and its x-, y-, and z-derivatives. Outside the source, all four theoretical values are zero. Within the homogeneous prism in Test No. 10, the second-order trace is $-4\pi G\rho$ in SI units, approximately -0.838717 E for the adopted density, while the differentiated traces remain zero away from source boundaries. Fig. 6 of the main text shows Test No. 8; Figs. S18 and S19 show Tests No. 9 and 10. The final column maps the absolute residual on a logarithmic colour scale, using E for the second-order trace and E km^{-1} for its three differentiated identities.

The statistics in Tables S19–S21 quantify the residuals over all 160,801 points. For the second-order trace, the proposed method gives RMSE values of 1.47×10^{-14}, 2.47×10^{-14}, and 1.56×10^{-12} E in Tests No. 8–10, respectively, which are approximately 2.0×10^8, 7.0×10^8, and 2.2×10^6 times smaller than the Fukushima values.

For the x- and y-derivative identities, the largest RMSE of the proposed method is 3.02×10^{-15}, 6.97×10^{-15}, and 4.48×10^{-13} E km^{-1} in Tests No. 8–10, compared with 2.41×10^{-5}, 2.46×10^{-5}, and 1.20×10^{-6} E km^{-1} for Fukushima's method. The most pronounced discrepancy is the z-derivative identity. In Tests No. 8 and 9, Fukushima's RMSE is approximately 8.89×10^{-2} E km^{-1}, its mean absolute residual is approximately 4.44×10^{-2} E km^{-1}, and its maximum residual reaches 1.84×10^{-1} and 1.83×10^{-1} E km^{-1}, respectively. The corresponding RMSE values of the proposed method are 5.87×10^{-12} and 2.71×10^{-14} E km^{-1}; in Test No. 10 they are 3.88×10^{-8} and 5.39×10^{-18} E km^{-1} for Fukushima's and the proposed method, respectively.

The spatially extensive Fukushima residual in the z-derivative identity indicates a systematic discrepancy in the tested calculation. The trace identities assess tensor consistency; the independent profile reference assesses individual-component errors.

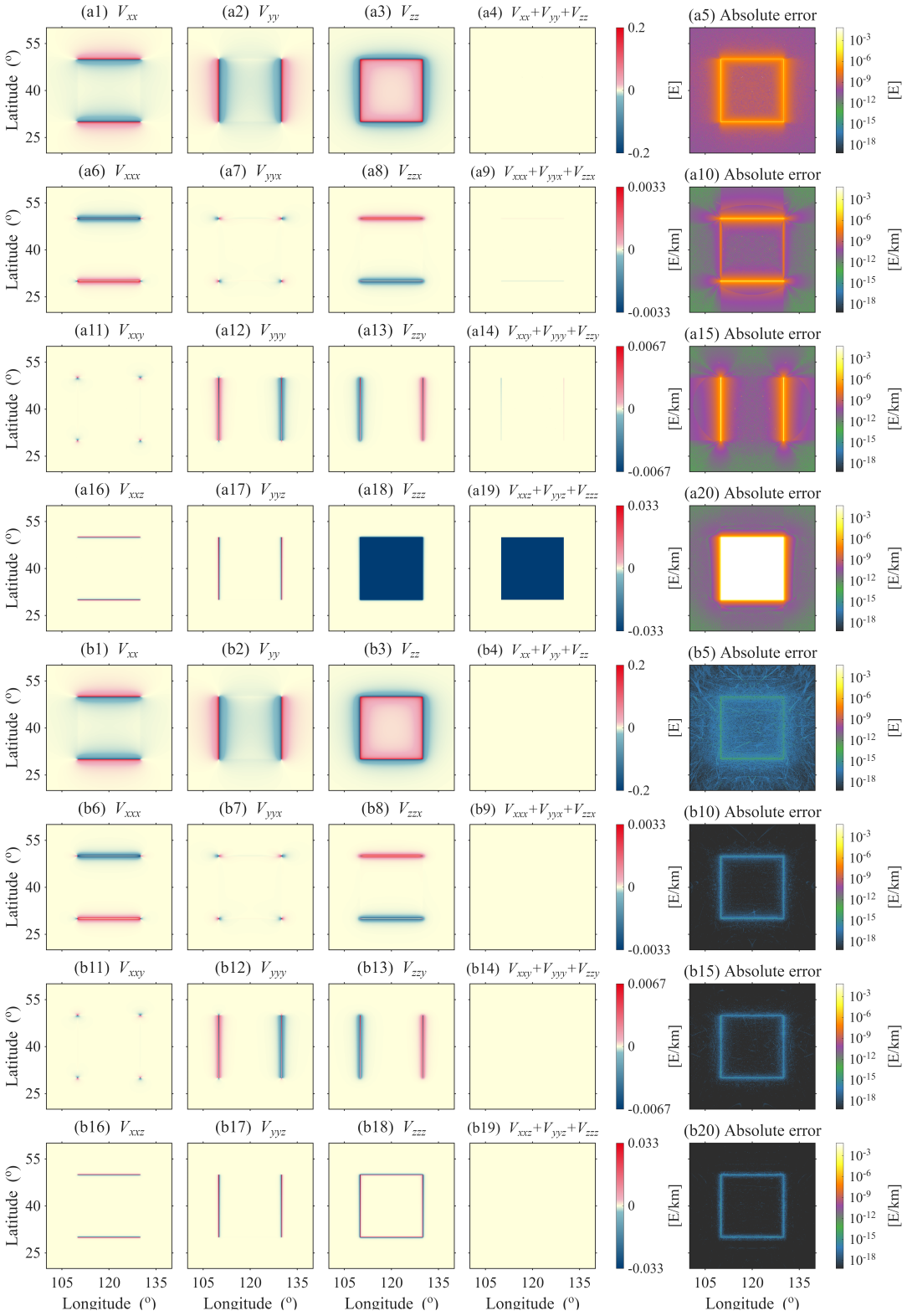


Figure S18: Numerical Test No. 9: verification of Laplace's equation 31 km below the prism's outer radial surface. For each method, the rows show the second-order trace and its x-, y-, and z-derivative identities. Columns 1–3 show the individual tensor components, column 4 their sum, and column 5 the absolute residual on a logarithmic colour scale (E in the first row and $\mathrm{E\,km^{-1}}$ in the remaining rows). Panels (a1)–(a20) show Fukushima's method and panels (b1)–(b20) show the proposed method.

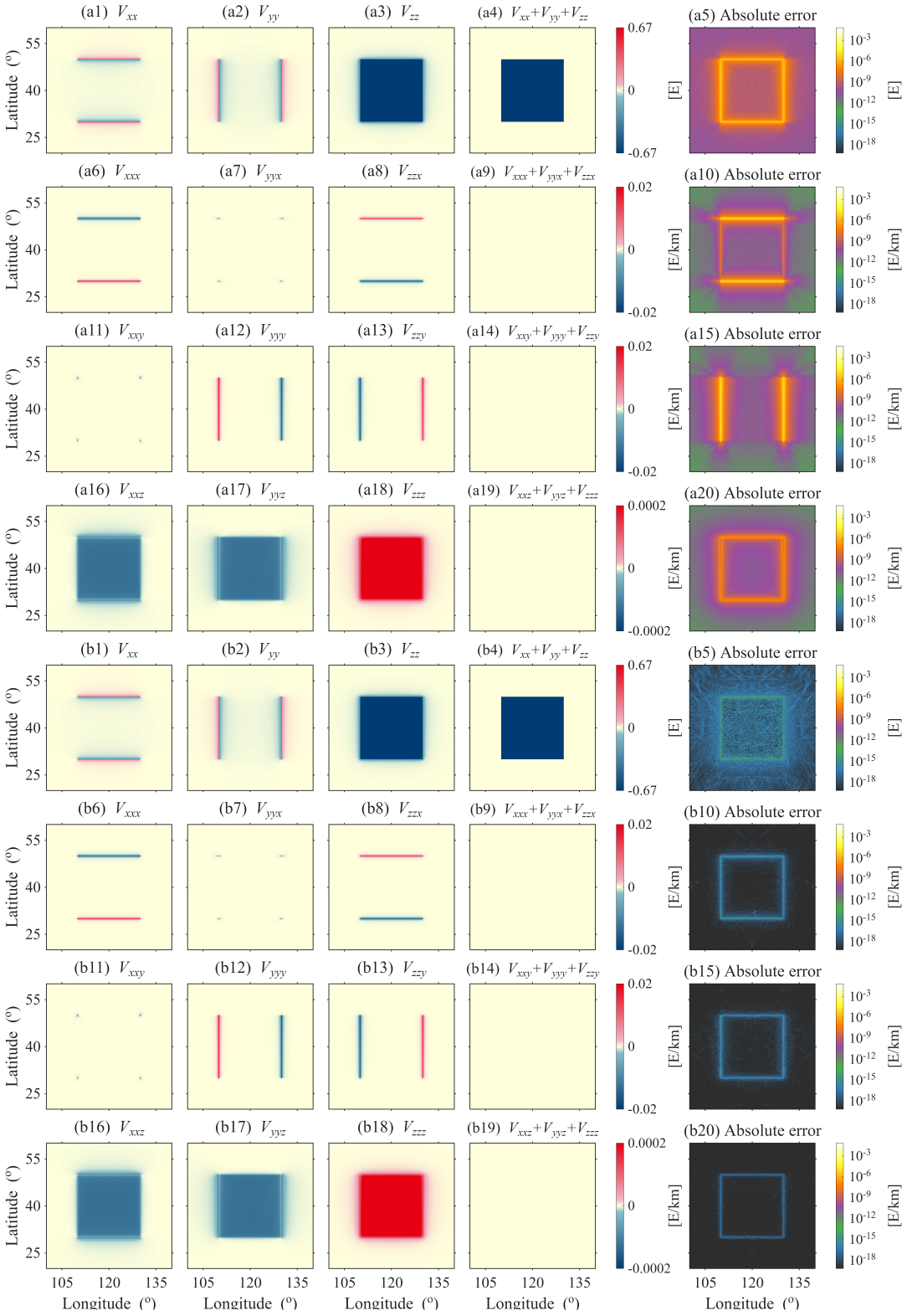


Figure S19: Numerical Test No. 10: verification of Poisson's equation on an observation grid within the radial extent of the spherical prism at $R_E - 15$ km. For each method, the four rows show the second-order trace and its x-, y-, and z-derivative identities. Columns 1–3 show the tensor components, column 4 their sum, and column 5 the absolute residual on a logarithmic colour scale (E in the first row and E km^{-1} in the remaining rows). Panels (a1)–(a20) show Fukushima's method and panels (b1)–(b20) show the proposed method.

Table S19: Absolute-residual statistics for the Laplace/Poisson identities in Test No. 8. Trace residuals are in E; differentiated residuals are in $\mathrm{E\,km^{-1}}$.

Identity	Method	Minimum	Maximum	Mean	RMSE
$V_{xx} + V_{yy} + V_{zz}$	Fukushima	$6.08 \cdot 10^{-16}$	$6.56 \cdot 10^{-4}$	$1.02 \cdot 10^{-7}$	$2.97 \cdot 10^{-6}$
$V_{xxx} + V_{yyx} + V_{zzx}$	Fukushima	$1.40 \cdot 10^{-16}$	$1.13 \cdot 10^{-4}$	$5.41 \cdot 10^{-7}$	$7.47 \cdot 10^{-6}$
$V_{xxy} + V_{yyy} + V_{zzy}$	Fukushima	$5.87 \cdot 10^{-18}$	$5.72 \cdot 10^{-4}$	$1.65 \cdot 10^{-6}$	$2.41 \cdot 10^{-5}$
$V_{xxz} + V_{yyz} + V_{zzz}$	Fukushima	$8.78 \cdot 10^{-17}$	$1.84 \cdot 10^{-1}$	$4.44 \cdot 10^{-2}$	$8.89 \cdot 10^{-2}$
$V_{xx} + V_{yy} + V_{zz}$	Proposed	$0.00 \cdot 10^{0}$	$3.58 \cdot 10^{-12}$	$4.65 \cdot 10^{-16}$	$1.47 \cdot 10^{-14}$
$V_{xxx} + V_{yyx} + V_{zzx}$	Proposed	$0.00 \cdot 10^{0}$	$1.13 \cdot 10^{-12}$	$3.25 \cdot 10^{-17}$	$3.02 \cdot 10^{-15}$
$V_{xxy} + V_{yyy} + V_{zzy}$	Proposed	$0.00 \cdot 10^{0}$	$1.69 \cdot 10^{-13}$	$2.11 \cdot 10^{-17}$	$6.08 \cdot 10^{-16}$
$V_{xxz} + V_{yyz} + V_{zzz}$	Proposed	$0.00 \cdot 10^{0}$	$2.35 \cdot 10^{-9}$	$1.50 \cdot 10^{-14}$	$5.87 \cdot 10^{-12}$

Table S20: Absolute-residual statistics for the Laplace/Poisson identities in Test No. 9. Trace residuals are in E; differentiated residuals are in $\mathrm{E\,km^{-1}}$.

Identity	Method	Minimum	Maximum	Mean	RMSE
$V_{xx} + V_{yy} + V_{zz}$	Fukushima	$4.13 \cdot 10^{-15}$	$4.55 \cdot 10^{-3}$	$1.76 \cdot 10^{-7}$	$1.72 \cdot 10^{-5}$
$V_{xxx} + V_{yyx} + V_{zzx}$	Fukushima	$4.44 \cdot 10^{-17}$	$1.14 \cdot 10^{-4}$	$5.52 \cdot 10^{-7}$	$7.63 \cdot 10^{-6}$
$V_{xxy} + V_{yyy} + V_{zzy}$	Fukushima	$1.17 \cdot 10^{-16}$	$5.84 \cdot 10^{-4}$	$1.68 \cdot 10^{-6}$	$2.46 \cdot 10^{-5}$
$V_{xxz} + V_{yyz} + V_{zzz}$	Fukushima	$4.92 \cdot 10^{-17}$	$1.83 \cdot 10^{-1}$	$4.44 \cdot 10^{-2}$	$8.89 \cdot 10^{-2}$
$V_{xx} + V_{yy} + V_{zz}$	Proposed	$0.00 \cdot 10^{0}$	$7.68 \cdot 10^{-12}$	$4.86 \cdot 10^{-16}$	$2.47 \cdot 10^{-14}$
$V_{xxx} + V_{yyx} + V_{zzx}$	Proposed	$0.00 \cdot 10^{0}$	$2.48 \cdot 10^{-12}$	$5.61 \cdot 10^{-17}$	$6.97 \cdot 10^{-15}$
$V_{xxy} + V_{yyy} + V_{zzy}$	Proposed	$0.00 \cdot 10^{0}$	$2.64 \cdot 10^{-12}$	$3.69 \cdot 10^{-17}$	$6.66 \cdot 10^{-15}$
$V_{xxz} + V_{yyz} + V_{zzz}$	Proposed	$0.00 \cdot 10^{0}$	$5.08 \cdot 10^{-12}$	$2.53 \cdot 10^{-16}$	$2.71 \cdot 10^{-14}$

Table S21: Absolute-residual statistics for the Laplace/Poisson identities in Test No. 10. Trace residuals are in E; differentiated residuals are in $\mathrm{E\,km^{-1}}$.

Identity	Method	Minimum	Maximum	Mean	RMSE
$V_{xx} + V_{yy} + V_{zz}$	Fukushima	$3.98 \cdot 10^{-15}$	$5.88 \cdot 10^{-4}$	$9.13 \cdot 10^{-8}$	$3.36 \cdot 10^{-6}$
$V_{xxx} + V_{yyx} + V_{zzx}$	Fukushima	$8.21 \cdot 10^{-17}$	$1.22 \cdot 10^{-4}$	$9.61 \cdot 10^{-8}$	$1.20 \cdot 10^{-6}$
$V_{xxy} + V_{yyy} + V_{zzy}$	Fukushima	$1.86 \cdot 10^{-18}$	$4.78 \cdot 10^{-5}$	$1.28 \cdot 10^{-7}$	$1.05 \cdot 10^{-6}$
$V_{xxz} + V_{yyz} + V_{zzz}$	Fukushima	$2.91 \cdot 10^{-16}$	$1.02 \cdot 10^{-5}$	$3.27 \cdot 10^{-9}$	$3.88 \cdot 10^{-8}$
$V_{xx} + V_{yy} + V_{zz}$	Proposed	$0.00 \cdot 10^{0}$	$4.41 \cdot 10^{-10}$	$5.86 \cdot 10^{-15}$	$1.56 \cdot 10^{-12}$
$V_{xxx} + V_{yyx} + V_{zzx}$	Proposed	$0.00 \cdot 10^{0}$	$1.79 \cdot 10^{-10}$	$1.17 \cdot 10^{-15}$	$4.48 \cdot 10^{-13}$
$V_{xxy} + V_{yyy} + V_{zzy}$	Proposed	$0.00 \cdot 10^{0}$	$3.10 \cdot 10^{-12}$	$4.32 \cdot 10^{-17}$	$8.02 \cdot 10^{-15}$
$V_{xxz} + V_{yyz} + V_{zzz}$	Proposed	$0.00 \cdot 10^{0}$	$8.61 \cdot 10^{-16}$	$4.71 \cdot 10^{-19}$	$5.39 \cdot 10^{-18}$

6 Checks using Fukushima quadruple precision

6.1 Accuracy of Fukushima's DP and QP implementations

Fukushima's double-precision (DP) and quadruple-precision (QP) implementations (Deng, 2023, 2025) are first compared with polar-axis analytical solutions evaluated at 128-digit working precision. Numerical Tests No. 1 and 2 use the same high-latitude spherical prism and zonal band, respectively. Because QP is computationally expensive, the comparison is restricted to 21 altitudes from -10 to 10 km. The implementations are compared for the eight non-zero polar-axis components V, V_z, V_{xx}, V_{yy}, V_{zz}, V_{xxz}, V_{yyz}, and V_{zzz}. The high and low parts of each QP output are retained separately during error evaluation to avoid loss through premature conversion to double precision.

The potential and the non-zero polar-axis acceleration component show the expected improvement. In Tests No. 1 and 2, the maximum QP relative errors are 6.47×10^{-33} and 4.81×10^{-33} for V, and 5.63×10^{-19} and 5.90×10^{-18} for V_z, respectively. These values are below double-precision machine epsilon and substantially smaller than the errors of the proposed double-precision calculation. By contrast, the maximum DP relative errors of V_z are 1.21×10^{-7} and 1.27×10^{-6}, and the near-surface DP curve is visibly distorted.

QP also improves V_{xx}, V_{zz}, V_{xxz}, and V_{zzz}, but neither precision reproduces the analytical V_{yy} and V_{yyz} values on the polar axis. The QP comparison below is restricted to the potential and acceleration vector. Gravity-gradient and third-order accuracy are assessed with the separate analytical and mpmath references.

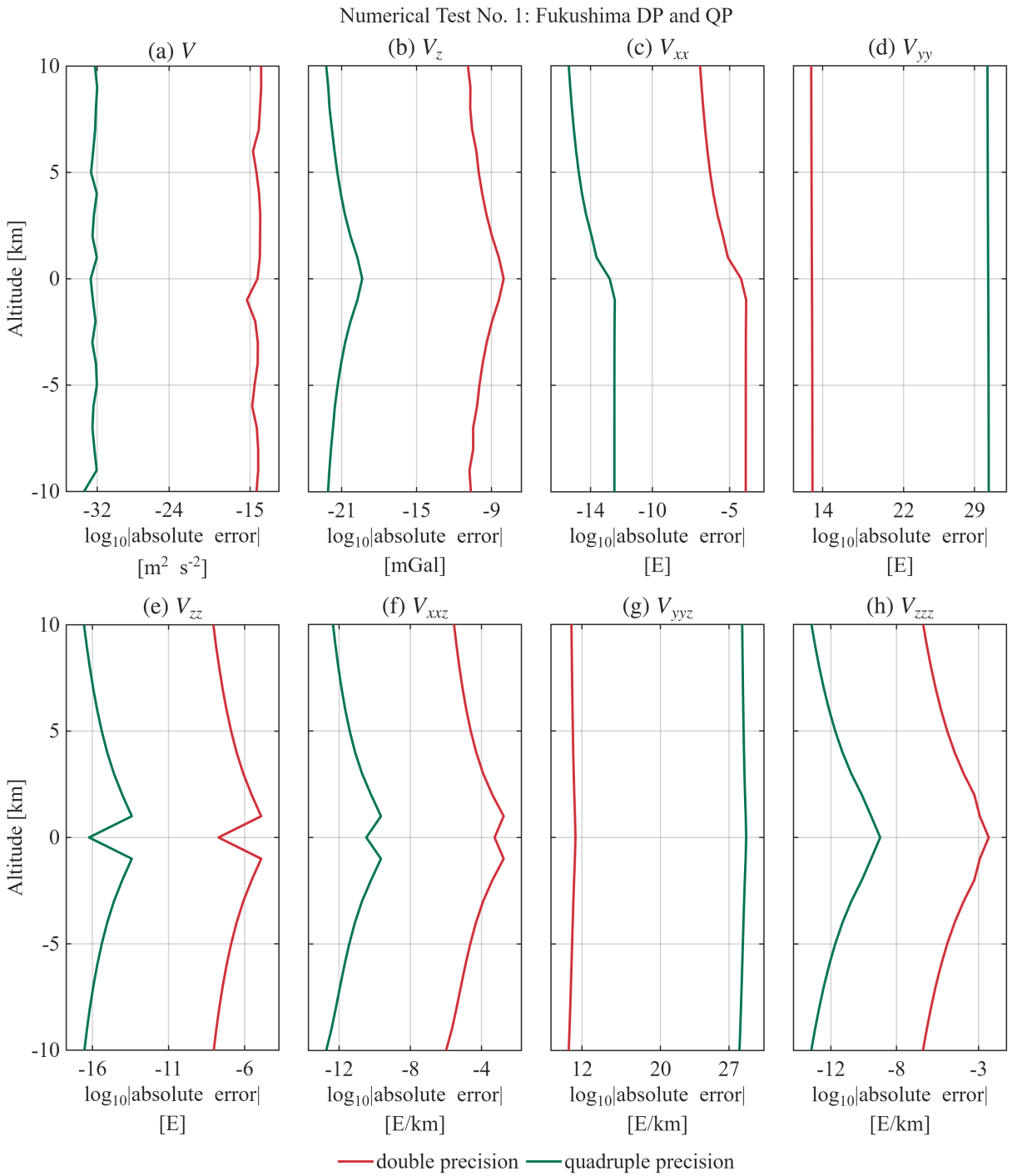


Figure S20: Numerical Test No. 1: absolute errors on logarithmic axes of Fukushima's DP (red) and QP (green) implementations relative to the polar-axis analytical solution. The anomalous V_{yy} and V_{yyz} errors show that these two components are not reliably evaluated on the polar axis. Physical units follow Eq. (86).

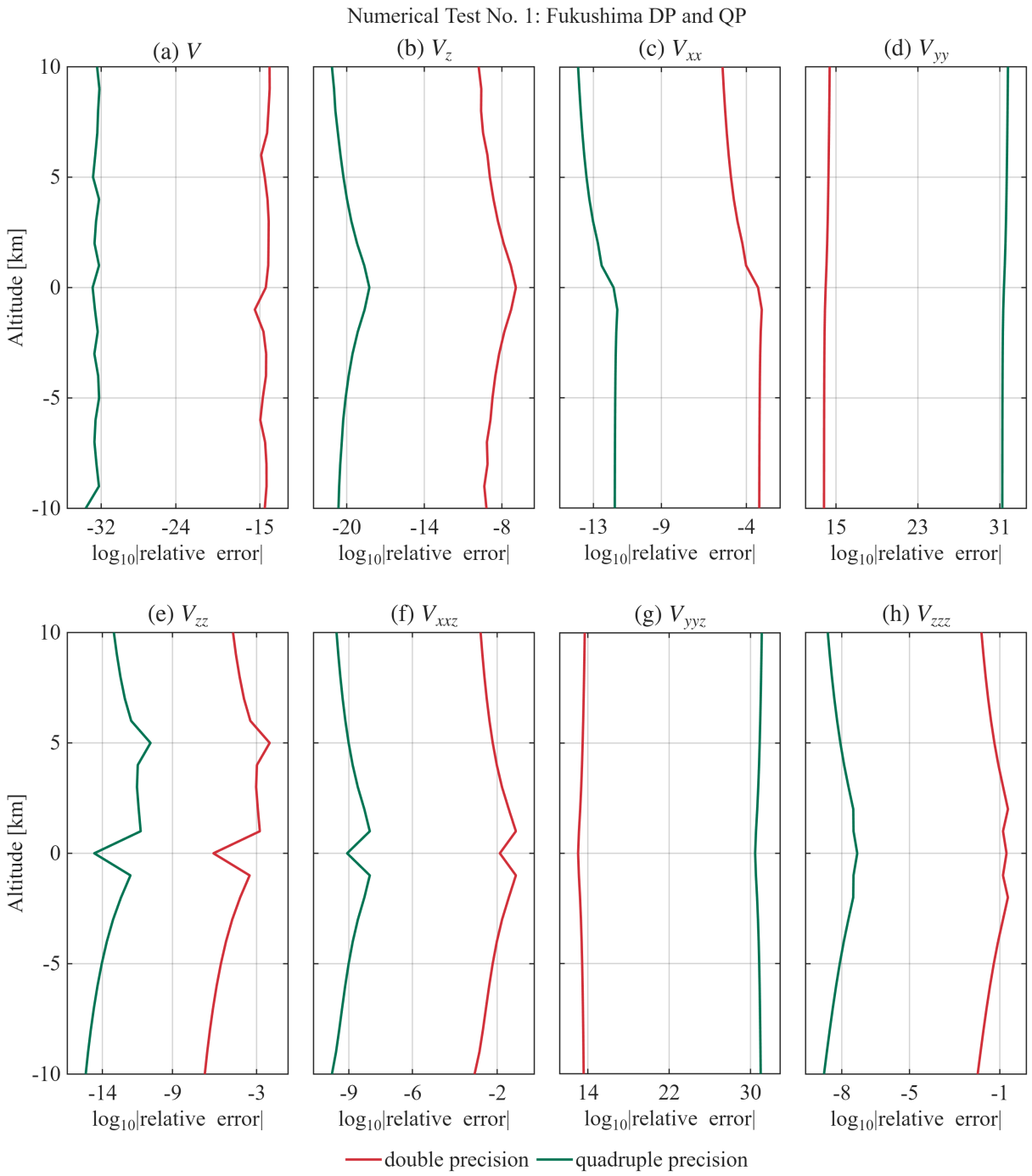


Figure S21: Numerical Test No. 1: logarithmic relative errors of Fukushima's DP (red) and QP (green) implementations. The QP errors for V and V_z are below double-precision machine epsilon, whereas the polar-axis results for V_{yy} and V_{yyz} do not converge to the analytical values and are therefore unreliable.

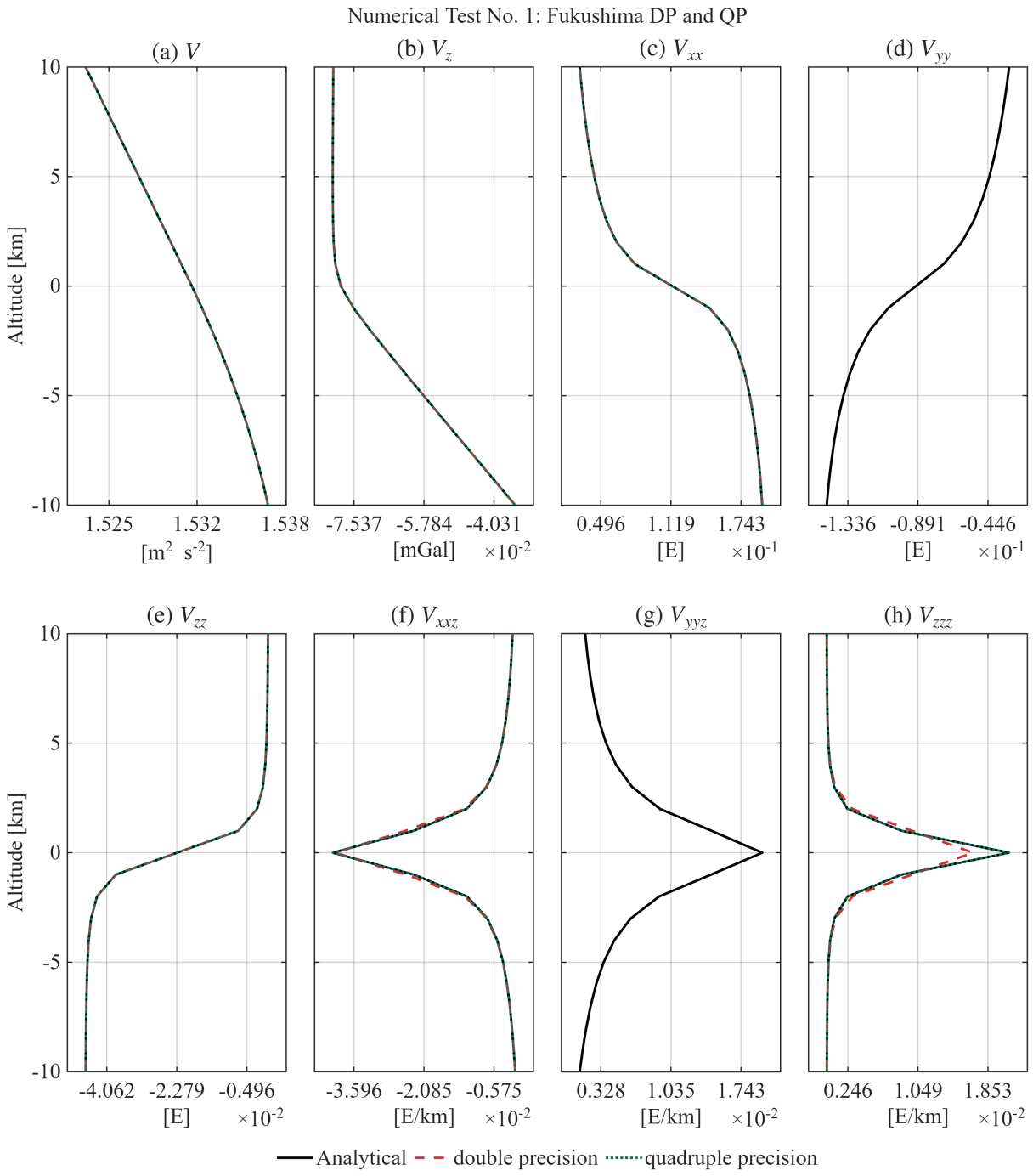


Figure S22: Numerical Test No. 1: analytical values (black) and Fukushima DP (red dashed) and QP (green dotted) values. The amplitudes of the Fukushima DP and QP results for V_{yy} and V_{yyz} are far outside the range of the analytical solutions and therefore lie outside the plotted horizontal ranges. Physical units follow Eq. (86).

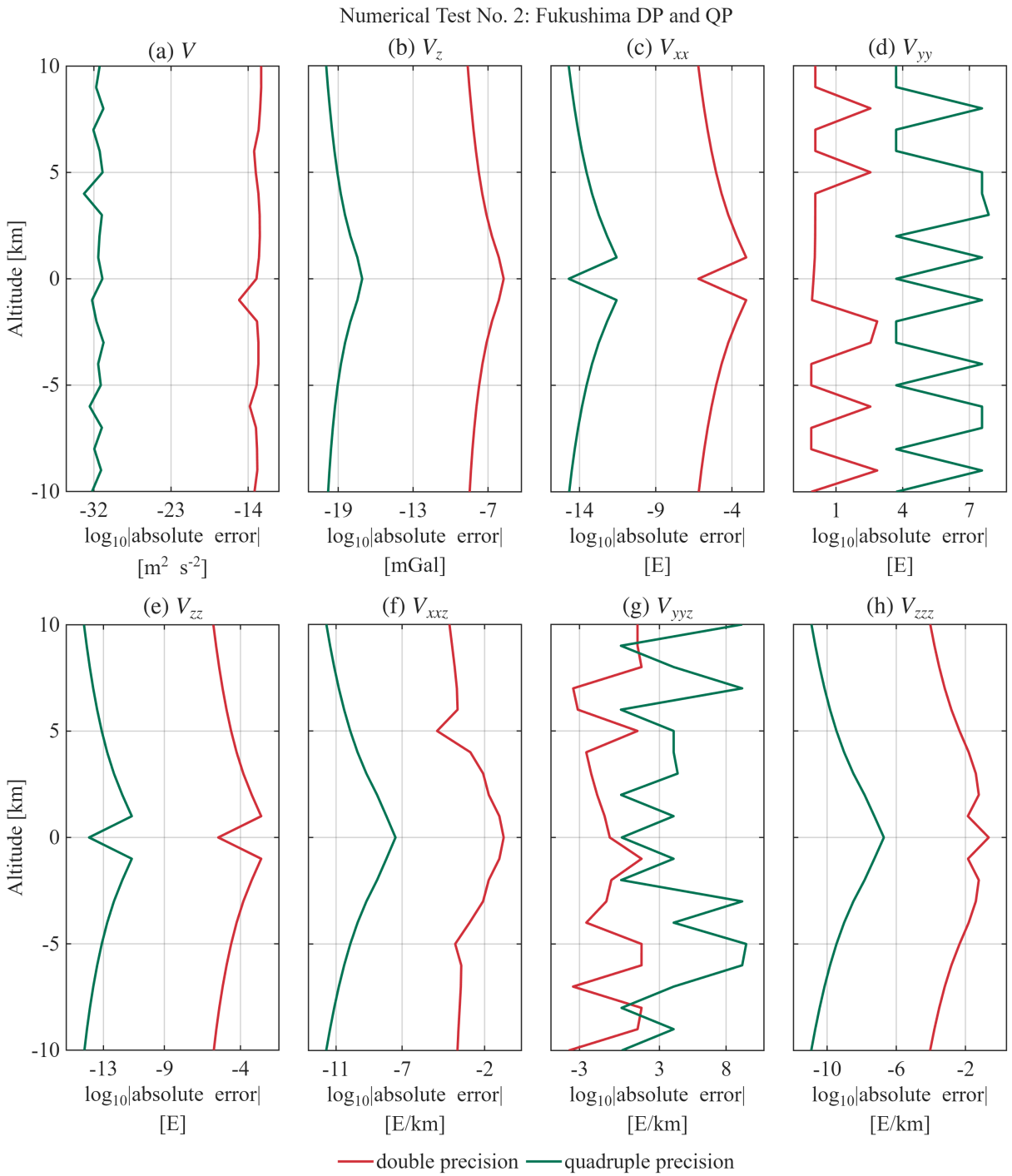


Figure S23: Numerical Test No. 2: absolute errors on logarithmic axes of Fukushima's DP (red) and QP (green) implementations relative to the polar-axis analytical solution. The anomalous V_{yy} and V_{yyz} errors show that these two components are not reliably evaluated on the polar axis. Physical units follow Eq. (86).

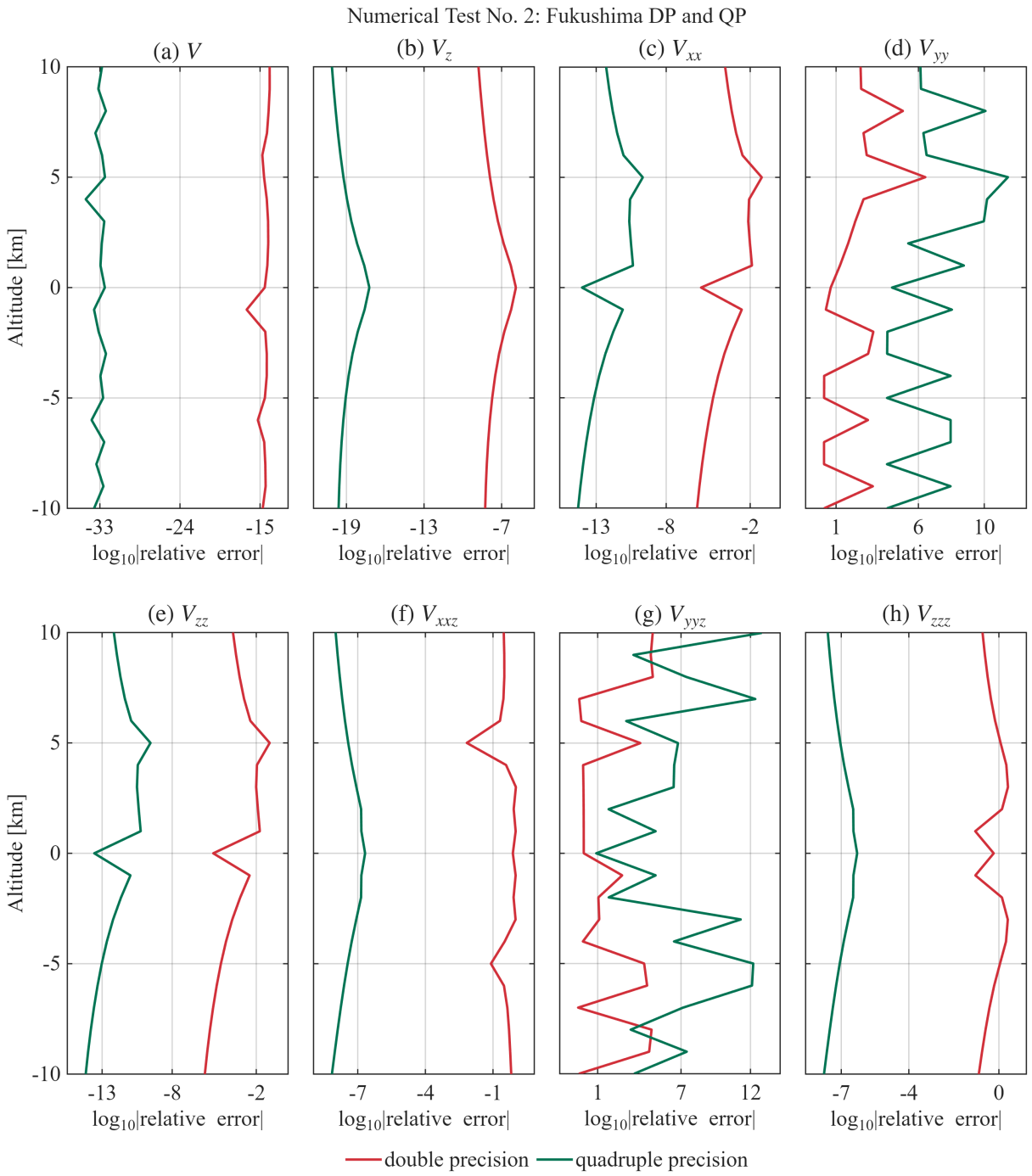


Figure S24: Numerical Test No. 2: logarithmic relative errors of Fukushima's DP (red) and QP (green) implementations. The polar-axis results for V_{yy} and V_{yyz} do not reproduce the analytical values and are therefore unreliable.

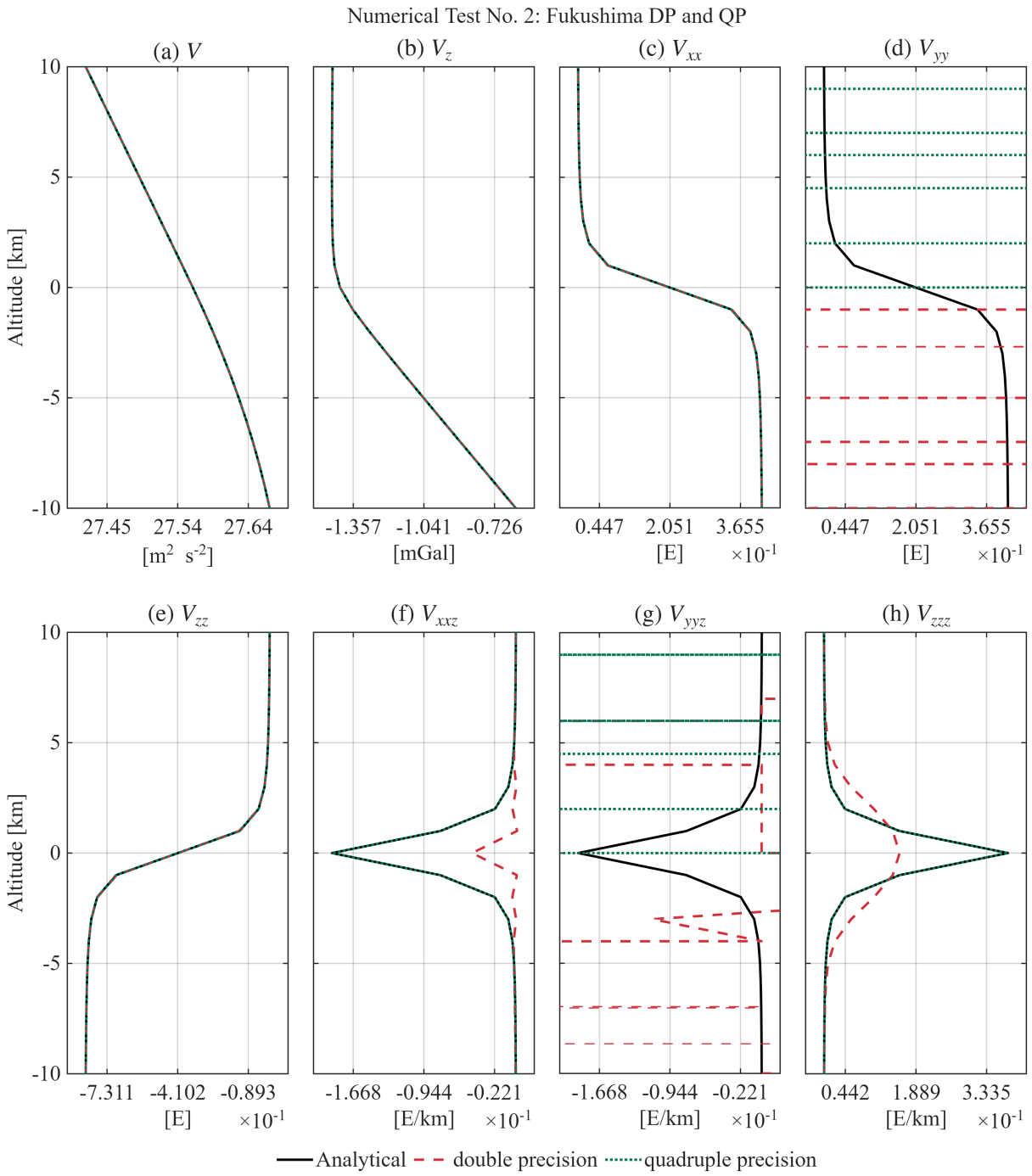


Figure S25: Numerical Test No. 2: analytical values (black) and Fukushima DP (red dashed) and QP (green dotted) values. The discontinuous DP/QP patterns of V_{yy} and V_{yyz} are numerical artifacts and do not reproduce the analytical polar-axis solutions. Physical units follow Eq. (86).

Table S22: Absolute-error statistics of Fukushima DP and QP for Numerical Test No. 1. Physical units follow Eq. (86).

Component	Precision	Minimum	Maximum	Mean	RMSE
V	DP	$4.42 \cdot 10^{-16}$	$1.72 \cdot 10^{-14}$	$8.03 \cdot 10^{-15}$	$9.29 \cdot 10^{-15}$
V	QP	$3.39 \cdot 10^{-34}$	$9.86 \cdot 10^{-33}$	$5.28 \cdot 10^{-33}$	$5.96 \cdot 10^{-33}$
V_z	DP	$1.29 \cdot 10^{-11}$	$9.55 \cdot 10^{-9}$	$1.02 \cdot 10^{-9}$	$2.44 \cdot 10^{-9}$
V_z	QP	$5.78 \cdot 10^{-23}$	$4.43 \cdot 10^{-20}$	$4.70 \cdot 10^{-21}$	$1.13 \cdot 10^{-20}$
V_{xx}	DP	$1.22 \cdot 10^{-7}$	$1.18 \cdot 10^{-4}$	$5.68 \cdot 10^{-5}$	$7.83 \cdot 10^{-5}$
V_{xx}	QP	$3.87 \cdot 10^{-16}$	$3.72 \cdot 10^{-13}$	$1.79 \cdot 10^{-13}$	$2.47 \cdot 10^{-13}$
V_{yy}	DP	$7.50 \cdot 10^{12}$	$9.89 \cdot 10^{12}$	$8.76 \cdot 10^{12}$	$8.80 \cdot 10^{12}$
V_{yy}	QP	$1.95 \cdot 10^{30}$	$2.57 \cdot 10^{30}$	$2.27 \cdot 10^{30}$	$2.28 \cdot 10^{30}$
V_{zz}	DP	$8.83 \cdot 10^{-9}$	$1.22 \cdot 10^{-5}$	$1.57 \cdot 10^{-6}$	$3.88 \cdot 10^{-6}$
V_{zz}	QP	$2.79 \cdot 10^{-17}$	$3.86 \cdot 10^{-14}$	$4.97 \cdot 10^{-15}$	$1.23 \cdot 10^{-14}$
V_{xxz}	DP	$1.02 \cdot 10^{-6}$	$1.76 \cdot 10^{-3}$	$2.54 \cdot 10^{-4}$	$5.72 \cdot 10^{-4}$
V_{xxz}	QP	$1.88 \cdot 10^{-13}$	$2.24 \cdot 10^{-10}$	$3.23 \cdot 10^{-11}$	$7.25 \cdot 10^{-11}$
V_{yyz}	DP	$4.46 \cdot 10^{10}$	$2.17 \cdot 10^{11}$	$1.17 \cdot 10^{11}$	$1.26 \cdot 10^{11}$
V_{yyz}	QP	$1.16 \cdot 10^{28}$	$5.63 \cdot 10^{28}$	$3.04 \cdot 10^{28}$	$3.27 \cdot 10^{28}$
V_{zzz}	DP	$4.10 \cdot 10^{-7}$	$4.16 \cdot 10^{-3}$	$3.81 \cdot 10^{-4}$	$9.95 \cdot 10^{-4}$
V_{zzz}	QP	$6.40 \cdot 10^{-14}$	$1.02 \cdot 10^{-9}$	$8.55 \cdot 10^{-11}$	$2.40 \cdot 10^{-10}$

Table S23: Relative-error statistics of Fukushima DP and QP for Numerical Test No. 1.

Component	Precision	Minimum	Maximum	Mean	RMSE
V	DP	$2.88 \cdot 10^{-16}$	$1.13 \cdot 10^{-14}$	$5.25 \cdot 10^{-15}$	$6.08 \cdot 10^{-15}$
V	QP	$2.20 \cdot 10^{-34}$	$6.47 \cdot 10^{-33}$	$3.45 \cdot 10^{-33}$	$3.90 \cdot 10^{-33}$
V_z	DP	$1.61 \cdot 10^{-10}$	$1.21 \cdot 10^{-7}$	$1.32 \cdot 10^{-8}$	$3.11 \cdot 10^{-8}$
V_z	QP	$7.18 \cdot 10^{-22}$	$5.63 \cdot 10^{-19}$	$6.11 \cdot 10^{-20}$	$1.44 \cdot 10^{-19}$
V_{xx}	DP	$3.98 \cdot 10^{-6}$	$8.08 \cdot 10^{-4}$	$3.36 \cdot 10^{-4}$	$4.52 \cdot 10^{-4}$
V_{xx}	QP	$1.26 \cdot 10^{-14}$	$2.55 \cdot 10^{-12}$	$1.06 \cdot 10^{-12}$	$1.43 \cdot 10^{-12}$
V_{yy}	DP	$6.73 \cdot 10^{13}$	$2.41 \cdot 10^{14}$	$1.27 \cdot 10^{14}$	$1.41 \cdot 10^{14}$
V_{yy}	QP	$1.75 \cdot 10^{31}$	$6.24 \cdot 10^{31}$	$3.29 \cdot 10^{31}$	$3.67 \cdot 10^{31}$
V_{zz}	DP	$2.03 \cdot 10^{-7}$	$8.61 \cdot 10^{-3}$	$6.93 \cdot 10^{-4}$	$1.96 \cdot 10^{-3}$
V_{zz}	QP	$6.47 \cdot 10^{-16}$	$2.72 \cdot 10^{-11}$	$2.19 \cdot 10^{-12}$	$6.20 \cdot 10^{-12}$
V_{xxz}	DP	$8.59 \cdot 10^{-4}$	$7.65 \cdot 10^{-2}$	$1.56 \cdot 10^{-2}$	$2.70 \cdot 10^{-2}$
V_{xxz}	QP	$1.59 \cdot 10^{-10}$	$9.74 \cdot 10^{-9}$	$2.17 \cdot 10^{-9}$	$3.59 \cdot 10^{-9}$
V_{yyz}	DP	$1.11 \cdot 10^{13}$	$4.98 \cdot 10^{13}$	$2.92 \cdot 10^{13}$	$3.12 \cdot 10^{13}$
V_{yyz}	QP	$2.88 \cdot 10^{30}$	$1.29 \cdot 10^{31}$	$7.59 \cdot 10^{30}$	$8.10 \cdot 10^{30}$
V_{zzz}	DP	$1.05 \cdot 10^{-2}$	$2.30 \cdot 10^{-1}$	$8.23 \cdot 10^{-2}$	$1.09 \cdot 10^{-1}$
V_{zzz}	QP	$1.63 \cdot 10^{-9}$	$4.85 \cdot 10^{-8}$	$1.41 \cdot 10^{-8}$	$1.95 \cdot 10^{-8}$

Table S24: Absolute-error statistics of Fukushima DP and QP for Numerical Test No. 2. Physical units follow Eq. (86).

Component	Precision	Minimum	Maximum	Mean	RMSE
V	DP	$8.62 \cdot 10^{-16}$	$3.30 \cdot 10^{-13}$	$1.41 \cdot 10^{-13}$	$1.67 \cdot 10^{-13}$
V	QP	$6.99 \cdot 10^{-34}$	$1.33 \cdot 10^{-31}$	$4.96 \cdot 10^{-32}$	$6.44 \cdot 10^{-32}$
V_z	DP	$2.36 \cdot 10^{-9}$	$1.80 \cdot 10^{-6}$	$1.91 \cdot 10^{-7}$	$4.59 \cdot 10^{-7}$
V_z	QP	$1.09 \cdot 10^{-20}$	$8.35 \cdot 10^{-18}$	$8.87 \cdot 10^{-19}$	$2.13 \cdot 10^{-18}$
V_{xx}	DP	$6.20 \cdot 10^{-7}$	$8.63 \cdot 10^{-4}$	$1.11 \cdot 10^{-4}$	$2.74 \cdot 10^{-4}$
V_{xx}	QP	$1.96 \cdot 10^{-15}$	$2.73 \cdot 10^{-12}$	$3.51 \cdot 10^{-13}$	$8.68 \cdot 10^{-13}$
V_{yy}	DP	$7.67 \cdot 10^{-1}$	$7.12 \cdot 10^{2}$	$1.36 \cdot 10^{2}$	$2.69 \cdot 10^{2}$
V_{yy}	QP	$5.05 \cdot 10^{3}$	$7.44 \cdot 10^{7}$	$1.95 \cdot 10^{7}$	$2.93 \cdot 10^{7}$
V_{zz}	DP	$1.66 \cdot 10^{-6}$	$2.30 \cdot 10^{-3}$	$2.97 \cdot 10^{-4}$	$7.33 \cdot 10^{-4}$
V_{zz}	QP	$5.25 \cdot 10^{-15}$	$7.28 \cdot 10^{-12}$	$9.38 \cdot 10^{-13}$	$2.31 \cdot 10^{-12}$
V_{xxz}	DP	$1.32 \cdot 10^{-5}$	$1.44 \cdot 10^{-1}$	$1.70 \cdot 10^{-2}$	$4.00 \cdot 10^{-2}$
V_{xxz}	QP	$2.52 \cdot 10^{-12}$	$3.99 \cdot 10^{-8}$	$3.36 \cdot 10^{-9}$	$9.42 \cdot 10^{-9}$
V_{yyz}	DP	$1.56 \cdot 10^{-4}$	$4.54 \cdot 10^{1}$	$1.52 \cdot 10^{1}$	$2.43 \cdot 10^{1}$
V_{yyz}	QP	$1.28 \cdot 10^{0}$	$3.23 \cdot 10^{9}$	$4.61 \cdot 10^{8}$	$9.97 \cdot 10^{8}$
V_{zzz}	DP	$9.02 \cdot 10^{-5}$	$2.22 \cdot 10^{-1}$	$2.35 \cdot 10^{-2}$	$5.36 \cdot 10^{-2}$
V_{zzz}	QP	$1.21 \cdot 10^{-11}$	$1.92 \cdot 10^{-7}$	$1.61 \cdot 10^{-8}$	$4.52 \cdot 10^{-8}$

Table S25: Relative-error statistics of Fukushima DP and QP for Numerical Test No. 2.

Component	Precision	Minimum	Maximum	Mean	RMSE
V	DP	$3.12 \cdot 10^{-17}$	$1.20 \cdot 10^{-14}$	$5.14 \cdot 10^{-15}$	$6.07 \cdot 10^{-15}$
V	QP	$2.54 \cdot 10^{-35}$	$4.81 \cdot 10^{-33}$	$1.80 \cdot 10^{-33}$	$2.34 \cdot 10^{-33}$
V_z	DP	$1.63 \cdot 10^{-9}$	$1.27 \cdot 10^{-6}$	$1.38 \cdot 10^{-7}$	$3.26 \cdot 10^{-7}$
V_z	QP	$7.55 \cdot 10^{-21}$	$5.90 \cdot 10^{-18}$	$6.40 \cdot 10^{-19}$	$1.51 \cdot 10^{-18}$
V_{xx}	DP	$1.61 \cdot 10^{-6}$	$6.73 \cdot 10^{-2}$	$5.42 \cdot 10^{-3}$	$1.53 \cdot 10^{-2}$
V_{xx}	QP	$5.10 \cdot 10^{-15}$	$2.13 \cdot 10^{-10}$	$1.72 \cdot 10^{-11}$	$4.86 \cdot 10^{-11}$
V_{yy}	DP	$1.85 \cdot 10^{0}$	$2.64 \cdot 10^{6}$	$1.31 \cdot 10^{5}$	$5.76 \cdot 10^{5}$
V_{yy}	QP	$1.23 \cdot 10^{4}$	$2.75 \cdot 10^{11}$	$1.49 \cdot 10^{10}$	$6.01 \cdot 10^{10}$
V_{zz}	DP	$2.15 \cdot 10^{-6}$	$9.04 \cdot 10^{-2}$	$7.29 \cdot 10^{-3}$	$2.06 \cdot 10^{-2}$
V_{zz}	QP	$6.78 \cdot 10^{-15}$	$2.85 \cdot 10^{-10}$	$2.30 \cdot 10^{-11}$	$6.49 \cdot 10^{-11}$
V_{xxz}	DP	$6.89 \cdot 10^{-3}$	$1.04 \cdot 10^{0}$	$5.33 \cdot 10^{-1}$	$6.19 \cdot 10^{-1}$
V_{xxz}	QP	$7.14 \cdot 10^{-9}$	$2.12 \cdot 10^{-7}$	$6.17 \cdot 10^{-8}$	$8.52 \cdot 10^{-8}$
V_{yyz}	DP	$3.91 \cdot 10^{-1}$	$9.34 \cdot 10^{4}$	$2.14 \cdot 10^{4}$	$3.90 \cdot 10^{4}$
V_{yyz}	QP	$7.87 \cdot 10^{0}$	$6.52 \cdot 10^{12}$	$5.64 \cdot 10^{11}$	$1.57 \cdot 10^{12}$
V_{zzz}	DP	$8.90 \cdot 10^{-2}$	$2.52 \cdot 10^{0}$	$8.78 \cdot 10^{-1}$	$1.19 \cdot 10^{0}$
V_{zzz}	QP	$1.71 \cdot 10^{-8}$	$5.08 \cdot 10^{-7}$	$1.48 \cdot 10^{-7}$	$2.05 \cdot 10^{-7}$

6.2 Comparison of the proposed method with Fukushima QP

We next use Fukushima QP as the reference for V, V_x, V_y, and V_z. The prism is the one used in Section 3.3 of the main text: $R_E - 30$ km $\leq r' \leq R_E$, $30^\circ \leq \varphi' \leq 50^\circ$, and $110^\circ \leq \lambda' \leq 130^\circ$. Three horizontal locations probe distinct near-zone geometries: a location inside the angular bounds at $(115^\circ\text{E}, 45^\circ\text{N})$, one just inside the western boundary at $(110.1^\circ\text{E}, 40^\circ\text{N})$, and one just outside the northwestern corner at $(109.9^\circ\text{E}, 50.1^\circ\text{N})$. At each location, altitude ranges from -10 to -1 km and from 1 to 10 km. The 0 km surface point is excluded because boundary-limit evaluation is outside the scope of the proposed formulation.

Across all three tests, the proposed method closely agrees with Fukushima QP. The largest RMSE values of the relative errors are 4.20×10^{-13}, 3.52×10^{-12}, and 1.77×10^{-14} at the angularly interior, western-inside, and northwestern-outside locations, respectively. The largest individual relative error is 9.54×10^{-12} for V_x at the western-inside location; for every other component and configuration it is no greater than 4.41×10^{-13}. These differences quantify agreement with the QP potential and acceleration vector for the three sampled locations.

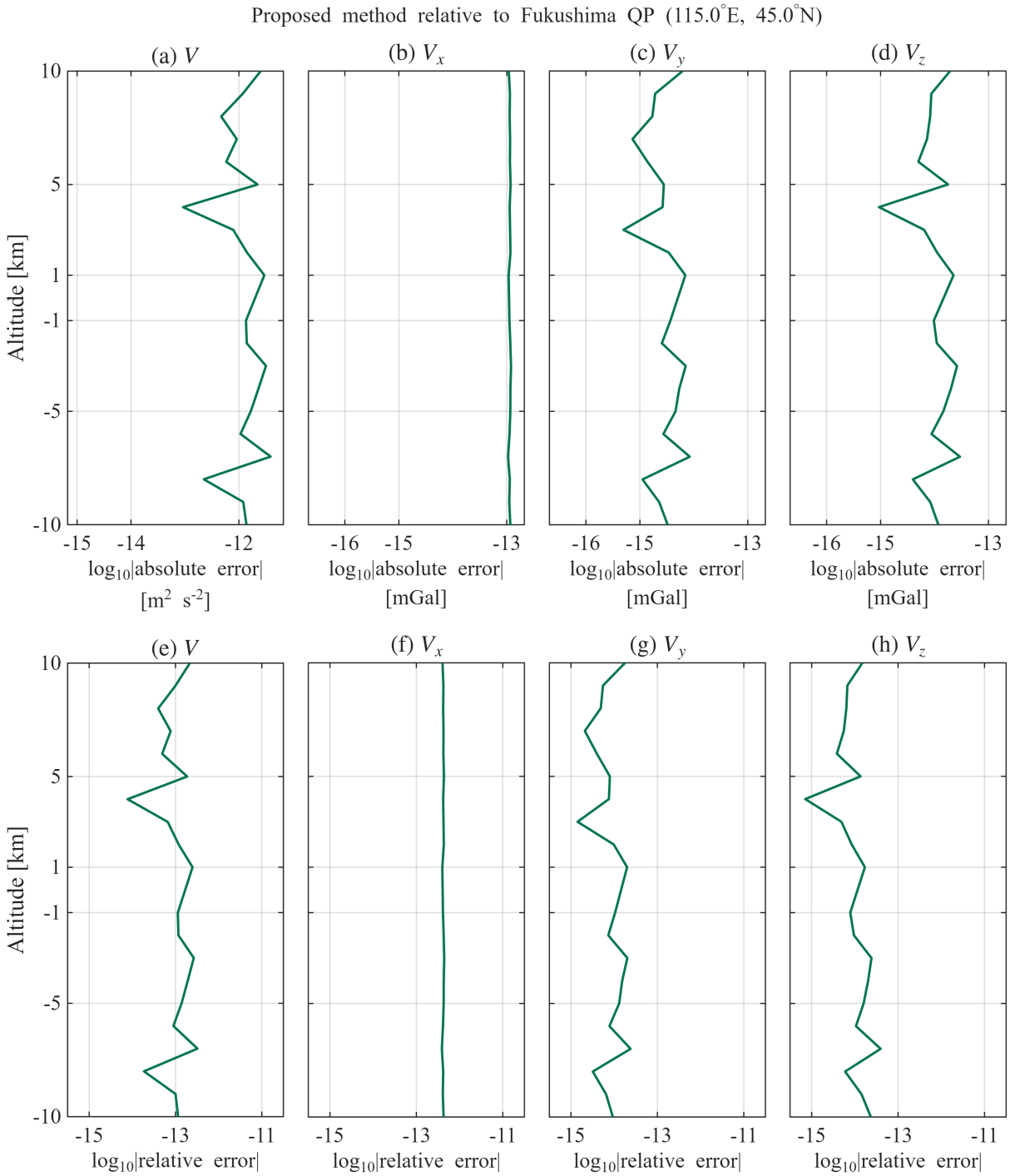


Figure S26: Comparison with Fukushima QP at (115°E, 45°N). Panels (a)–(d) show absolute errors on logarithmic axes of the proposed method for V, V_x, V_y, and V_z; panels (e)–(h) show the corresponding logarithmic relative errors. Physical units follow Eq. (86).

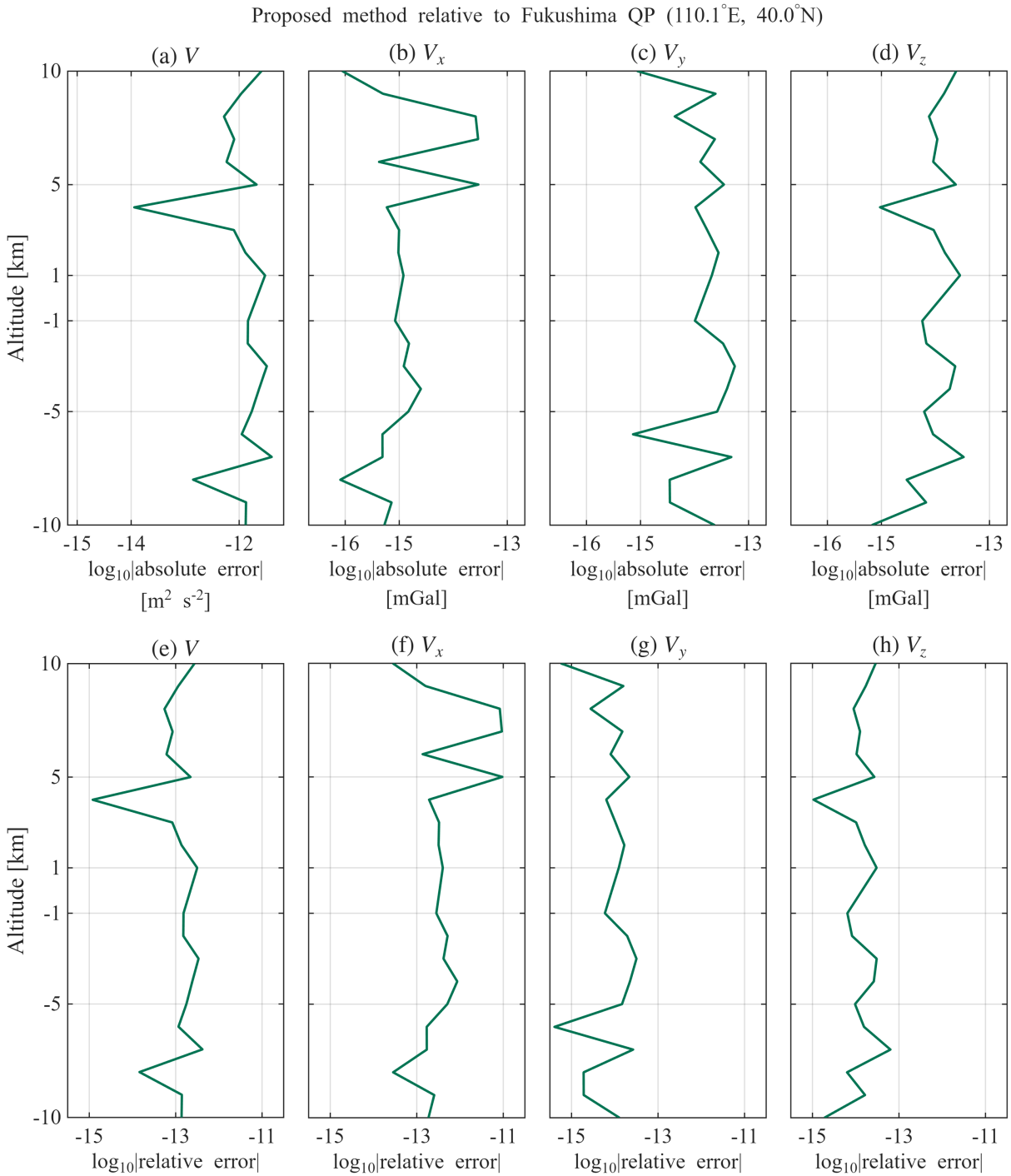


Figure S27: Proposed method relative to Fukushima QP at the location just inside the western boundary, (110.1°E, 40°N). Panel order is the same as in Fig. S26.

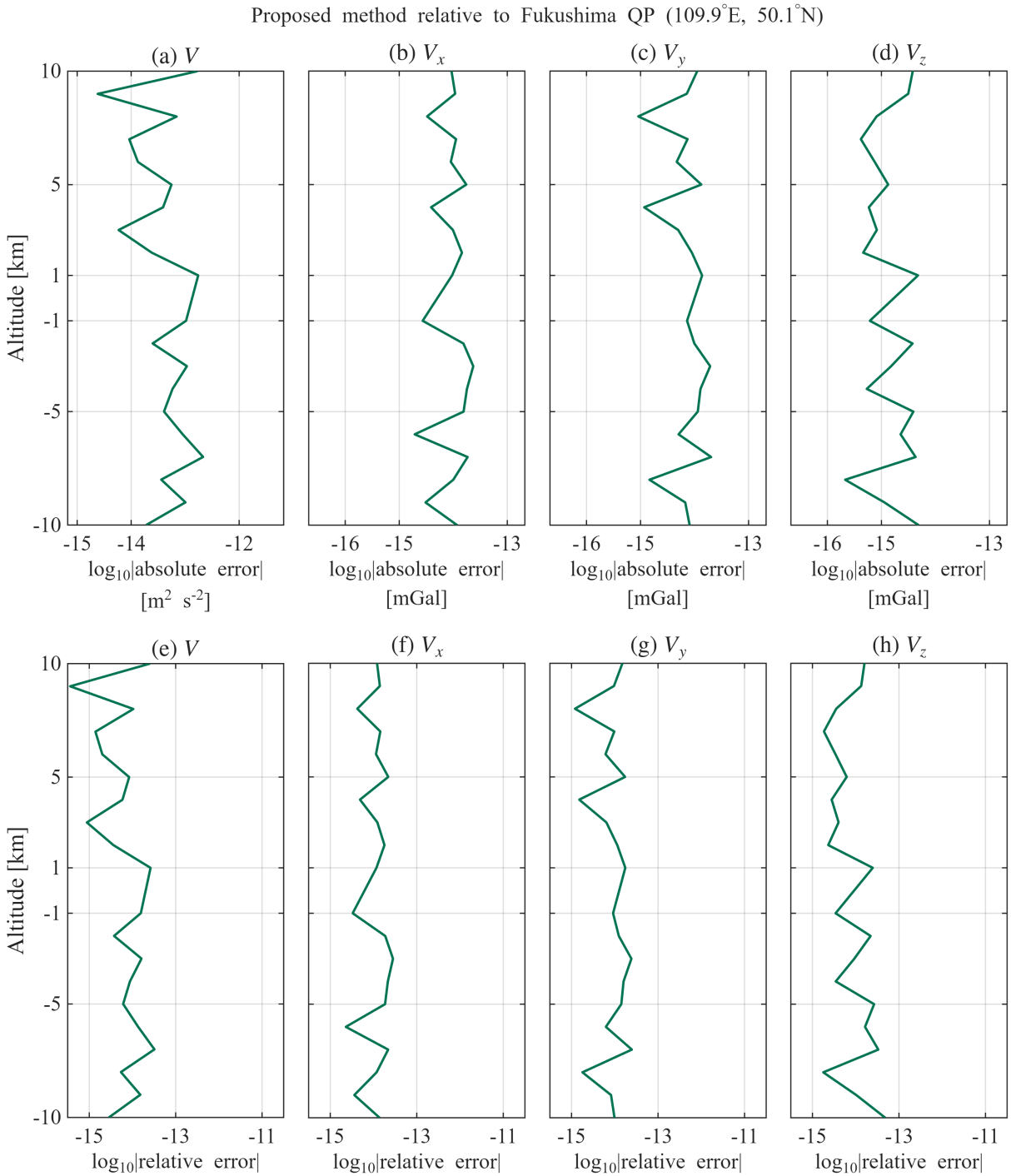


Figure S28: Proposed method relative to Fukushima QP at the location just outside the northwestern corner, (109.9°E, 50.1°N). Panel order is the same as in Fig. S26.

Table S26: Absolute- and relative-error statistics of the proposed method relative to Fukushima QP at (115°E, 45°N). Absolute rows use $\mathrm{m^2\,s^{-2}}$ for V and mGal for the acceleration components; relative rows are dimensionless.

Component	Error	Minimum	Maximum	Mean	RMSE
V	Absolute	$9.19 \cdot 10^{-14}$	$3.87 \cdot 10^{-12}$	$1.54 \cdot 10^{-12}$	$1.83 \cdot 10^{-12}$
V	Relative	$7.74 \cdot 10^{-15}$	$3.23 \cdot 10^{-13}$	$1.29 \cdot 10^{-13}$	$1.53 \cdot 10^{-13}$
V_x	Absolute	$1.06 \cdot 10^{-13}$	$1.20 \cdot 10^{-13}$	$1.14 \cdot 10^{-13}$	$1.14 \cdot 10^{-13}$
V_x	Relative	$3.89 \cdot 10^{-13}$	$4.41 \cdot 10^{-13}$	$4.20 \cdot 10^{-13}$	$4.20 \cdot 10^{-13}$
V_y	Absolute	$4.96 \cdot 10^{-16}$	$8.40 \cdot 10^{-15}$	$3.46 \cdot 10^{-15}$	$4.11 \cdot 10^{-15}$
V_y	Relative	$1.42 \cdot 10^{-15}$	$2.40 \cdot 10^{-14}$	$9.90 \cdot 10^{-15}$	$1.17 \cdot 10^{-14}$
V_z	Absolute	$9.34 \cdot 10^{-16}$	$2.95 \cdot 10^{-14}$	$1.26 \cdot 10^{-14}$	$1.46 \cdot 10^{-14}$
V_z	Relative	$7.11 \cdot 10^{-16}$	$3.94 \cdot 10^{-14}$	$1.27 \cdot 10^{-14}$	$1.55 \cdot 10^{-14}$

Table S27: Absolute- and relative-error statistics of the proposed method relative to Fukushima QP at (110.1°E, 40°N). Absolute rows use $\mathrm{m^2\,s^{-2}}$ for V and mGal for the acceleration components; relative rows are dimensionless.

Component	Error	Minimum	Maximum	Mean	RMSE
V	Absolute	$1.14 \cdot 10^{-14}$	$4.02 \cdot 10^{-12}$	$1.55 \cdot 10^{-12}$	$1.86 \cdot 10^{-12}$
V	Relative	$1.19 \cdot 10^{-15}$	$4.17 \cdot 10^{-13}$	$1.61 \cdot 10^{-13}$	$1.94 \cdot 10^{-13}$
V_x	Absolute	$8.11 \cdot 10^{-17}$	$2.93 \cdot 10^{-14}$	$4.93 \cdot 10^{-15}$	$1.09 \cdot 10^{-14}$
V_x	Relative	$2.79 \cdot 10^{-14}$	$9.54 \cdot 10^{-12}$	$1.60 \cdot 10^{-12}$	$3.52 \cdot 10^{-12}$
V_y	Absolute	$7.25 \cdot 10^{-16}$	$5.57 \cdot 10^{-14}$	$2.11 \cdot 10^{-14}$	$2.62 \cdot 10^{-14}$
V_y	Relative	$4.03 \cdot 10^{-16}$	$3.17 \cdot 10^{-14}$	$1.24 \cdot 10^{-14}$	$1.52 \cdot 10^{-14}$
V_z	Absolute	$6.64 \cdot 10^{-16}$	$3.32 \cdot 10^{-14}$	$1.28 \cdot 10^{-14}$	$1.58 \cdot 10^{-14}$
V_z	Relative	$1.05 \cdot 10^{-15}$	$6.36 \cdot 10^{-14}$	$1.74 \cdot 10^{-14}$	$2.23 \cdot 10^{-14}$

Table S28: Absolute- and relative-error statistics of the proposed method relative to Fukushima QP at (109.9°E, 50.1°N). Absolute rows use $\mathrm{m^2\,s^{-2}}$ for V and mGal for the acceleration components; relative rows are dimensionless.

Component	Error	Minimum	Maximum	Mean	RMSE
V	Absolute	$2.40 \cdot 10^{-15}$	$2.15 \cdot 10^{-13}$	$6.80 \cdot 10^{-14}$	$9.05 \cdot 10^{-14}$
V	Relative	$3.61 \cdot 10^{-16}$	$3.22 \cdot 10^{-14}$	$1.02 \cdot 10^{-14}$	$1.36 \cdot 10^{-14}$
V_x	Absolute	$1.93 \cdot 10^{-15}$	$2.34 \cdot 10^{-14}$	$1.10 \cdot 10^{-14}$	$1.24 \cdot 10^{-14}$
V_x	Relative	$2.29 \cdot 10^{-15}$	$2.82 \cdot 10^{-14}$	$1.34 \cdot 10^{-14}$	$1.52 \cdot 10^{-14}$
V_y	Absolute	$9.09 \cdot 10^{-16}$	$2.03 \cdot 10^{-14}$	$8.82 \cdot 10^{-15}$	$1.03 \cdot 10^{-14}$
V_y	Relative	$1.20 \cdot 10^{-15}$	$2.48 \cdot 10^{-14}$	$1.11 \cdot 10^{-14}$	$1.29 \cdot 10^{-14}$
V_z	Absolute	$2.13 \cdot 10^{-16}$	$4.88 \cdot 10^{-15}$	$2.00 \cdot 10^{-15}$	$2.57 \cdot 10^{-15}$
V_z	Relative	$1.77 \cdot 10^{-15}$	$4.82 \cdot 10^{-14}$	$1.27 \cdot 10^{-14}$	$1.77 \cdot 10^{-14}$

7 Independent arbitrary-precision profile validation

Tests No. 11–14 evaluate all 20 quantities at 1 km above the source, along profiles through the centre of its angular projection:

(1) Test No. 11 (Profile A): east–west;

(2) Test No. 12 (Profile B): south–north;

(3) Test No. 13 (Profile C): southwest–northeast;

(4) Test No. 14 (Profile D): northwest–southeast.

The prism spans 30°–50°N, 110°–130°E, and radii 6341.2–6371.2 km. The profiles are at radius 6372.2 km, with the unit-density convention in Eq. (86).

The independent reference uses the six-face representation of Zhong et al (2019). All six faces are integrated numerically with mpmath (The mpmath development team, 2023), without the further analytical reduction of four faces to edge integrals. Refinement starts with 40 decimal digits and 16 Gauss–Legendre nodes per panel direction, then increases to $(60, 24)$, $(80, 32)$, and $(100, 40)$, where each pair denotes decimal digits and nodes. The configured schedule allows further stages through $(160, 64)$; none was needed for the main profile reference. The calculation uses neither the proposed radial moments nor finite-difference derivatives. The experiment comprises 260 unique points and 263 profile samples; symmetry reflections are applied after the canonical reference has converged. Of the 185 canonical reference jobs, 14 stopped at $(60, 24)$, 108 at $(80, 32)$, and 63 at $(100, 40)$. For every component, successive values $X^{(a)}$ and $X^{(b)}$ must satisfy

$$|X^{(b)} - X^{(a)}| \leq 10^{-30} + 10^{-20} \max(|X^{(b)}|, |X^{(a)}|). \tag{87}$$

This criterion is applied to the unscaled kilometre-based numerical outputs before conversion to physical units. Odd east-index components at the symmetry meridian are set to their exact zero values; their zeros are imposed by symmetry, rather than established by numerical convergence. Input coordinates are promoted exactly from the same binary64 values used by the production calculations.

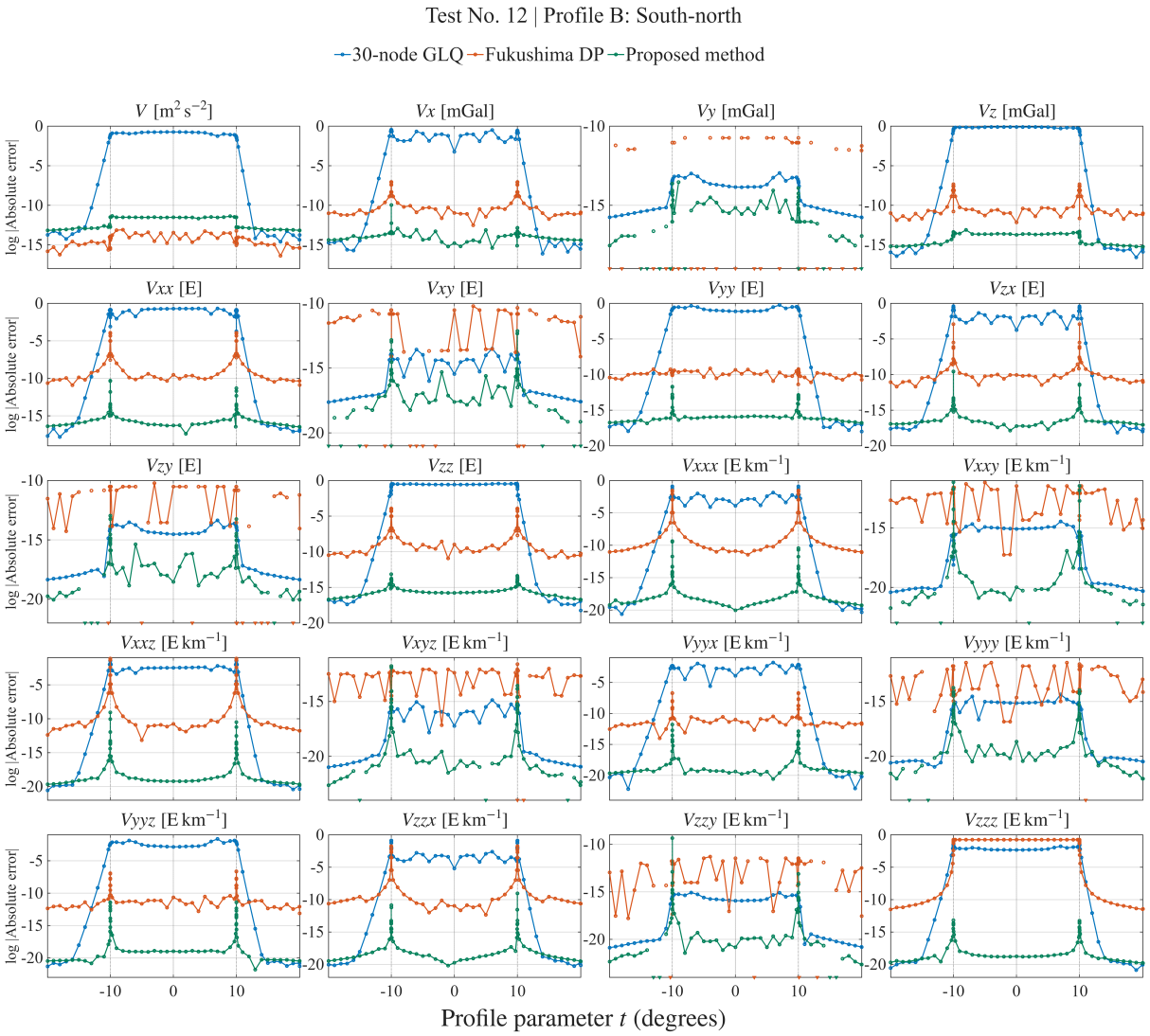
Three pilot calculations were repeated with altered angular partitions; their largest relative difference, normalized by the larger magnitude of each pair, was 6.46×10^{-32}. A separate near-edge check of V_{xxx} at $(110.001°\text{E}, 40°\text{N}, 6372.2\ \text{km})$ used 100- and 120-digit six-face calculations and an independently derived radially reduced volume calculation. The highest-order results from the two representations agreed to a relative difference of 4.47×10^{-40}; the refined six-face value differed from the original profile reference by 1.39×10^{-29}. These are targeted numerical checks, not rigorous error bounds or a claim that every profile sample was evaluated at 120 digits.

7.1 Absolute errors

Main-text Fig. 7 shows Profile A; Figs. S29–S31 show Profiles B–D. Errors are reported in $\text{m}^2\,\text{s}^{-2}$, mGal, E, and $\text{E}\,\text{km}^{-1}$ for derivative orders zero through three. Absolute errors retain meaningful comparisons at zero crossings and symmetry zeros. These componentwise references complement the polar-axis, complete-shell, and identity tests.

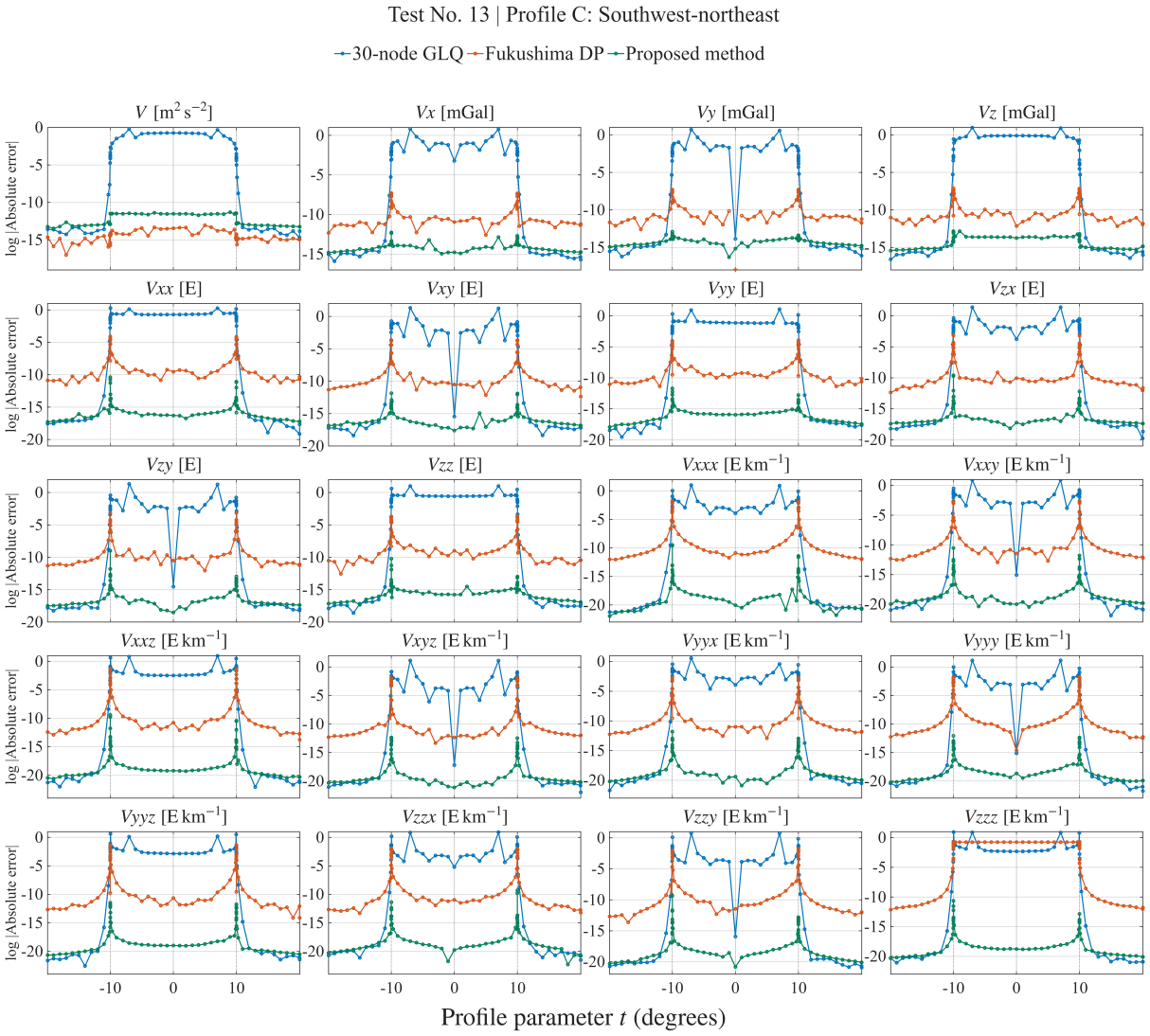
7.2 Reference and computed field values

Figures S32–S35 show the signed reference and computed values. Seven components with an odd number of y derivatives vanish by symmetry on Profile B. The proposed curves follow the reference through the projected source-edge extrema; GLQ misses several extrema. Fukushima's lower-order curves closely follow the reference, while several third derivatives have distorted extrema or interior levels, notably V_{yyz} and V_{zzz} on Profile A and V_{xxz} and V_{zzz} on Profile B. The absolute-error plots resolve differences where field-value curves overlap.



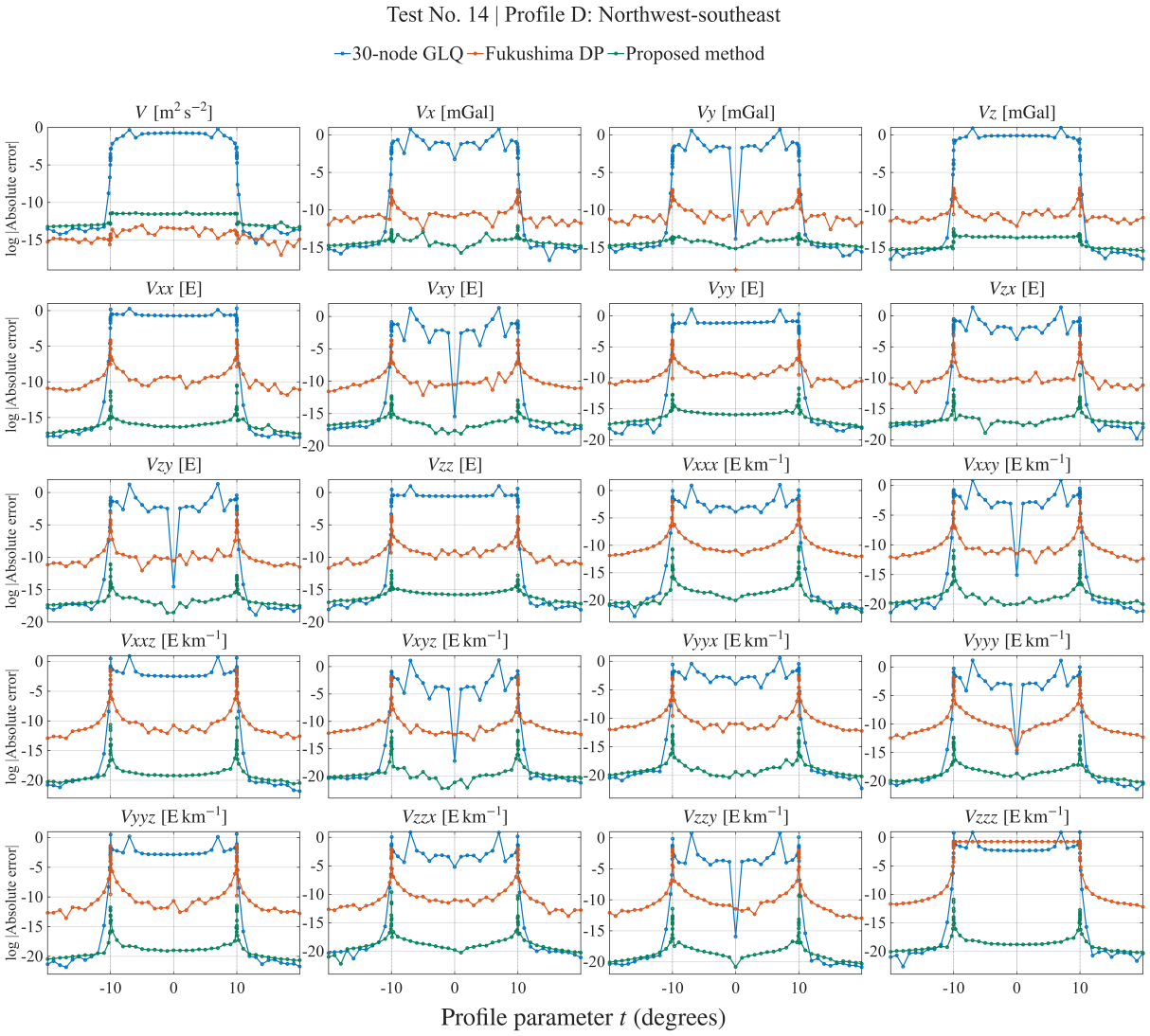
Dashed lines: source edges; units: m² s⁻², mGal, E, E km⁻¹.

Figure S29: Test No. 12 (Profile B): independent arbitrary-precision validation along the south–north profile at 1 km above the source. The 20 panels show physical absolute errors relative to the independently converged six-face high-precision reference. Dashed lines mark projected source edges.



Dashed lines: source edges; units: m² s⁻², mGal, E, E km⁻¹.

Figure S30: Test No. 13 (Profile C): independent arbitrary-precision validation along the southwest–northeast profile at 1 km above the source. The 20 panels show physical absolute errors relative to the independently converged six-face high-precision reference. Dashed lines mark projected source edges.



Dashed lines: source edges; units: m² s⁻², mGal, E, E km⁻¹.

Figure S31: Test No. 14 (Profile D): independent arbitrary-precision validation along the northwest–southeast profile at 1 km above the source. The 20 panels show physical absolute errors relative to the independently converged six-face high-precision reference. Dashed lines mark projected source edges.

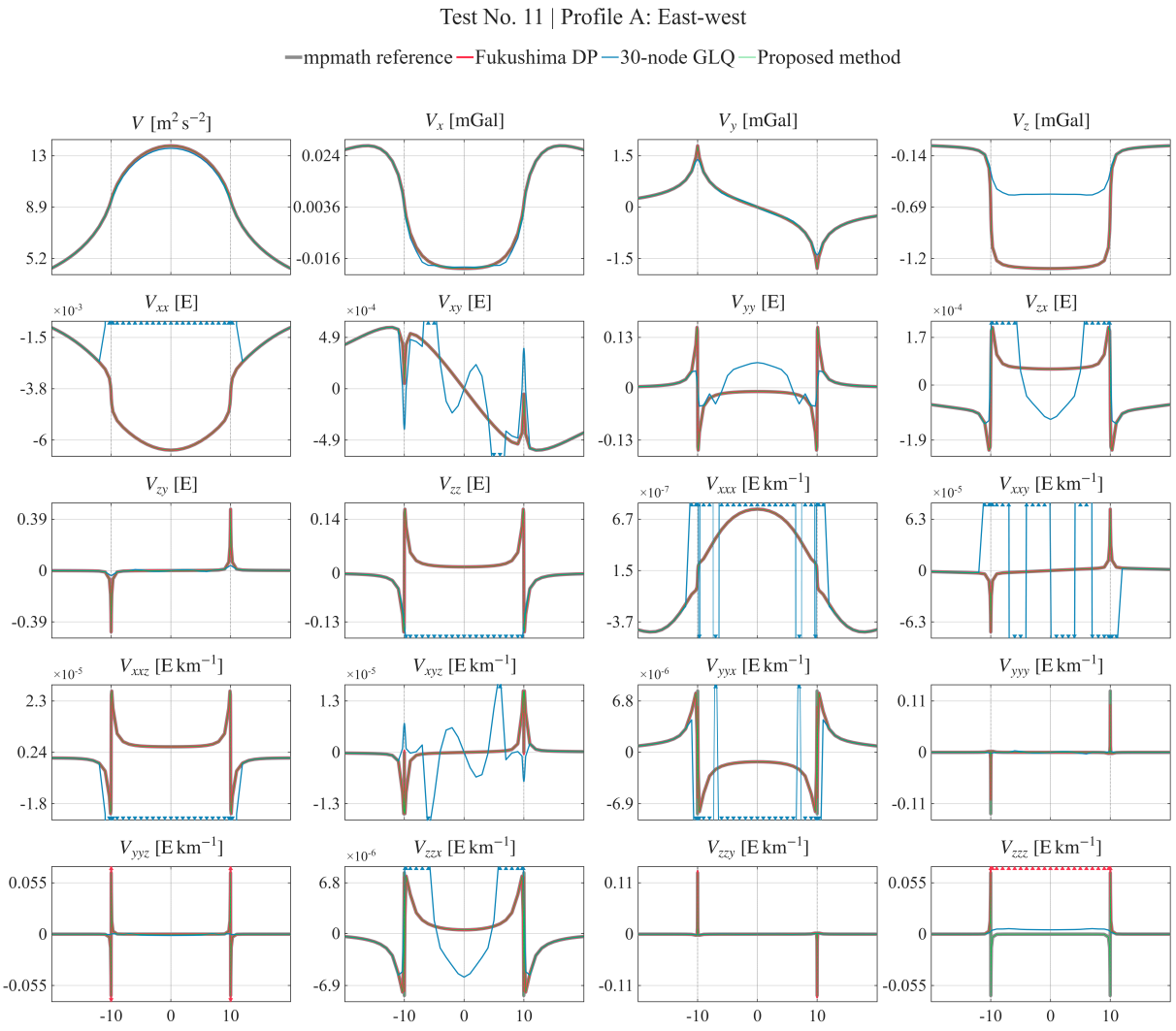


Figure S32: Test No. 11 (Profile A): signed physical field values along the east–west profile. Grey: independently converged mpmath reference; red: Fukushima double precision; blue: 30-node GLQ; green: proposed C++ method. Dashed lines mark projected source edges and triangles indicate values outside the limits.

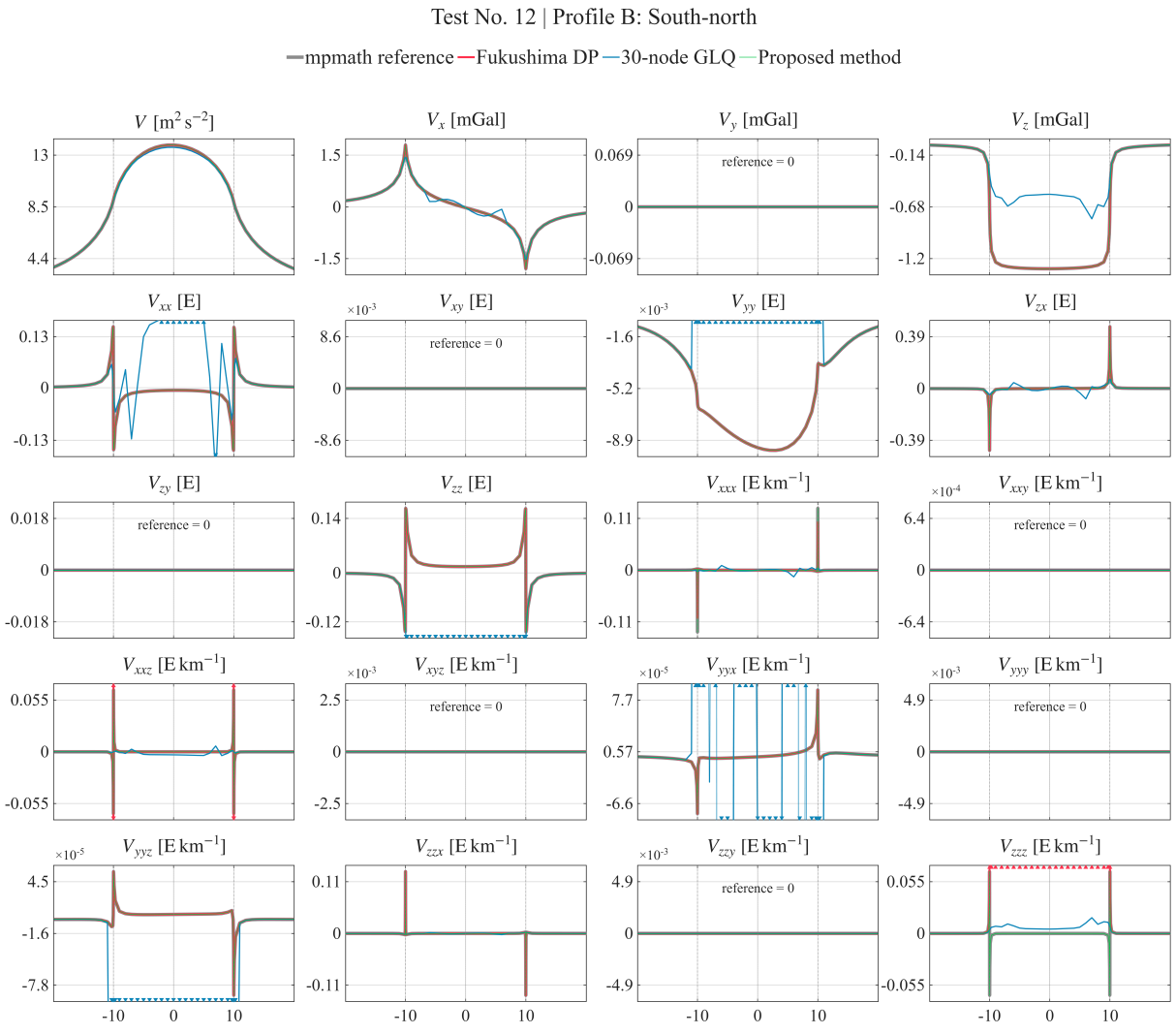
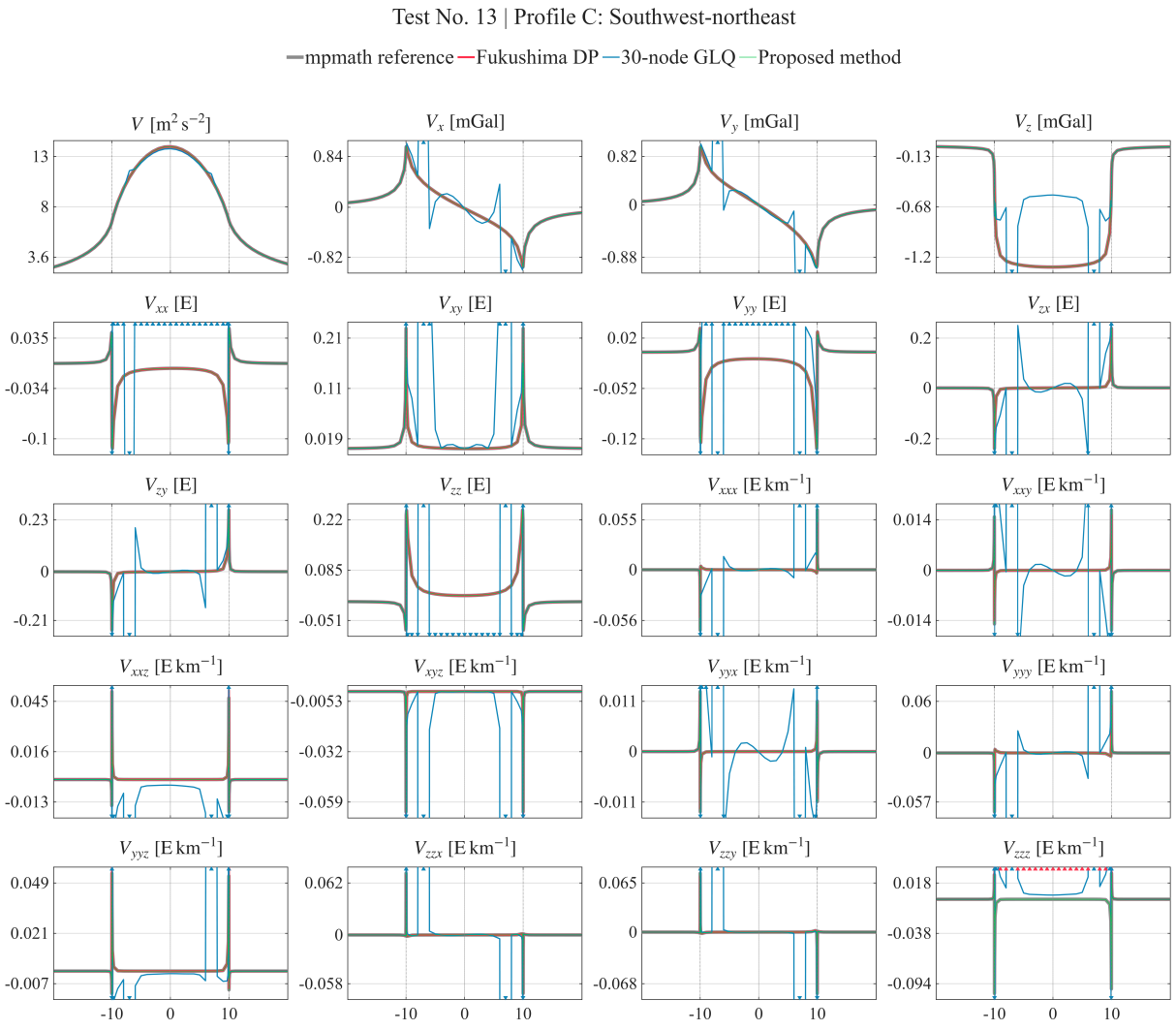


Figure S33: Test No. 12 (Profile B): signed physical field values along the south–north profile. The component order, curve styles, and boundary markers are as in Fig. S32. Panels labelled "reference = 0" contain components that vanish identically by symmetry.



Dashed lines: source edges.
Triangles: values outside the reference-based limits.

Figure S34: Test No. 13 (Profile C): signed physical field values along the southwest–northeast profile. The component order, curve styles, and boundary markers are as in Fig. S32.

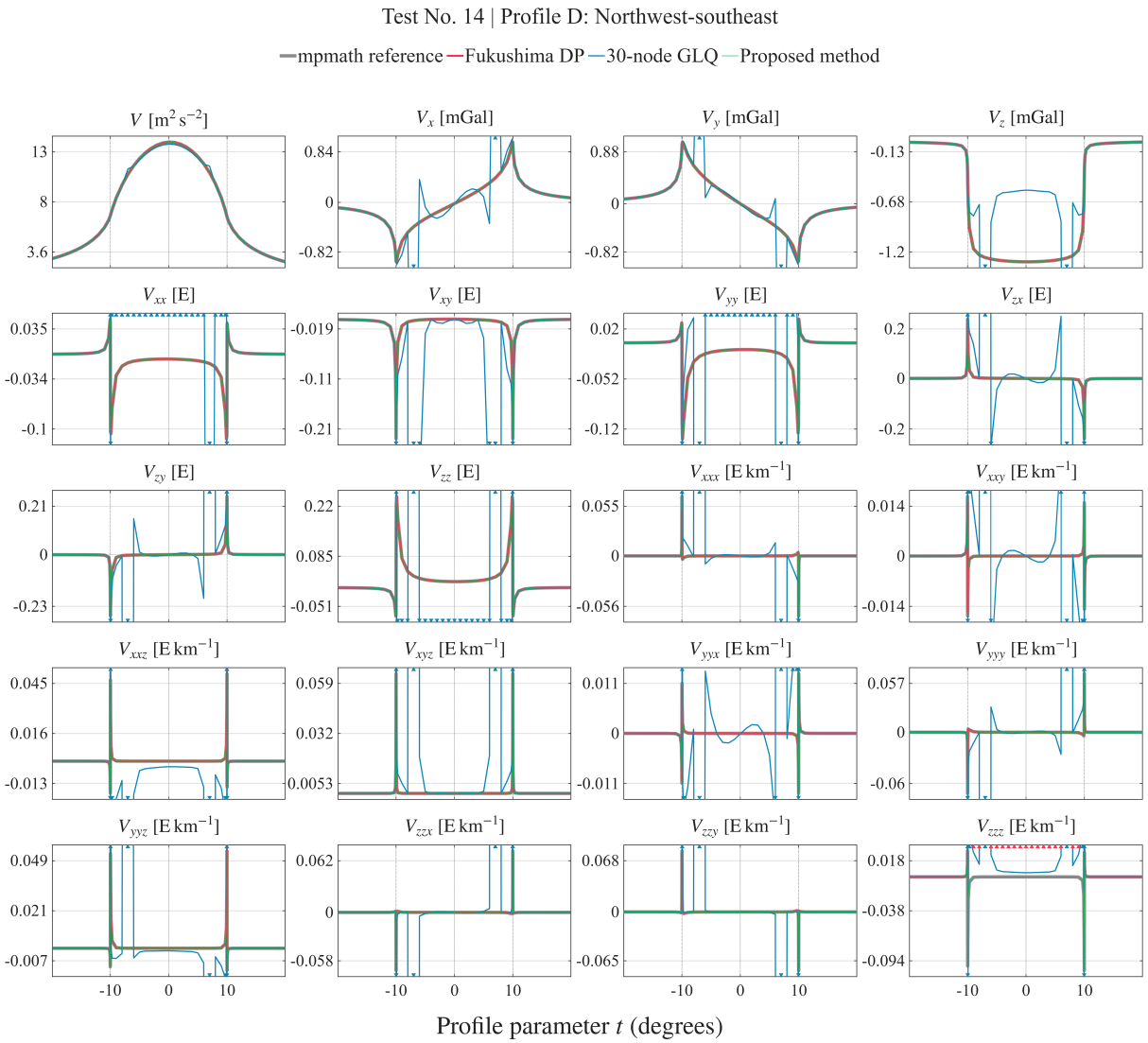


Figure S35: Test No. 14 (Profile D): signed physical field values along the northwest–southeast profile. The component order, curve styles, and boundary markers are as in Fig. S32.

8 C++ implementation and data interfaces

The C++ package contains cell-by-cell spherical-prism programs and MMSI programs for interpolated mass and magnetization shells. Both distribute observation points with MPI. Rank zero reads and broadcasts the model, gathers the field arrays, and writes the result file. Models, observation coordinates, and integration tolerances are supplied at run time.

Integration settings and diagnostics. The C++ adaptive integrator raises the relative input tolerance to at least $100\epsilon_{\mathrm{mach}} \simeq 2.22\times10^{-14}$. Its stopping logic includes estimated-error tests and resource or minimum-subregion limits; the default limit is 1,000 vectorized integrand calls per subintegral. The saved profile outputs do not include every internal termination flag. A finite output therefore does not certify that every subintegral met its requested tolerance. Accuracy claims in this work rest on the reported analytical and independent-reference discrepancies.

8.1 Software requirements

The programs require a C++17-compatible compiler, an MPI implementation, OpenMP support, and the header-only Eigen linear-algebra library. The supplied programs were developed and tested with the Intel oneAPI compiler and Intel MPI. Other C++17 compilers require compatible MPI wrappers and OpenMP support. Eigen is available from https://gitlab.com/libeigen/eigen; its header directory must be added to the compiler include path because the adaptive-quadrature and interpolation routines use both fixed-size and dynamic Eigen matrices. The relevant source and header files should be compiled together with optimization enabled. No external numerical-integration library is required.

All binary files use native C++ scalar representations: 32-bit signed integers for counts and 64-bit IEEE double-precision values for coordinates, model parameters, tolerances, and computed quantities. A file prepared on another architecture must use the byte order expected by the executable. Angular coordinates are geocentric latitude and longitude in degrees; all radii within a run use one consistent length unit. Arrays are stored contiguously in the order specified below. The implementation assumes $r_2 \geq r_1$, $\varphi_2 \geq \varphi_1$, and a consistently ordered longitude interval for every prism.

8.2 MATLAB interfaces for parameter and result files

MATLAB scripts for preparing binary parameter files and reading the computed results accompany both code sets. The cell-by-cell spherical-prism package includes:

- `Work1_SetGravityParameters.m`: writes `TFG.ForPar`;
- `Work1_SetMagneticParameters.m`: writes `TFM.ForPar`;
- `Work2_ReadGravityForwardResults.m`: reads the gravity output and exports a matrix data file;
- `Work2_ReadMagneticForwardResults.m`: reads the magnetic output and exports a matrix data file.

The MMSI package includes:

- `Work1_SetMSIGParameters.m`: creates the mass-shell parameter file;

- `Work1_SetMSIMParameters.m`: creates the magnetization-shell parameter file;
- `Work2_ReadMSIGResults.m`: reads `ResultG.dat`, retains component names, and exports the gravity arrays;
- `Work2_ReadMSIMResults.m`: reads `ResultM.dat`, retains component names, and exports the magnetic arrays.

In these script names, "Work1" and "Work2" denote parameter-writing and result-reading stages; they do not denote alternative forward solvers.

The scripts are intentionally explicit: each `fwrite` or `fread` statement corresponds to one field in the binary layouts documented below. They therefore serve as reference implementations for Python, Julia, Fortran, R, or other languages. Any reimplementation must preserve byte order, scalar widths, array ordering, and the sequence of field quantities, and should be verified against the supplied example parameter and result files.

8.3 Cell-by-cell spherical-prism programs

The cell-by-cell code set contains two executable entry points:

- `Work1_GravityEstimateMPI.cpp` computes the gravitational potential and all derivatives through third order;
- `Work1_MagneticEstimateMPI.cpp`:
  computes the magnetic potential, three-component magnetic-field vector, and magnetic-field-gradient tensor.

Their common numerical implementation is organized as follows:

- `_integral2.h`: vectorized adaptive two-dimensional quadrature, embedded error estimation, and recursive subdivision of the angular integration domain;
- `_tesseroid_integralkernel.h`: the surface-integral kernels and radial master moments used to construct the potential and its derivatives;
- `_tesseroid_estimate.h`: spherical-prism assembly, near-zone shell decomposition, peak-aware partitioning, coordinate transformations, and serial, OpenMP, and MPI interfaces.

The gravity program reads `TFG.ForPar`. Its binary fields, in order, are

`AbsTol`, `RelTol`; `NumModel`;
$\varphi_1[]$, $\varphi_2[]$, $\lambda_1[]$, $\lambda_2[]$, $r_1[]$, $r_2[]$, $\rho[]$;
`NumPoints`; $\lambda[]$, $\varphi[]$, $r[]$.

Here, the two tolerances are doubles, the two counts are 32-bit integers, every model array has `NumModel` doubles, and every observation array has `NumPoints` doubles. The output `Result_TFG.dat` starts with `NumPoints`, followed by λ, φ, and r, and then

$$V, V_x, V_y, V_z, V_{xx}, V_{xy}, V_{yy}, V_{zx}, V_{zy}, V_{zz}, V_{xxx}, V_{xxy}, V_{xxz}, V_{xyz}, V_{yyx}, V_{yyy}, V_{yyz}, V_{zzx}, V_{zzy}, V_{zzz}.$$

Each symbol denotes a complete array of `NumPoints` doubles rather than interleaved point records.

The magnetic program reads `TFM.ForPar`. Its layout is identical except that the density array is replaced by the three source-centred magnetization arrays M_x, M_y, and M_z, in that order. The output `Result_TFM.dat` contains the coordinate arrays followed by

$$V, V_x, V_y, V_z, V_{xx}, V_{xy}, V_{yy}, V_{zx}, V_{zy}, V_{zz}.$$

The magnetization components must follow the local Cartesian convention used in the main text. The code transforms each source magnetization vector into the local frame of the observation point before assembling the magnetic quantities.

8.4 MMSI programs

The MMSI code set contains the common headers `_integral2.h`, `_sphinterp.h`, `_tesseroid_surfacekernel.h`, and `_tesseroid_MSI.h`. The interpolation header evaluates the gridded shell interfaces and physical-property fields at quadrature points; the surface-kernel header provides the radially integrated kernels; and `_tesseroid_MSI.h` performs shell assembly, adaptive surface integration, peak treatment, and MPI/OpenMP parallelization.

Two complete-field entry points are supplied:

- `Work1_MassShellIntegrationMPI.cpp`: all 20 gravitational quantities for a mass shell;
- `Work1_MagnetizationShellIntegrationMPI.cpp`: all 10 magnetic quantities for a magnetization shell.

The mass-shell programs read `MSIG.ForPar`; the magnetization-shell programs read `MSIM.ForPar`. Both formats begin with

$$\texttt{AbsTol}, \texttt{RelTol}, \lambda_s, \lambda_e, \varphi_s, \varphi_e, \Delta\lambda, \Delta\varphi, \texttt{NumLon}, \texttt{NumLat}.$$

The first eight entries are doubles and the grid dimensions are 32-bit integers. The shell contains `NumLon`×`NumLat` nodes, with latitude varying fastest in the contiguous arrays. For `MSIG.ForPar`, the header is followed by $r_2[\,]$, $r_1[\,]$, and $\rho[\,]$. For `MSIM.ForPar`, it is followed by $r_2[\,]$, $r_1[\,]$, $M_x[\,]$, $M_y[\,]$, and $M_z[\,]$. Each file then stores `NumPoints` and the observation arrays $\lambda[\,]$, $\varphi[\,]$, and $r[\,]$.

The mass-shell result `ResultG.dat` uses the same coordinate and 20-quantity ordering as `Result_TFG.dat`. The magnetization-shell result `ResultM.dat` uses the same coordinate and 10-quantity ordering as `Result_TFM.dat`. Because complete arrays are written sequentially, a reader should first obtain the point count, allocate the arrays, and then read each block using exactly that count.

8.5 Units and conversion examples

The C++ programs return gravity arrays with density and radius-normalization factors included, before multiplication by G and conversion of length and output units. Let R_n denote a saved rank-n array, $n = 0, 1, 2, 3$, calculated with radii in kilometres and density in $\mathrm{kg\,m^{-3}}$.

With $\mathbf{x} = L\boldsymbol{\xi}$ and $L = 1000$ m, Newton's potential and its Cartesian derivatives satisfy

$$V = G\rho L^2 \mathcal{I}_0, \qquad D^{(n)}V = G\rho L^{2-n}\mathcal{I}_n, \tag{88}$$

where $\mathcal{I}_n$ is the corresponding geometric integral in the dimensionless coordinates. The volume element supplies L^3, the potential kernel supplies L^{-1}, and each spatial derivative supplies L^{-1}.

Using $g = G/(1\ \mathrm{m^3\,kg^{-1}\,s^{-2}}) = 6.67430 \times 10^{-11}$, the numerical values in the reported units are

$$\begin{aligned} V\ [\mathrm{m^2\,s^{-2}}] &= 10^6 g R_0, & 10^6 g &= 6.67430 \times 10^{-5}, \\ V_i\ [\mathrm{mGal}] &= 10^8 g R_1, & 10^8 g &= 6.67430 \times 10^{-3}, \\ V_{ij}\ [\mathrm{E}] &= 10^9 g R_2, & 10^9 g &= 6.67430 \times 10^{-2}, \\ V_{ijk}\ [\mathrm{E\,km^{-1}}] &= 10^9 g R_3, & 10^9 g &= 6.67430 \times 10^{-2}. \end{aligned} \tag{89}$$

Here 1 mGal $= 10^{-5}$ m s^{-2}, 1 E $= 10^{-9}$ s^{-2}, and 1 E km^{-1} $= 10^{-12}$ m^{-1} s^{-2}. Density is already present in R_n and is not applied again. If the radius inputs are in metres, the four coefficients are g, $10^5 g$, $10^9 g$, and $10^{12} g$ instead.

For a positive conversion coefficient C_n, the absolute error satisfies

$$\epsilon_{\mathrm{abs},n} = C_n |R_{n,\mathrm{num}} - R_{n,\mathrm{ref}}|. \tag{90}$$

Field values, absolute-error statistics, standard deviations, and RMSE use the same coefficient; relative errors and same-component error ratios are unchanged.

For magnetism, specify M_x, M_y, and M_z in A m^{-1}. In the magnetic output arrays, the legacy names V, V_i, and V_{ij} denote U, B_i, and $\partial_j B_i$, respectively, with the sign convention $\mathbf{B} = \nabla U$ defined in Appendix A of the main text. The magnetic conversions use the conventional approximation

$$\frac{\mu_0}{4\pi} \simeq 10^{-7}\ \mathrm{T\,m\,A^{-1}}.$$

With the recommended radius unit of kilometres, the useful direct conversions are

$$\begin{aligned} V_{\mathrm{mT\,m}} &= 0.1 V, & [V_{\mathrm{mT\,m}}] &= \mathrm{mT\,m}, \\ (V_i)_{\mathrm{nT}} &= 100 V_i, & [(V_i)_{\mathrm{nT}}] &= \mathrm{nT}, \\ (V_{ij})_{\mathrm{pT/m}} &= 100 V_{ij}, & [(V_{ij})_{\mathrm{pT/m}}] &= \mathrm{pT\,m^{-1}}. \end{aligned}$$

For example, a raw value of $V_z = 2.5$ corresponds to 250 nT, and $V_{zz} = 2.5$ corresponds to 250 pT m^{-1}. If all radii are instead specified in metres, the conversion factors for V, V_i, and V_{ij} are 10^{-4}, 100, and 10^5, respectively, for the same output units. Use the conversion factors for the radius unit supplied to the calculation.

8.6 Numerical and physical conventions

(1) **Tolerances.** `AbsTol` and `RelTol` control the embedded error test for the internal geometric integrals before physical output conversion.

(2) **Geometry.** Longitude, latitude, and radius must describe geocentric spherical coordinates. The source and observation radii must use the same length unit. Longitudes should be normalized consistently, and the gridded MMSI model must cover the stated interval without a mismatch between its dimensions and angular increments.

(3) **Model ordering.** Cell-by-cell model arrays use the same index for all boundaries and physical properties of a spherical prism. MMSI arrays use latitude-fast ordering, equivalent to column-major storage of a `NumLat`×`NumLon` matrix.

(4) **Output units.** The numerical routines evaluate the kernel combinations using the supplied radii, density, or magnetization. Users must apply the factors listed above, or rederive them when using a different radius unit. A physical constant or unit conversion must not be applied twice when post-processing saved results.

(5) **Parallel output.** Only MPI rank zero writes the final result. The ordering of observation points

in the output is identical to that in the parameter file, irrespective of the number of MPI processes.

(6) **Validation.** Before a large production calculation, the parameter-file reader and unit conversion should be checked using a small model whose output can be compared with one of the analytical or numerical tests reported in the main text and this Supplementary Information.

References

Deng XL (2023) Corrections to: "Accurate computation of gravitational field of a tesseroid" by Fukushima (2018) in J. Geod. 92(12):1371–1386. J Geod 97:8. https://doi.org/10.1007/s00190-022-01673-2

Deng XL (2025) Accurate computation of gravitational curvature of a tesseroid. J Geod 99:34. https://doi.org/10.1007/s00190-025-01958-2

Deng XL, Sneeuw N (2023) Analytical solutions for gravitational potential up to its third-order derivatives of a tesseroid, spherical zonal band, and spherical shell. Surv Geophys 44:1125–1173. https://doi.org/10.1007/s10712-023-09774-z

The mpmath development team (2023) mpmath: a Python library for arbitrary-precision floating-point arithmetic (version 1.3.0). Available at https://mpmath.org/

Zhong Y, Ren Z, Chen C, et al (2019) A new method for gravity modeling using tesseroids and 2D Gauss-Legendre quadrature rule. J Appl Geophys 164:53–64. https://doi.org/10.1016/j.jappgeo.2019.03.003